Pediatric Nursing & Newborn

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The Art of Pediatric Nursing

- The term pediatric is derived from the Greek words
- Pedia → child
- Iatrike → treatment
- Ics → branch of science

- Nursing of infants, children, and adolescents is consistent with the American Nurses Association (2010) definition of nursing as the protection, promotion, and optimization of health and abilities, prevention of illness and injury, alleviation of suffering through the diagnosis and treatment of human response, and advocacy in the care of individuals, families, and populations.

Definition Pediatric nursing

- Pediatric means the science of childcare and scientific treatment of childhood disease. Pediatric is synonymous with child health

- Pediatric nursing is the science of child care and scientific treatment of childhood. This branch of medical science deals with the care of children from conception to adolescence in health care

Role of the pediatric nurse

- The pediatric nurse is responsible for promoting the health and well-being of the child and family.
- The role of pediatric nurse is both caring and curing
- Caring is a continuous process in both wellness and illness. It refers a helping, guiding and counseling.
- Curing refers to the act of diagnosis and management, usually during illness.
- Pediatric nurse have the responsibilities of providing nursing care in hospital, home, clinical, school and community where children and their parents have health and counseling needs.

- Primary caregiver
- Health educator
- Nurse- counselor
- Social worker
- Team coordinator and collaborator
- Manager
- Child care advocate
- Recreationist
- Nurse consultant
- Nurse researcher

Objectives of pediatric nursing

1. Identify normal and abnormal growth and development of the child.
2. Identify health care needs of children (all ages, including infants and adolescents)
3. Establish nursing diagnosis for the client in the acute care arena as well as in the Community.
4. Identify the nursing process in caring for the child utilizing evidence-based practice.
5. Identify teaching strategies for the client and the family about the care required for their disease process.
6. Discuss ethical and legal principles when caring for the client.

Important vital statistical rates (Indicators)

- One way to measure the health status of a nation is to determine mortality rates of infants and children.
- Birth rate
- Specific death rate
- Perinatal mortality rate
- Neonatal mortality rate
- Postnatal mortality rate
- Infant mortality rate
- Under five mortality rate

Neonatal mortality rate

- Neonatal mortality rate is the number of neonates dying before reaching 28 days of age, per 1,000 live births in a given year.

$$\text{Neonatal mortality rate} = \frac{\text{Number of neonatal deaths}}{\text{Total number of live births}} \times 1000$$

Nursing Process

Nursing Process

- The nursing process is a method of problem identification and problem solving that describes what the nurse actually does. The nursing process model includes assessment, diagnosis outcomes identification, planning, implementation, and evaluation

- The nursing process is a proven form of problem solving based on the scientific method

The nursing process components

- Assessment

Dynamic way to collect and analyze data about a client. Assessment includes physiological, psychological, sociocultural, spiritual, economic, and life-style factors as well.

Assessment is a continuous process that operates at all phases of problem solving and is the foundation for decision-makin

Assessment involves multiple nursing skills and consists of the purposeful collection, classification, and analysis of data from a variety of sources

To provide an accurate and comprehensive assessment, the nurse must consider information about the patient's biophysical, psychologic, sociocultural, and spiritual background.

- Diagnosis

The diagnosis is the basis for the nurse’s care plan.

The next stage of the nursing process is problem identification and nursing diagnosis

At this point, the nurse must interpret and make decisions about the data gathered

Not all children have actual health problems; some have a potential health problem, which is a risk state that requires nursing intervention to prevent the development of an actual problem

Potential health problems may be indicated by risk factors, or signs and predispose a child and family to a dysfunctional health pattern and are limited to individuals at greater risk than the population as a whole

Nursing interventions are directed toward reducing risk factors

- Outcomes / Planning

Based on the assessment and diagnosis, the nurse sets measurable and achievable short- and long-range goals

The goal for outcomes identification is to establish priorities and select expected patient outcomes or goals

- Planning

After identifying specific patient- and family-focused goals, the nurse develops a care plan specific to the identified outcomes.

The outcome is the projected or expected change in a patient's health status, clinical condition, or behavior that occurs after nursing interventions have been instituted.

The care plan must be established before specific nursing interventions are developed and implemented.

- Implementation

Nursing care is implemented according to the care plan, so continuity of care for the patient during hospitalization and in preparation for discharge needs to be assured. Care is documented in the patient’s record.

The implementation phase begins when the nurse puts the selected intervention into action and accumulates feedback data regarding its effects (or the patient's response to the intervention).

The feedback returns in the form of observation and communication and provides a database on which to evaluate the outcome of the nursing intervention

Throughout the implementation stage, the main concerns are the patient's physical safety and psychologic comfort in terms of atraumatic care

- Evaluation

Both the patient’s status and the effectiveness of the nursing care must be continuously evaluated, and the care plan modified as needed.

Evaluation is the last step in the nursing care process

The nurse gathers, sorts, and analyzes data to determine whether (1) the established outcome has been met, (2) the nursing interventions were appropriate, (3) the plan requires modification, or (4) other alternatives should be considered. The evaluation phase either completes the nursing process (outcome is met) or serves as the basis for selecting alternative interventions to solve the specific problem.

With the current focus on patient outcomes in health care, the patient's care is evaluated not only at discharge but thereafter as well to ensure that the outcomes are met and there is adequate care for resolving existing or potential health problems

- Documentation

Although documentation is not one of the steps of the nursing process, it is essential for evaluation.

The nurse can assess, diagnose, and identify problems; plan; and implement without documentation; however, evaluation is best performed with written evidence of progress toward outcomes

Normal Newborn

Neonatal Period

A period from birth 4 weeks postnatal. After the initial observation for neonatal condition requiring immediate intervention, the baby is sent to the normal newborn nursery or maternity floor for the purpose of follow up and stabilization.

Early Neonatal Period: from delivery to the first 7 days of delivery (0-7 days)

Late Neonatal Period: from 7 days of delivery to 28 days (7-28 days)

Immediate Care of the Newborn

- Immediately after birth, the most common problems are low temperature and rapid breathing.

- A baby begins to lose heat immediately after birth. Heat loss can be prevented and low temperature can be avoided by drying the baby immediately after birth and removing wet towels or clothing. The baby should be placed naked between the mother’s breasts, and the mother and baby should be covered with a clean, dry drape or cloth. The baby’s head should be covered. Monitor the baby’s temperature about every 15-30 minutes by feeling the skin. Measure the temperature if the skin is cool.

- Skin-to-skin care should begin at birth and should continue for at least one hour. If a mother is not well, other family members can provide skin-to-skin care.

- Rapid breathing after birth often occurs because of delay in absorption of lung fluid and usually resolves rapidly. However, rapid breathing may persist or be caused by a more serious problem that would require advanced care. Breathing in all babies should be monitored every 15 minutes for rapid breathing (>60/min) and chest indrawing until the time of the first complete exam. This can be done by observation alone without handling if the baby appears well.

Immediate Care of the Newborn

1. Clear airway (suction mouth, pharynx)
2. Established respiration.
3. Maintenance of temperature (dry, wrap).
4. Clamping the cord.
5. Provide eye care
6. Identification.
7. Vit K to prevent hemorrhage
8. Anthropometric measurement

Maintain a Patent Airway

Establishing a patent airway is a primary objective in the delivery room. When the newborn is supine, a neutral neck position (i.e., avoiding neck flexion or hyperextension) is critical to achieving and maintaining a patent airway.

Bulb syringe is kept near the infant and is used if suctioning is required. If more forceful removal of secretions is required, mechanical suction is used.

The use of the properly sized catheter and correct suctioning technique is essential to prevent mucosal damage and edema.

Gentle suctioning is necessary to prevent reflex bradycardia, laryngospasm, and cardiac arrhythmias from vagal stimulation.

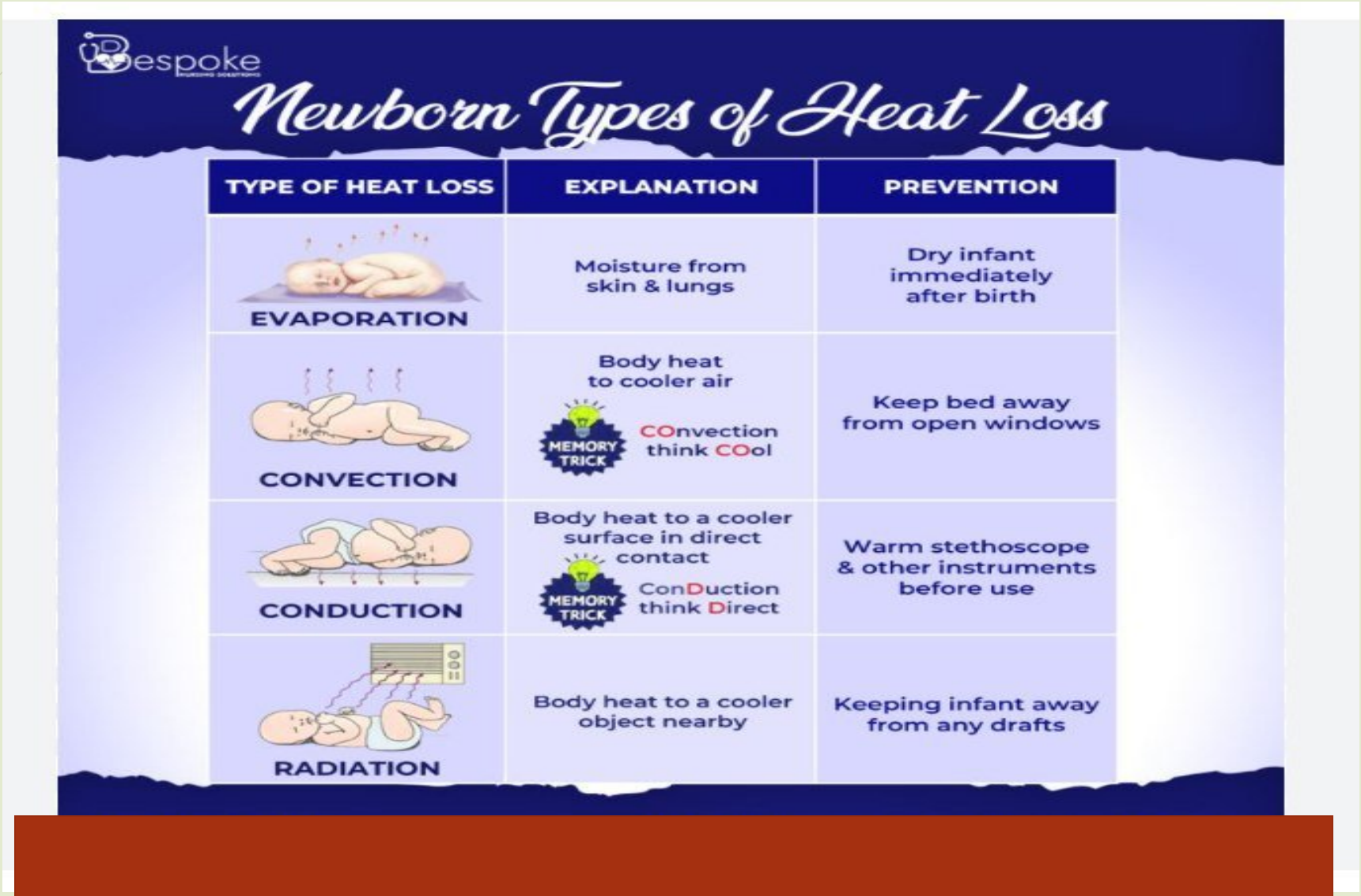
Oropharyngeal suctioning is performed for up to 5 seconds, with sufficient time between each attempt to allow the infant to reoxygenate.

- The cardinal signs of respiratory distress in a newborn include tachypnea, nasal flaring, grunting, intercostal retractions, and cyanosis.

Established respiration

- Monitor for rapid breathing (>60 breathes/ minute) and chest indrawing every 15 minutes until first complete exam.
- Many babies who breathe fast but without increased effort gradually improve.
- Babies with severe breathing problems or babies with mild breathing problems who do not improve need advanced care.

Maintenance of temperature
Methods of heat loss



At birth, a major cause of heat loss is evaporation, the loss of heat through moisture. The amniotic fluid that bathes the infant's skin favors evaporation, especially when combined with the cool atmosphere of the delivery room

- Heat loss through evaporation is minimized by rapidly drying the skin and hair with a warmed towel and placing the infant in skin-to-skin contact with the mother, covered by a blanket.
- Prevention: Carefully dry the infant after delivery or after bathing.

Radiation, the loss of heat to cooler solid objects in the environment that are not in direct contact with the infant. Loss of heat through radiation increases as these solid objects become colder and closer to the infant.

- To prevent this, place cribs as far away as possible from exterior walls, windows, and ventilating units

Conduction involves loss of heat from the body because of direct contact of skin with a cooler solid object.

- Placing the infant on a padded, covered surface and providing insulation through clothes and blankets rather than directly on a cool hard table can minimize heat loss.
- Placing the newborn skin to- skin with the mother on her chest or abdomen immediately after delivery is physically beneficial in terms of conserving heat, as well as fostering maternal attachment and breastfeeding.
- Prevention: Warm all objects before the infant comes into contact with them.

Convection is similar to conduction except that heat loss is aided by surrounding air currents. For example, placing the infant in the direct flow of air from a fan or air-conditioner vent will cause rapid heat loss through convection.

- Transporting the neonate in a crib with solid sides reduces airflow around the infant.
- Prevention:· Keep infant out of drafts· Close one end of heat shield in incubator to reduce velocity of air.

Protect from Infection and Injury

The most important practice for preventing cross-infection is thorough hand washing of all individuals involved in the infant's care. Other procedures to prevent infection include eye care, umbilical care, bathing, and care of the circumcision.

Identification

Proper identification of the newborn is absolutely essential. The nurse must verify that identifying bands are securely fastened and verify the information (name, gender, mother's admission number, date, and time of birth) against the birth records and the child's actual gender. This identification process should take place optimally in the delivery room

Eye Care

Prophylactic eye treatment against ophthalmia neonatorum, infectious conjunctivitis of the newborn, includes the use of (1) silver nitrate (1%) solution, (2) erythromycin (0.5%) ophthalmic ointment or drops, or (3) tetracycline (1%) ophthalmic ointment or drops (preferably in single-dose ampules or tubes).

All three are effective against gonococcal conjunctivitis. Chlamydia trachomatis is the major cause of ophthalmia neonatorum in the United States; topical antibiotics (tetracycline and erythromycin) and silver nitrate are not effective in the prevention and treatment of chlamydial conjunctivitis

Eye Care

Herpes simplex virus may also cause neonatal conjunctivitis; treatment in such cases involves the use of topical and systemic antiviral medications.

Recent publications have explored alternate substances for ophthalmia neonatorum prevention, with a focus on growing concerns about the development of antimicrobial resistance. Colostrum (Ghaemi, Navaei, Rahimirad, et al, 2014) and povidone iodine (2.5%) (David, Rumelt, and Weintraub, 2011; Meyer, 2014) have been studied with small samples of infants.

Vitamin K Administration

Shortly after birth, vitamin K is administered to prevent hemorrhagic disease of the newborn. Normally, vitamin K is synthesized by the intestinal flora. However, because infants' intestines are relatively sterile at birth and because breast milk contains low levels of vitamin K, the supply is inadequate for at least the first 3 or 4 days.

The major function of vitamin K is to catalyze the synthesis of prothrombin in the liver, which is needed for blood clotting. The vastus lateralis muscle is the traditionally recommended injection site, but the ventrogluteal (not the dorso-gluteal) muscle can be used.

Current recommendations are that vitamin K be given to all newborns as a single intramuscular dose of 0.5 to 1.0 mg

Hepatitis B Vaccine Administration

To decrease the incidence of hepatitis B virus in children and its serious consequences (cirrhosis and liver cancer) in adulthood, the first of three doses of hepatitis B vaccine are recommended soon after birth and before hospital discharge for all newborns born to hepatitis B surface antigen (HBsAg)- negative mothers

Care of the Umbilicus

Because the umbilical stump is an excellent medium for bacterial growth, various methods of cord care have been practiced to prevent infection. Some methods popular in the past include the use of an antimicrobial agent (such as, bacitracin or triple dye) and agents (such as alcohol or povidone iodine). The use of antiseptic agents has been shown to prolong cord drying and separation

Bathing

Bath time is an opportunity for the nurse to accomplish much more than general hygiene. It is an excellent time for observing the infant's behavior, state of arousal, alertness, and muscular activity.

Older studies suggested that healthy full-term newborns with a stable body temperature could be safely bathed as early as 1 hour of age without experiencing problems

More recent studies have demonstrated that early bathing (within the first hour of life), interferes with skin-to-skin holding and breastfeeding, compromising basic protection against neonatal infection and possibly contributing to problems such as hypothermia and hypoglycemia

One of the most important considerations in skin cleansing is preservation of the skin's acid mantle, which is formed from the uppermost horny layer of the epidermis; sweat; superficial

fat; metabolic products; and external substances, such as amniotic fluid.



Circumcision

Circumcision, the surgical removal of the foreskin on the glans penis

- Benefits of Neonatal Circumcision

1. Prevention of penile cancer and posthitis (inflammation of prepuce)
2. Decreased incidence of balanitis (inflammation of glans), urinary tract infections in male infants, and some sexually transmitted infections later in life (herpes, syphilis, gonorrhea)
3. Decreased incidence of human immunodeficiency virus (HIV) infection, human papillomavirus (HPV), and cervical cancer (in female partner)

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Breastfeeding

Human milk is the preferred form of nutrition for all infants.

Healthy People 2020 has a goal to increase breastfeeding rates in the United States to 81.9% in early postpartum and to 61% for mothers who continue to breastfeed for at least 6 months

Studies exploring breastfeeding mothers' reasons for early cessation of breastfeeding

suggest several factors contribute to this decision, such as problems with lactation,

concerns with newborn or maternal health, and lower maternal education and

professional and social support

The Baby-Friendly Initiative (BFI) is a joint effort of the WHO and the UNICEF to

Encourage, promote, and support breastfeeding as the model for optimum infant

nutrition.

Ten Steps to Successful Breastfeeding

- Every facility providing maternity services and care for newborn infants should:

1. Have a written breastfeeding policy that is routinely communicated to all health care staff.
2. Train all health care staff in skills necessary to implement this policy.
3. Inform all pregnant women about the benefits and management of breastfeeding.
4. Help mothers initiate breastfeeding within hour of birth.
5. Show mothers how to breastfeed and how to maintain lactation even if they should be separated from their infants.
6. Give newborn infants no food or drink other than breast milk unless medically indicated.
7. Practice rooming-in—allowing mothers and infants to remain together—24 hours a day.
8. Encourage breastfeeding on demand.
9. Give no artificial teats or pacifiers (also called dummies or soothers) to breastfeeding infants.
10. Foster the establishment of breastfeeding support groups, and refer mothers to them on discharge from the hospital or clinic.

Breastfeeding

Human milk is the most economical form of feeding.

It is always available, ready to serve at body temperature, and free of contamination.

Although human milk is not sterile, healthy full-term infants can tolerate varying amounts of nonpathogenic and pathogenic organisms.

Human milk's protection against infection can provide additional cost

savings in terms of fewer medical visits

Breastfed infants, especially beyond 2 to 3 months old, tend to grow at a satisfactory but slower rate than bottle-fed infants.

Breastfeeding: Contraindications to breastfeeding include the following

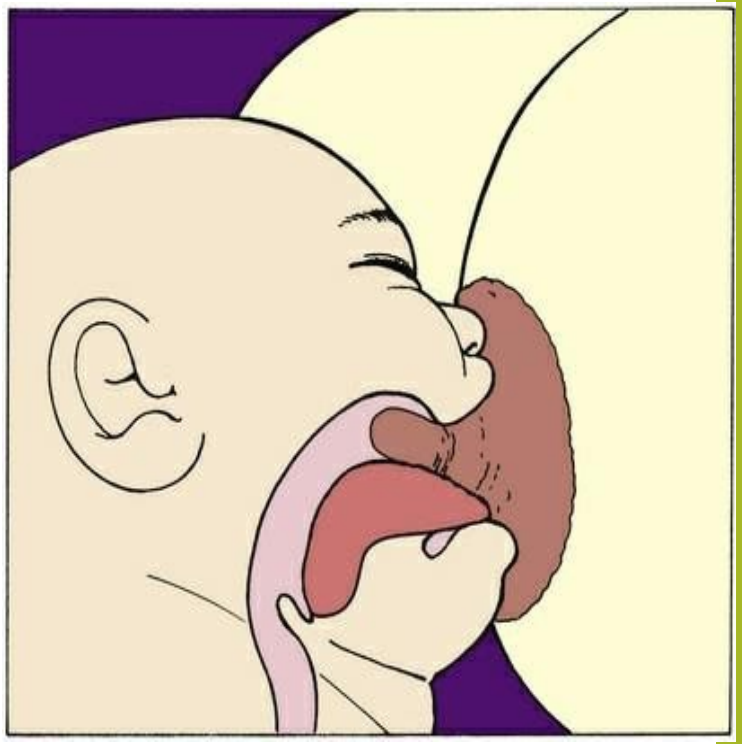
1. Maternal chemotherapy antimetabolites and certain antineoplastic drugs
2. Active tuberculosis not under treatment in the mother
3. HIV in the mother
4. Galactosemia in the infant
5. Maternal herpes simplex lesion on a breast
6. Cytomegalovirus (CMV): May be a risk to extremely low birth weight preterm infants (<1500 gm). CMV is not a risk for full-term infants whose mother is seropositive for CMV.
7. Maternal substance abuse with street drugs (e.g., phencyclidine [PCP], cocaine, and cannabis)
8. Human T-cell leukemia virus types I and II
9. Mothers who are receiving diagnostic or radioactive isotopes or who have had exposure to radioactive materials (for as long as there is radioactivity in milk)

Three main criteria have been proposed as essential in promoting positive breastfeeding:

1. Absence of a rigid feeding schedule;
2. Correct positioning of the infant at the breast to achieve a deep, areolar latch; and
3. Correct suckling technique. Correct suckling for breastfeeding is defined as a wide-open mouth, tongue under the areola, and expression of milk by effective alveolar compression

The following interventions promote breastfeeding:

- Frequent and early breastfeeding, especially during the first hour of life; immediate skin-to-skin contact; non-separation of mother and infant; and feeding on demand
- Increased information and support to mothers after discharge, including phone follow-up
- Early breast pumping every 2 to 3 hours for 10 to 15 minutes bilaterally if the newborn is unable to breastfeed immediately (increases oxytocin production and thus milk production)



Early Newborn Discharge Criteria

- It was a singleton birth between 38 and 42 weeks of gestation.
- Baby was delivered by uncomplicated vaginal delivery.
- Birth weight is appropriate for gestational age (AGA).
- Physical examination was normal.
- Vital signs are within normal range and stable for the 12 hours preceding discharge.
- Infant has urinated and passed at least one stool.
- Infant has completed at least two successful feedings.
- Clinical significance of jaundice, if present, has been determined and appropriate management or follow-up plans put in place.
- Appropriate maternal and infant blood tests have been performed.
- Appropriate neonatal immunizations have been administered.
- Newborn hearing screening has been completed per hospital protocol and state regulations.
- Family, environmental, and social risk factors have been assessed.
- Documentation is in place that the mother has received usual infant care training and has demonstrated competency.
- Support persons are available to assist mother and her infant after discharge.
- Continuing medical care is planned, including that infants discharged sooner than 48 hours be examined within 48 hours of discharge from the hospital.

Nursing Care of the Newborn and Family

- Assessment

Newborns require thorough, skilled observation to ensure a satisfactory adjustment to extrauterine life. Physical assessment after delivery can be divided into four phases:

1. The initial assessment, which includes the Apgar scoring system
2. Transitional assessment during the periods of reactivity
3. Assessment of gestational age
4. Systematic physical examination

Initial Assessment: Apgar Scoring

- The most frequently used method to assess newborns' immediate adjustment to extrauterine life is the Apgar scoring system, which is based on newborn heart rate, respiratory effort, muscle tone, reflex irritability, and color.
- Each item is given a score of 0, 1, or 2. Evaluations of all five categories are made at 1 and 5 minutes after birth and repeated until the infant's condition stabilizes
- Total scores of 0 to 3 represent severe distress, scores of 4 to 6 signify moderate difficulty, and scores of 7 to 10 indicate absence of difficulty in adjusting to extrauterine life.
- The Apgar score is affected by the degree of physiologic immaturity, infection, congenital malformations, maternal sedation or analgesia, and neuromuscular disorders.

APGAR scoring system (1&5 min)

1. Appearance (skin color)
2. Pulse (heart rate)
3. Grimace response (reflexes)
4. Activity (muscle tone)
5. Respiration (breathing rate and effort)

APGAR Score

Score / Item	2	1	zero
Heart beats	> 100 b/min Strong	< 100 b/min Or weak beats	No heart beats
Cry & breathing	Strong crying	weak crying / irregular breathing	No cry / breathing
Color	Pink body & face	Pink body & blue extremities	Pale or blue body
Movement & tone	Active	Some movements	Flaccid
Grimace	Try to keep cath. away	Grimace of face	No response

Activate Windo
Go to Settings to act

APGAR scoring chart

SIGN	0	1	2	1 min	5 min
Heart Rate	Absent	Less Than 100	Over 100	2	2
Respiratory Effort	Absent	Slow, Irregular	Good Cry	1	2
Muscle Tone	Limp	Some Flexion	Active Motion	1	2
Reflex Irritability	No Response	Grimace	Cry	1	2
Color	Pale	Body Pink, Extr. Blue	All Pink	1	2
TOTAL SCORE				6	10

Activate Windows

- Assessment of APGAR score
- 7-10 well adjusted
- 5-7 mild asphyxia
- 3-4 moderate asphyxia
- < 3 severe asphysxia

Clinical Assessment of Gestational Age

- Assessment of gestational age is an important criterion because perinatal morbidity and mortality are related to gestational age and birth weight. A frequently used method of determining gestational age is the New Ballard Scale (NBS)

- Each sign has a number score, and the cumulative score correlates with a maturity rating of 20 to 44 weeks of gestation.

ESTIMATION OF GESTATIONAL AGE BY MATURITY RATING

NEUROMUSCULAR MATURITY

	-1	0	1	2	3	4	5
Posture							
Square Window (wrist)	> 90°	90°	60°	45°	30°	0°	
Arm Recoil		180°	140°- 180°	110° 140°	90°- 110°	< 90°	
Popliteal Angle	180°	160°	140°	120°	100°	90°	< 90°
Scarf Sign							
Heel to Ear							

PHYSICAL MATURITY

Skin	sticky friable transparent	gelatinous red, translucent	smooth pink, visible veins	superficial peeling &/or rash, few veins	cracking pale areas rare veins	parchment deep cracking no vessels	leathery cracked wrinkled
Lanugo	none	sparse	abundant	thinning	bald areas	mostly bald	
Plantar Surface	heel-toe 40-50 mm: –1 <40 mm: –2	>50 mm no crease	faint red marks	anterior transverse crease only	creases ant. 2/3	creases over entire sole	
Breast	imperceptible	barely perceptible	flat areola no bud	stippled areola 1-2 mm bud	raised areola 3-4 mm bud	full areola 5-10 mm bud	
Eye/Ear	lids fused loosely: –1 tightly: –2	lids open pinna flat stays folded	slightly curved pinna; soft; slow recoil	well-curved pinna; soft but ready recoil	formed & firm instant recoil	thick cartilage ear stiff	
Genitals (male)	scrotum flat, smooth	scrotum empty faint rugae	testes in upper canal rare rugae	testes descending few rugae	testes down good rugae	testes pendulous deep rugae	
Genitals (female)	clitoris prominent labia flat	prominent clitoris small labia minora	prominent clitoris enlarging minora	majora & minora equally prominent	majora large minora small	majora cover clitoris & minora	

MATURITY RATING

score	weeks
–10	20
–5	22
0	24
5	26
10	28
15	30
20	32
25	34
30	36
35	38
40	40
45	42
50	44

A

A, Ballard scale for newborn maturity rating. Expanded scale includes extremely premature infants and has been refined to improve accuracy in more mature infants. (A, From Ballard JL, Khoury JC, Wedig K, et al: New Ballard score expanded to include extremely premature infants, *J Pediatr* 119:417, 1991.)

General Measurements

Head circumference

- Several important measurements of newborns have significance when compared with each other and when recorded over time on a graph. For full-term infants, average head circumference is between 33 and 35.5 cm (13 and 14 inches).
- Head circumference may be somewhat less immediately after birth because of the molding process that occurs during vaginal deliveries. Usually by the second or third day, the skull is normal in size and contour.

Measurement of infant length.

- Head-to-heel length is also measured. Because of the usual flexed position of infants, it is important to extend the legs completely when measuring total body length. The average length of newborns is 48 to 53 cm (19 to 21 inches).

Body weight

Body weight should be measured soon after birth because weight loss occurs fairly rapidly.

Normally, neonates lose about 10% of their birth weight by 3 to 4 days of age because of loss of extracellular fluid and meconium, as well as limited food intake, especially in breastfed infants.

The birth weight is usually regained by the tenth to fourteenth day of life. Most newborns weigh 2700 to 4000 g (6 to 9 pounds), the average weight being about 3400 g (7.5 pounds).

Accurate birth weights and lengths are important because they provide a baseline for assessment of future growth.

Vital Signs

Another category of measurements is vital signs. Axillary temperatures are taken because insertion of a thermometer into the rectum can potentially cause perforation of the mucosa if performed incorrectly.

Core body temperature varies according to the periods of reactivity but is usually 36.5° to 37.6° C (97.7° to 99.7° F). Skin temperature is slightly lower than core body temperature

General Appearance

Before each body system is assessed, it is important to describe the general posture and behavior of the newborn. The overall appearance yields valuable clues to the infant's physical status.

In full-term neonates, the posture is one of complete flexion as a result of in utero position. Most infants are born in a vertex presentation with the head flexed and the chin resting on the upper chest, the arms flexed with the hands clenched, the legs flexed at the knees and hips, and the feet dorsiflexed.

- Skin

The texture of the newborn's skin is velvety smooth and puffy, especially about the eyes, the legs, the dorsal aspect of the hands and the feet, and the scrotum or labia. Skin color depends on racial and familial background and varies greatly among newborns. In general, white infants are usually pink to red.

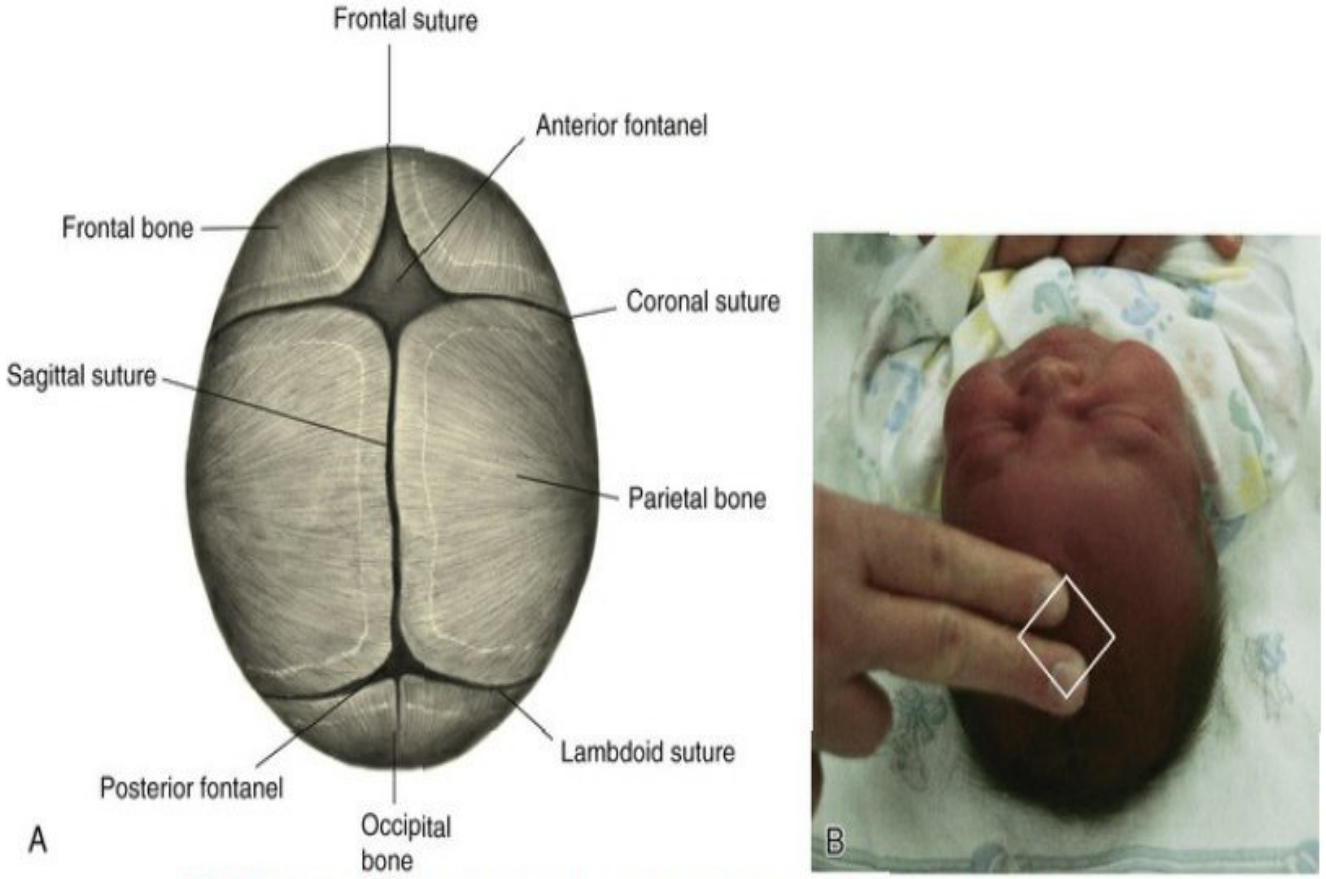


FIG 7-6 A, Location of sutures and fontanels. B, Palpating the anterior fontanel.

- Eyes

  Because newborns tend to have their eyes tightly closed, it is best to begin the examination of the eyes by observing the eyelids for edema, which is normally present for the first 2 days after delivery. The eyes are observed for symmetry. Tears may be present at birth, but purulent discharge from the eyes shortly after birth is abnormal.

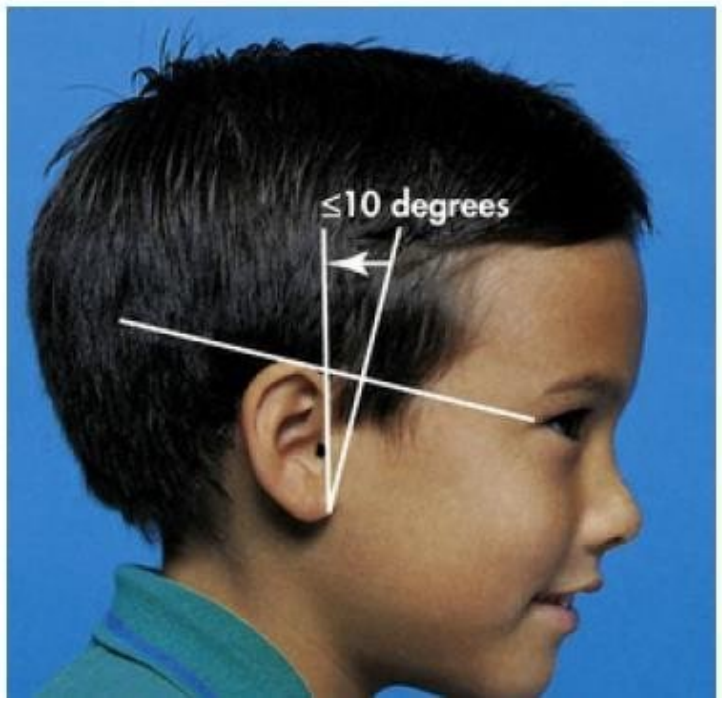
- Ears

  The ears are examined for position, structure, and auditory function. The top of the pinna should lie in a horizontal plane to the outer canthus of the eye. The pinna is often flattened against the side of the head from pressure in utero. An otoscopic examination may be difficult to perform if the canals are filled with vernix caseosa and amniotic fluid, making visualization of the tympanic membrane difficult.

Ears

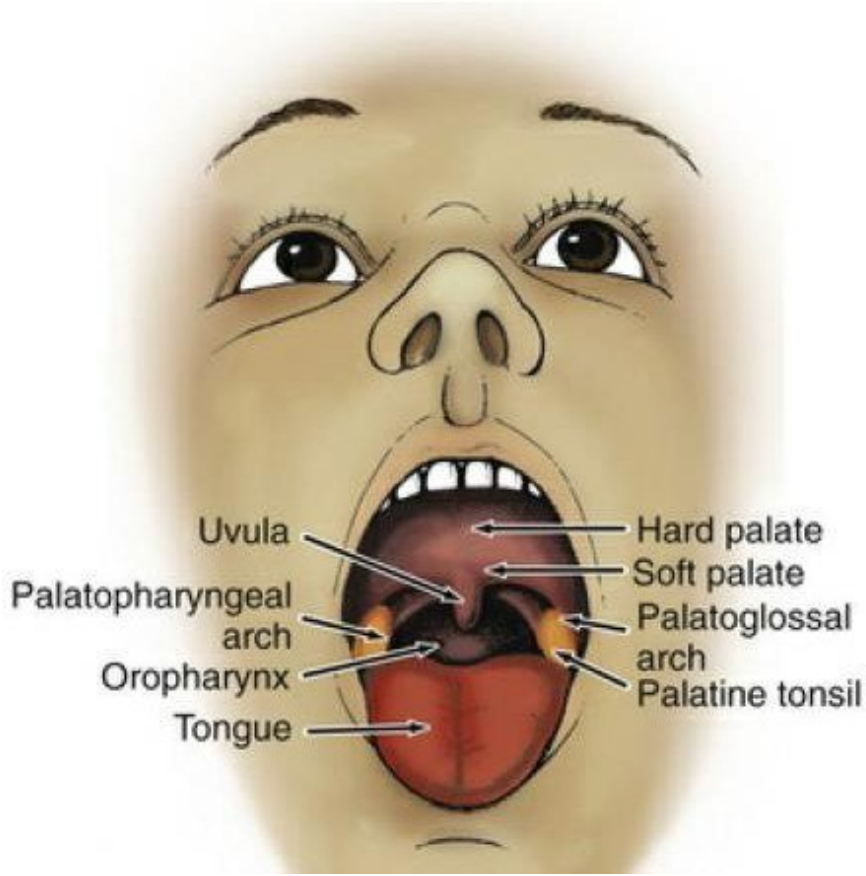
Inspection of External Structures

The entire external ear is called the **pinna**, or **auricle**; one is located on each side of the head. Measure the height alignment of the pinna by drawing an imaginary line from the outer orbit of the eye to the occiput, or most prominent protuberance of the skull. The top of the pinna should meet or cross this line. Low-set ears are commonly associated with renal anomalies or cognitive impairment. Measure the angle of the pinna by drawing a perpendicular line from the imaginary horizontal line and aligning the pinna next to this mark. Normally the pinna lies within a 10-degree angle of the vertical line (Fig. 4-20). If it falls outside this area, record the deviation and look for other anomalies.



Inspection of Internal Structures

The major structures that are visible within the oral cavity and oropharynx are the mucosal lining of the lips and cheeks, gums (or gingiva), teeth, tongue, palate, uvula, tonsils, and posterior oropharynx (Fig. 4-27). Inspect all areas lined with **mucous membranes** (inside the lips and cheeks, gingiva, underside of the tongue, palate, and back of the pharynx) for color, any areas of white patches or ulceration, bleeding, sensitivity, and moisture. The membranes should be bright pink, smooth, glistening, uniform, and moist.



- Nose

The nose is usually flattened after birth, and bruises are common. Patency of the nasal canals can be assessed by holding a hand over the infant's mouth and one canal and noting the passage of air through the unobstructed opening. If nasal patency is questionable, report it because most newborns are obligatory nose breathers and are unable to breathe orally in response to nasal occlusion. Sneezing and thin white mucus are common up to several hours after birth.

- Mouth and Throat

An external defect of the mouth (such as, cleft lip) is readily apparent; however, the internal structures require careful inspection. The palate is normally highly arched and somewhat narrow. Rarely, teeth may be present. A common finding is Epstein pearls, small, white, epithelial cysts along both sides of the midline of the hard palate. They are insignificant and disappear in several weeks.

- Neck

Because the newborn's neck is short and covered with folds of tissue, adequate assessment of the neck requires allowing the head to fall gently backward in hyperextension while the back is supported in a slightly raised position. Observe for range of motion, shape, and any abnormal masses and palpate each clavicle for possible fractures.

- Chest

The shape of the newborn's chest is almost circular because the anteroposterior and lateral diameters are equal. The ribs are flexible, and slight intercostal retractions are normally seen on inspiration. The xiphoid process is commonly visible as a small protrusion at the end of the sternum. The sternum is generally raised and slightly curved.

- Back and Rectum

Inspect the spine with the infant prone. The shape of the spine is gently rounded, with none of the characteristic S-shaped curves seen later in life. Any abnormal openings, masses, dimples, or soft areas are noted. A protruding sac anywhere along the spine, but most commonly in the sacral area, indicates some type of spina bifida.

Meconium

Meconium, the first stool of the newborn, is a thick black tarry substance composed of dead cells, mucus, and bile that collects in the rectum of the fetus. Passage of meconium should occur within the first 24 to 48 hours and confirms the presence of a patent anus.

Vernix Caseosa

- Vernix serves as:

  Vernix serves is the waxy or cheese-like white substance found coating the skin of newborn

  It is a lubricant and is protective against infections; therefore, it is best to gently massage the vernix into the skin and allow it to wear off naturally.

Desquamation

(Peeling of the hands and feet) begins in the second and third weeks. Usually the fingertips peel first followed by the feet. The peeling is painless, but the new skin may be tender

commonly called skin peeling is the shedding of the outermost membrane or layer of a tissue, such as the skin.

The term is from Latin desquamare, meaning 'to scrape the scales off a fish.

Physiological Jaundice

A newborn's immature liver often can't remove bilirubin quickly enough, causing an excess of bilirubin. Jaundice due to these normal newborn conditions is called physiologic jaundice, and it typically appears on the second or third day of life.

Milia

Small white or yellow pinpoint spots. - Common on the nose, forehead, & chin of the newborn infants due to accumulations of secretions from the sweat & sebaceous glands that have not yet drain normally. They will disappear within 1-2 weeks, they should not expressed.

Head

General observation of the contour of the head is important because molding occurs in almost all vaginal deliveries. In a vertex delivery, the head is usually flattened at the forehead, with the apex rising and forming a point at the end of the parietal bones and the posterior skull or occiput dropping abruptly.

The Anterior fontanel: is diamond in shape, located at the junction of 2 parietal & frontal bones. It is 2-3 cm in width & 3-4 cm in length. It closes between 12-18 months of age. The posterior fontanel: is triangular in shape, located between the parietal & occipital bones. It closes by the 2nd month of age.

Fontanels should be flat, soft, & firm. It bulge when the baby cries or if there is increased in ICP. Two conditions may appear in the head: Caput succedaneum & cephlhemtoma

Caput succedaneum

An edematous swelling on the presenting portion of the scalp of an infant during birth, caused by the pressure of the presenting part against the dilating cervix.

Cephalhematoma

Cephalhematoma is a subperiosteal collection of blood secondary to rupture of blood vessels between the skull and the periosteum, in which bleeding is limited by suture lines (never cross the suture lines).

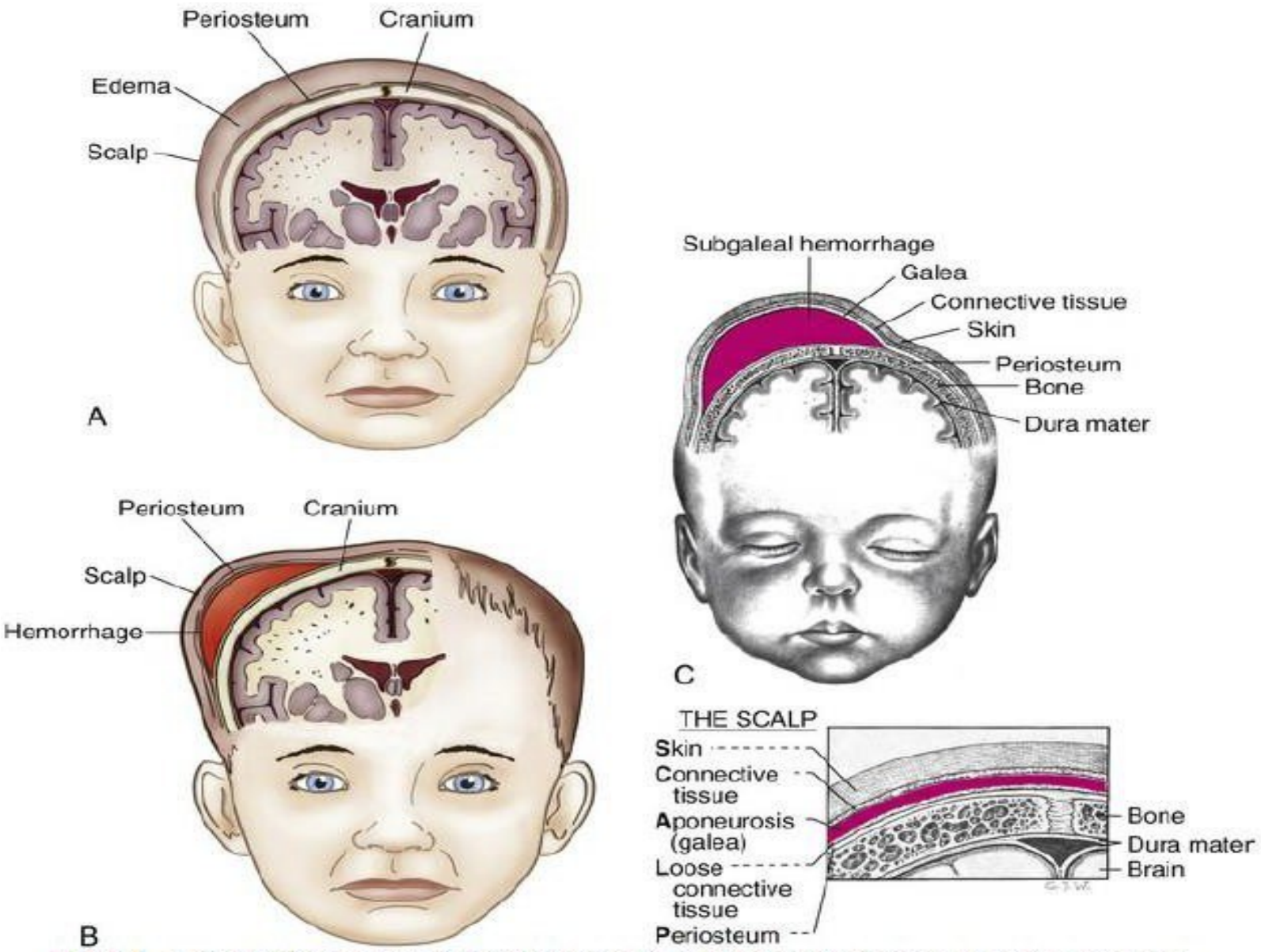


FIG 8-1 A, Caput succedaneum. B, Cephalhematoma. C, Subgaleal hemorrhage. (A and B, From Seidel HM, Ball JM, Davis JE, et al: *Mosby's guide to physical examination*, ed 6, St Louis, 2006, Mosby.)

Cleft Palate

When a baby is born with an opening (a cleft) in the roof of the mouth. This leaves a hole between the nose and the mouth

Cleft Lip

An opening or split in the upper lip that occurs when developing facial structures in an unborn baby don't close completely

Ankyloglossia

A condition present at birth that restricts the tongue's range of motion. With tongue-tie, an unusually short, thick or tight band of tissue (lingual frenulum) tethers the bottom of the tongue's tip to the floor of the mouth, so it may interfere with breast-feeding

Endocrine system

Swollen breasts: Appears on 3rd day in both sex, & lasts for 2-3 weeks and gradually disappears without treatment. N.B: The breasts should not be expressed as this may result in infection or tissue damage.

Infantile menstruation

During pregnancy, a surge in maternal estrogen levels can stimulate a female fetus's uterus. Within the first week of life, it's not uncommon for baby girls to have a mini period in which the uterus sheds a little blood.

Facial Paralysis

Pressure on the facial nerve (cranial nerve VII) during delivery may result in injury to that nerve.

The primary clinical manifestations are loss of movement on the affected side, such as an inability to completely close the eye, drooping of the corner of the mouth, and absence of wrinkling of the forehead and nasolabial fold (Fig. 8-2). The paralysis is most noticeable when the infant cries.

The mouth is drawn to the unaffected side, the wrinkles are deeper on the normal side, and the eye on the involved side remains open.

No medical intervention is necessary.

The paralysis usually disappears spontaneously in a few days but may take as long as several months



FIG 8-2 Facial nerve palsy. (From Clark-Gambelunghe MB, Clark D: Sensory development, *Pediatr Clin North Am* 62[2]:367–384, 2015.)

Brachial Palsy

Plexus injury results from forces that alter the normal position and relationship of the arm, shoulder, and neck. Erb palsy (Erb-Duchenne paralysis) is caused by damage to the upper plexus and usually results from stretching or pulling away of the shoulder from the head, as might occur with shoulder dystocia or with a difficult vertex or breech delivery.

Other identified risk factors include an infant with birth weight of more than 4000 g, multiparous pregnancy, a vacuum-assisted extraction, prolonged labor, and a previous history of brachial plexus injury.

Treatment of the affected arm is aimed at preventing contractures of the paralyzed muscles and maintaining correct placement of the humeral head within the glenoid fossa of the scapula.

Complete recovery from stretched nerves usually takes 3 to 6 months. Full recovery is expected in 88% to 92% of infants



Birthmarks

Discolorations of the skin are common findings in newborn infants

Most, such as mongolian spots or telangiectatic nevi, involve no therapy other than reassurance to parents of the benign nature of these discolorations.

However, some can be a manifestation of a disease that suggests further examination of the child and other family members (e.g., the multiple light brown café-au-lait spots that often characterize the autosomal dominant hereditary disorder neurofibromatosis and are common findings in Albrightsyndrome).



FIG 8-5 A, Port-wine stain. B, Strawberry hemangioma. (From Zitelli BJ, McIntire SC, Nowalk AJ: *Zitelli and Davis'*

Newborn classification

- Newborn are classified based on:

1. Gestational age
2. Birth weight

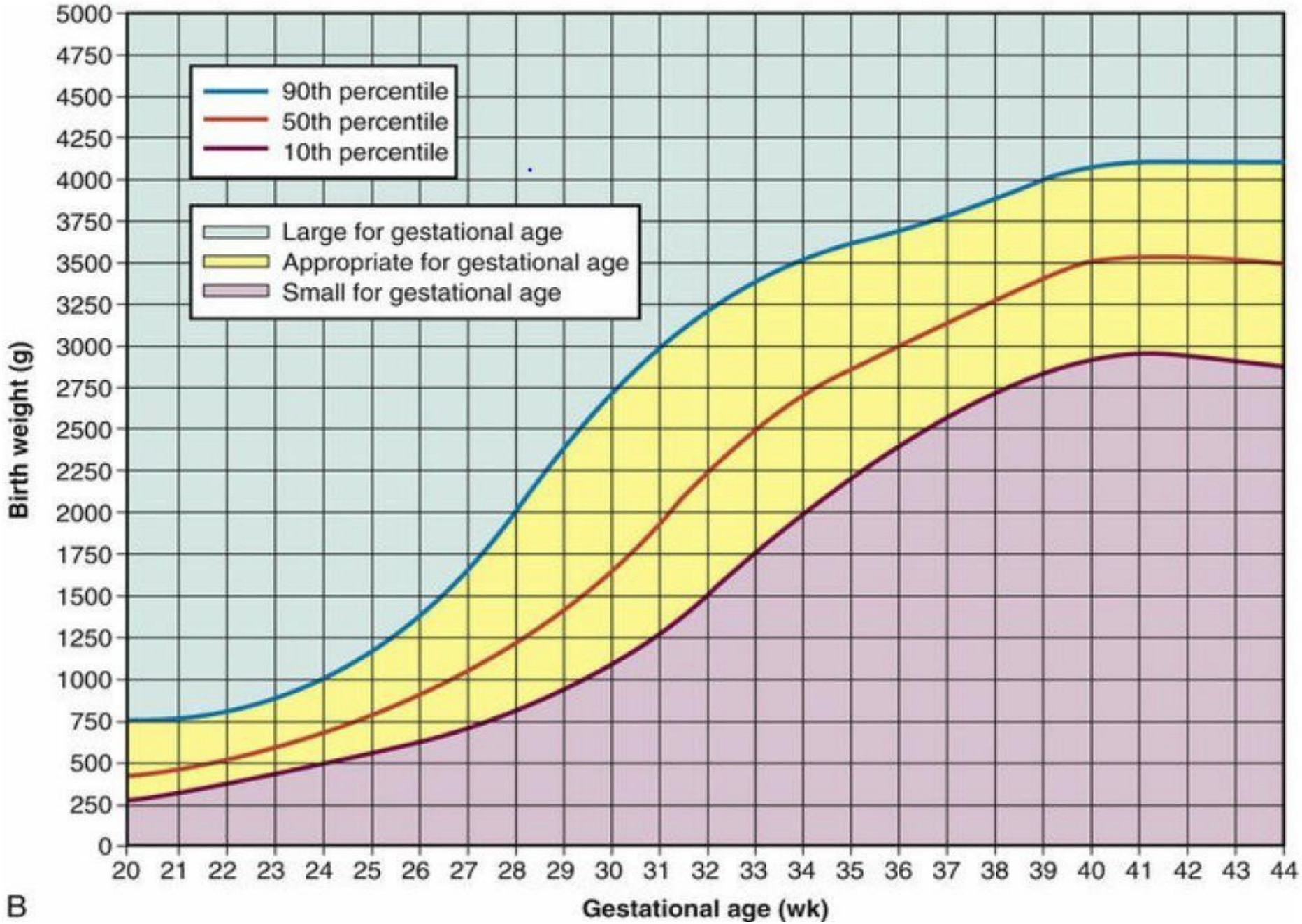
Gestational age

- Preterm: gestational age < 37 wk
- Term: complete of 37 wk to 42 wk
- Post term: gestational age above 42 wk



Birth weight

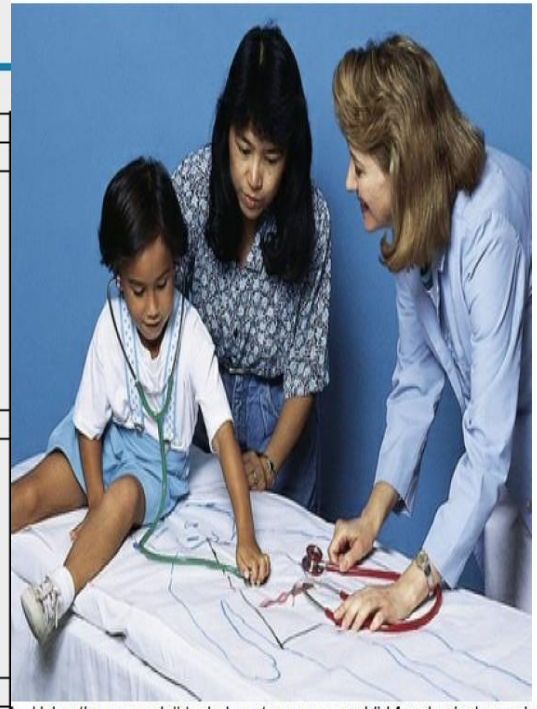
- Average birth weight 2700-4000 is normal, 2500 is OK
- Less than 2500gm – low birth weight (LBW)
- < 1500 gm – very LBW (VLBW)
- SGA: infants who are Small for gestational age (SGA) (below the 10th percentile) can be assumed to have intrauterine growth restriction or delay.
- LGA: Infants who are large for gestational age (LGA) (above the 90th percentile) can be presumed to have grown at an accelerated rate during fetal life
- AGA: Infants whose weight is appropriate for gestational age (AGA) (between the 10th and 90th percentiles) can be presumed to have grown at a normal rate regardless of the time of birth—preterm, term, or post-term


90th percentile
50th percentile
10th percentile
Large for gestational age
Appropriate for gestational age
Small for gestational age
Birth weight (g)
5000
4750
4500
4250
4000
3750
3500
3250
3000
2750
2500
2250
2000
1750
1500
1250
1000
750
500
250
0
20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44
Gestational age (wk)

B

Age-Specific Approaches to Physical Examination During Childhood

Position	Sequence	Preparation
Infant		
Before able to sit alone—supine or prone, preferably in parent's lap; before 4 to 6 months, can place on examining table After able to sit alone—sitting in parent's lap whenever possible; if on table, place with parent in full view	If quiet, auscultate heart, lungs, and abdomen. Record heart and respiratory rates. Palpate and percuss same areas. Proceed in usual head-to-toe direction. Perform traumatic procedures last (eyes, ears, mouth [while crying]). Elicit reflexes as body part is examined. Elicit Moro reflex last.	Completely undress if room temperature permits. Leave diaper on male infant. Gain cooperation with distraction, bright objects, rattles, talking. Smile at infant; use soft, gentle voice. Pacify with bottle of sugar water or feeding. Enlist parent's aid for restraining to examine ears, mouth. Avoid abrupt, jerky movements.
Toddler		
Sitting or standing on or by parent Prone or supine in parent's lap	Inspect body area through play: "Count fingers," "tickle toes." Use minimum physical contact initially. Introduce equipment slowly. Auscultate, percuss, palpate whenever quiet. Perform traumatic procedures last (same as for infant).	Have parent remove outer clothing. Remove underwear as body part is examined. Allow toddler to inspect equipment; demonstrating use of equipment is usually ineffective. If uncooperative, perform procedures quickly. Use restraint when appropriate; request parent's assistance. Talk about examination if cooperative; use short phrases. Praise for cooperative behavior.
Preschool Child		
Prefer standing or sitting Usually cooperative prone or supine Prefer parent's closeness	If cooperative, proceed in head-to-toe direction. If uncooperative, proceed as with toddler.	Request self-undressing. Allow to wear underpants if shy. Offer equipment for inspection; briefly demonstrate use. Make up story about procedure (e.g., "I'm seeing how strong your muscles are" [blood pressure]). Use paper-doll technique. Give choices when possible. Expect cooperation; use positive statements (e.g., "Open your mouth").
School-Age Child		
Prefer sitting Cooperative in most positions Younger child prefers parent's presence Older child may prefer privacy	Proceed in head-to-toe direction. May examine genitalia last in older child.	Respect need for privacy. Request self-undressing. Allow to wear underpants. Give gown to wear. Explain purpose of equipment and significance of procedure, such as otoscope to see eardrum, which is necessary for hearing. Teach about body function and care.
Adolescent		
Same as for school-age child Offer option of parent's presence	Same as older school-age child. May examine genitalia last.	Allow to undress in private. Give gown. Expose only area to be examined. Respect need for privacy. Explain findings during examination (e.g., "Your muscles are firm and strong"). Matter-of-factly comment about sexual development (e.g., "Your breasts are developing as they should be"). Emphasize normalcy of development. Examine genitalia as any other body part; may leave to end.



Using the paper-doll technique to prepare a child for physical examination.



Recommended Temperature Screening Routes in Infants and Children

Birth to 2 Years Old

Axillary

Rectal—if definitive temperature reading is needed for infants older than 1 month of age

2 to 5 Years Old

Axillary

Tympanic

Oral—when child can hold thermometer under tongue

Rectal—if definitive temperature reading is needed

Older Than 5 Years Old

Oral

Axillary

Tympanic



Place tip under tongue in Rt or Lt posterior sublingual pocket, not in front of tongue





Insert small infrared probe deeply into canal to allow sensor to obtain measurement. Size of probe (most are 8 mm)



Rectal
Place well-lubricated tip at maximum 2.5 cm (1 inch) into rectum for children and 1.5 cm



Temporal Artery
An infrared sensor probe scans across forehead, capturing heat from arterial blood flow. Temporal artery is only artery close enough to skin's surface to provide access for accurate temperature measurement.

Grading of Pulses

Grade	Description
0	Not palpable
+1	Difficult to palpate, thready, weak, easily obliterated with pressure
+2	Difficult to palpate, may be obliterated with pressure
+3	Easy to palpate, not easily obliterated with pressure (normal)
+4	Strong, bounding, not obliterated with pressure

Respiration

- Count the respiratory rate in children in the same manner as for adult patients.

- However, in infants, observe abdominal movements, because respirations are primarily diaphragmatic. Because the movements are irregular, count them for 1 full minute for accuracy

Age	Width (cm)	Length (cm)	Maximum Arm Circumference (cm)*
Newborn	4	8	10
Infant	6	12	15
Child	9	18	22
Small adult	10	24	26
Adult	13	30	34
Large adult	16	38	44
Thigh	20	42	52

*Calculated so that largest arm would still allow bladder to encircle arm by at least 80%.

From National High Blood Pressure Education Program Working Group on High Blood Pressure in Children and Adolescents: The fourth report on the diagnosis, evaluation, and treatment of high blood pressure in children and adolescents, *Pediatrics* 114(2 Suppl 4th Report):555–576, 2004.

Blood Pressure

BP should be measured annually in children 3 years old through adolescence and in children with symptoms of hypertension, children in emergency departments and intensive care units, and high risk infants

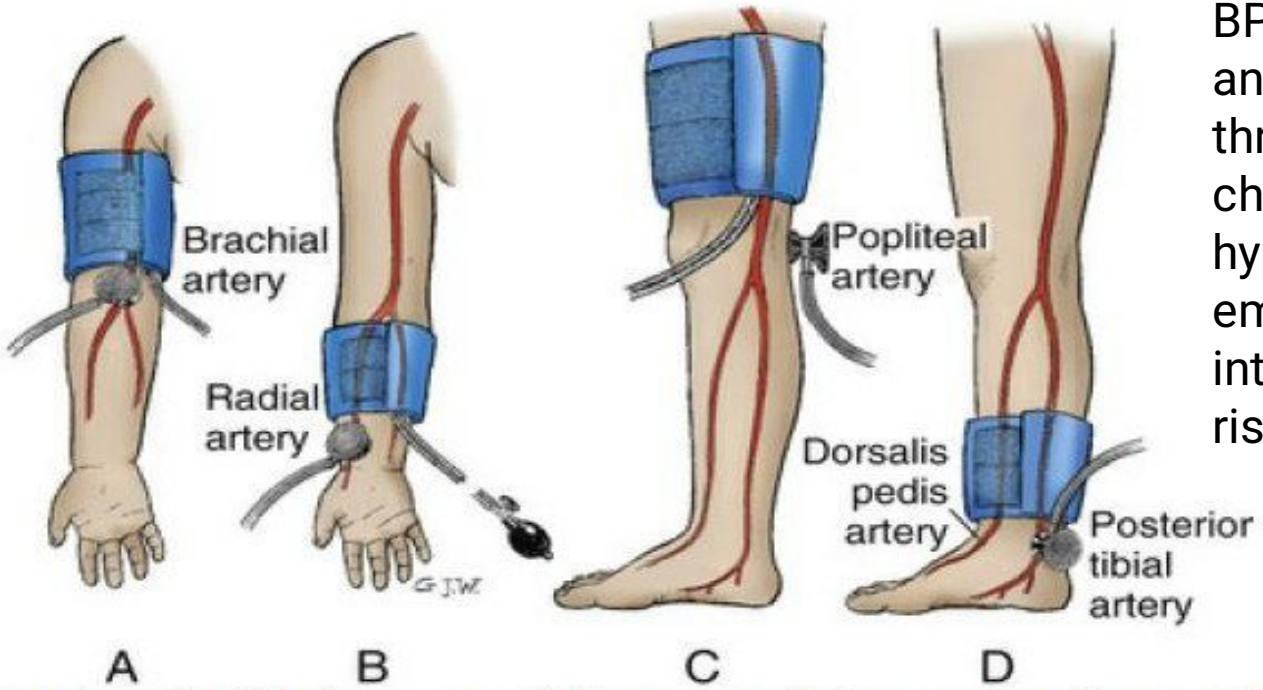


FIG 4-13 Sites for measuring blood pressure. **A,** Upper arm. **B,** Lower arm or forearm. **C,** Thigh. **D,** Calf or ankle.

Lymph Nodes

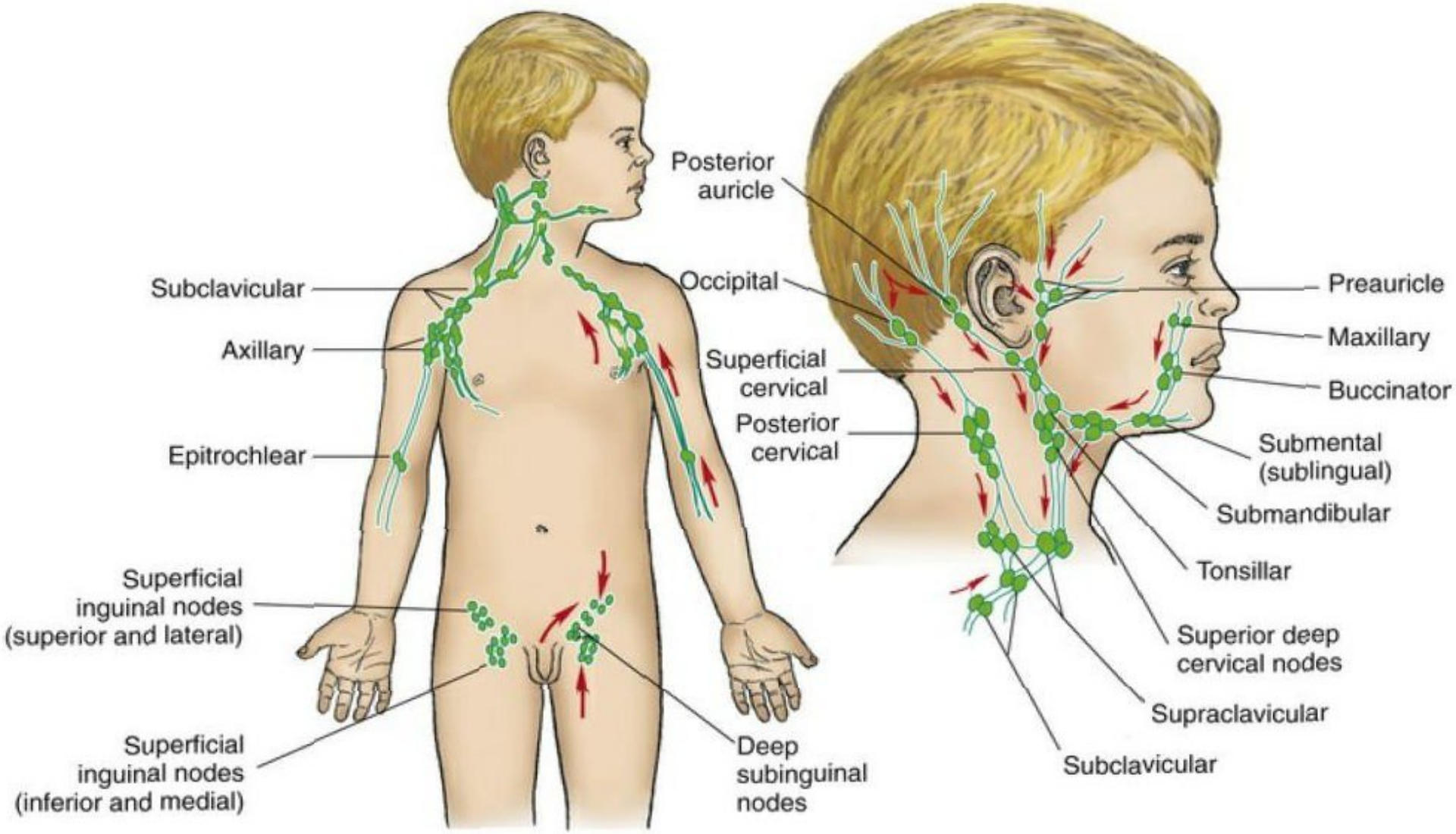


FIG 4-15 Location of superficial lymph nodes. *Arrows* indicate directional flow of lymph.

Questions ???

Thank You

Pediatric Nursing & Newborn

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January, 2025

Family-Centered Care of the Newborn

- Health Promotion of the Newborn and Family
- Health Problems of Newborns



Health Promotion of the Newborn and Family

Adjustment to Extrauterine Life

The most profound physiologic change required of neonates is transition from fetal or placental circulation to independent respiration.

The loss of the placental connection means the loss of complete metabolic support, especially the supply of oxygen and the removal of carbon dioxide.

Factors that interfere with this normal transition or that interfere with fetal oxygenation (including conditions such as hypoxemia, hypercapnia, and acidosis) affect the fetus's adjustment to extrauterine life.

Immediate Adjustments: Respiratory System

The most critical and immediate physiologic change required of newborns is the onset of breathing (The stimuli that help initiate the first breath are primarily chemical and thermal).

Chemical factors in the blood (low oxygen, high carbon dioxide, and low pH) initiate impulses that excite the respiratory center in the medulla.

The primary thermal stimulus is the sudden chilling of the infant, who leaves a warm environment and enters a relatively cooler atmosphere. This abrupt change in temperature excites sensory impulses in the skin that are transmitted to the respiratory center.

Immediate Adjustments: Respiratory System

Tactile stimulation may assist in initiating respiration. Descent through the birth canal and normal handling during delivery help stimulate respiration in uncompromised infants.

Acceptable methods of tactile stimulation include tapping or flicking the soles of the feet or gently rubbing the newborn's back, trunk, or extremities.

Slapping the newborn's buttocks or back is a harmful technique.

Prolonged tactile stimulation, beyond one or two taps or flicks to the soles of the feet or rubbing the back once or twice, can waste precious time in the event of respiratory difficulty and can cause additional damage in infants.

Immediate Adjustments: Respiratory System

The initial entry of air into the lungs is opposed by the surface tension of the fluid that filled the fetal lungs and the alveoli.

Some lung fluid is removed during the normal forces of labor and delivery

In the alveoli, the surface tension of the fluid is reduced by surfactant, a substance produced by the alveolar epithelium that coats the alveolar surface.

Circulatory System

Through the blood vessels in the umbilical cord, the fetus gets all needed nutrition and oxygen. The fetus gets life support from the mother through the placenta.

Waste products and carbon dioxide from the fetus are sent back through the umbilical cord and placenta to the mother's circulation to be removed.

The fetal circulatory system uses 3 shunts. These are small passages that direct blood that needs to be oxygenated.

The shunt that bypasses the lungs is called the foramen ovale. This shunt moves blood from the right atrium of the heart to the left atrium.

The ductus arteriosus moves blood from the pulmonary artery to the aorta.

As important as the initiation of respiration are the circulatory changes that allow blood to flow through the lungs.

The transition from fetal to postnatal circulation involves the functional closure of the fetal shunts: the foramen ovale, the ductus arteriosus, and the ductus venosus.

As the pulmonary vessels receive blood, the pressure in the right atrium, right ventricle, and pulmonary arteries decreases.

Left atrial pressure increases above right atrial pressure, with subsequent foramen ovale closure.

With the increase in pulmonary blood flow and dramatic reduction of pulmonary vascular resistance, the ductus arteriosus begins to close.

The most important factors controlling ductal closure are the increased oxygen concentration of the blood and the fall in endogenous prostaglandins.

The foramen ovale closes functionally at or soon after birth. The ductus arteriosus is closed functionally by the fourth day.

Failure of the ductus arteriosus or foramen ovale to close results in persistence of fetal shunting of blood away from the lungs.

Because of the reversible flow of blood through the ductus during the early neonatal period, functional murmur occasionally may be heard.

In conditions such as crying or straining, the increased pressure shunts deoxygenated blood from the right side of the heart across the ductal opening, which may cause transient cyanosis

Physiologic Status of Other Systems
Thermoregulation

Next to establishing respiration, heat regulation is most critical to the newborn's survival.

<u>Three factors predispose newborns to excessive heat loss:</u>

1. The newborn's large surface area facilitates heat loss to the environment, although this is partially compensated for by the newborn's usual position of flexion, which decreases the amount of surface area exposed to the environment.
2. The newborn's thin layer of subcutaneous fat provides poor insulation for conservation of heat.
3. The newborn's mechanism for producing heat is different from that of the adult, who can increase heat production through shivering. A chilled neonate cannot shiver but produces heat through non shivering thermogenesis (NST), which involves increased metabolism and oxygen consumption.

   The principal thermogenic sources are the heart, liver, and brain.

   Additional source, once believed to be unique to newborns is known as brown adipose tissue, or brown fat.

Brown fat, which owes its name to its larger content of mitochondrial cytochromes, has a greater capacity for heat production through intensified metabolic activity than ordinary adipose tissue

Because of these factors predisposing infants to loss of body heat, it is essential that newly born infants are:

- Quickly dried
- Placed skin-to-skin with their mother
- Provided with warm, dry blankets after delivery.

Although newborns' ability to conserve heat is usually a matter of concern, they may also have difficulty dissipating heat in an overheated environment, which increases the risk of hyperthermia.

Hematopoietic System

The blood volume of the newborn depends on the amount of placental transfer of blood.

The blood volume of a full-term infant is about 80 to 85 ml/kg of body weight.

Immediately after birth, the total blood volume averages 300 ml, but depending on how long umbilical cord clamping is delayed or if the umbilical cord is milked, as much as 100 ml can be added to the blood volume

Fluid and Electrolyte Balance

Changes occur in the total body water volume, extracellular fluid volume, and intracellular fluid volume during the transition from fetal to postnatal life.

1. At birth, the total weight of an infant is 73% fluid compared with 58% in an adult. Infants have a proportionately higher ratio of extracellular fluid than adults.
2. An infant's rate of metabolism is twice that of an adult in relation to body weight. As a result, twice as much acid is formed, leading to more rapid development of acidosis.
3. Immature kidneys cannot sufficiently concentrate urine to conserve body water.

These three factors make infants more prone to dehydration

Gastrointestinal System

Enzymes are adequate to handle proteins and simple carbohydrates (monosaccharides and disaccharides), but deficient production of pancreatic amylase impairs use of complex carbohydrates (polysaccharides).

Deficiency of pancreatic lipase limits absorption of fats, especially with ingestion of foods with high saturated fatty acid content, such as cow's milk. Human milk, despite its high fat content, is easily digested because the milk itself contains enzymes (such as lipase), which assist in digestion.

The liver is the most immature of the gastrointestinal organs. The activity of the enzyme glucuronyl transferase is reduced, which affects the conjugation of bilirubin with glucuronic acid and contributes to physiologic jaundice of newborns.

The liver is also deficient in forming plasma proteins. The decreased plasma protein concentration probably plays a role in the edema usually seen at birth.

Prothrombin and other coagulation factors are also low.

The liver stores less glycogen at birth than later in life. Consequently, newborns are prone to hypoglycemia, which may be prevented by early and effective feeding, ideally breastfeeding.

Some salivary glands are functioning at birth, but the majority do not begin to secrete saliva until about age 2 to 3 months, when drooling is frequent.

Newborn stomach capacity is about 20 ml at birth, thus, infants require small feedings at 1 hr.

The colon also has a small volume; newborns may have a bowel movement after each feeding.

Newborns who breastfeed usually have more frequent feedings and more frequent stools than infants who receive formula.

An infant's intestine is longer in relation to body size than that of the adult.

Therefore, there are a larger number of secretory glands and a larger surface area for absorption compared with an adult’s intestine.

Infants have rapid peristaltic waves and simultaneous nonperistaltic waves along the entire esophagus, which propel nutrients forward. The relative immaturity of the peristaltic waves combined with decreased lower esophageal sphincter (LES) pressure, inappropriate relaxation of the LES, and delayed gastric emptying make regurgitation a common occurrence.

Box 7-1

Change in Stooling Patterns of Newborns

Meconium

Infant's first stool; composed of amniotic fluid and its constituents, intestinal secretions, shed mucosal cells, and possibly blood (ingested maternal blood or minor bleeding of alimentary tract vessels).

Passage of meconium should occur within the first 24 to 48 hours, although it may be delayed up to 7 days in very low birth weight infants.

Transitional Stools

Usually appear by third day after initiation of feeding; greenish brown to yellowish brown, thin, and less sticky than meconium; may contain some milk curds.

Milk Stool

Usually appears by fourth day.

In **breastfed infants,** stools are yellow to golden, are pasty in consistency, and have an odor similar to that of sour milk.

In **formula-fed infants,** stools are pale yellow to light brown, are firmer in consistency, and have a more offensive odor.

Renal System

All structural components are present in the renal system, but there is a functional deficiency in the kidneys' ability to concentrate urine and to cope with conditions of fluid and electrolyte stress, such as dehydration or a concentrated solute load.

Total volume of urine per 24 hours is about 200 to 300 ml by the end of the first week. However, the bladder voluntarily empties when stretched by a volume of 15 ml, resulting in as many as 20 voidings per day.

The first voiding should occur within 24 hours. The urine is colorless and odorless and has a specific gravity of about 1.020.

Integumentary System

At birth, all of the structures within the skin are present, but many of the functions of the integument are immature. The outer two layers of the skin, the epidermis and dermis, are loosely bound to each other and very thin.

The sebaceous glands are active late in fetal life and in early infancy because of the high levels of maternal androgens. They are most densely located on the scalp, face, and genitalia and produce the greasy vernix caseosa that covers infants at birth.

The growth phases of hair follicles usually occur simultaneously at birth. During the first few months, the synchrony between hair loss and regrowth is disrupted, and there may be overgrowth of hair or temporary alopecia.

Because the amount of melanin is low at birth, newborns are lighter skinned than they will be as children. Consequently, they are more

Musculoskeletal System

At birth, the skeletal system contains more cartilage than ossified bone, although the process of ossification is fairly rapid during the first year. The nose, for example, is predominantly cartilage at birth and may be temporarily flattened or asymmetric because of the force of delivery.

Unlike the skeletal system, the muscular system is almost completely formed at birth. Growth in size of muscular tissue is caused by hypertrophy, rather than hyperplasia, of cells.

Defenses Against Infection

Infants are born with several defenses against infection. The first line of defense is the skin and mucous membranes, which protect the body from invading organisms.

The mature neonatal intestinal mucosal (gut) barrier also plays a vital role as an important defense mechanism against antigens.

The second line of defense is the macrophage system, which produces several types of cells capable of attacking a pathogen.

The third line of defense is the formation of specific antibodies to an antigen. Exposure to various foreign agents is necessary for antibody production to occur

Endocrine System

Ordinarily, the endocrine system of newborns is adequately developed, but its functions are immature. For example, the posterior lobe of the pituitary gland produces limited quantities of antidiuretic hormone, or vasopressin, which inhibits diuresis. This renders young infants highly susceptible to dehydration.

The effect of maternal sex hormones is particularly evident in newborns. The labia are hypertrophied, and the breasts of both genders may be engorged and secrete milk from the first few days of life to as long as 2 months of age. Female newborns may have pseudomenstruation (more often seen as a milky secretion than actual blood) from a sudden drop in progesterone and estrogen levels.

Neurologic System

At birth, the nervous system is incompletely integrated but sufficiently developed to sustain extrauterine life. Most neurologic functions are primitive reflexes. The autonomic nervous system is crucial during transition because it stimulates initial respirations, helps maintain acid–base balance, and partially regulates temperature control.

Myelin is necessary for rapid and efficient transmission of some, but not all, nerve impulses along the neural pathway

Sensory Functions
Vision

At birth, the eye is structurally incomplete. The fovea centralis is not yet completely differentiated from the macula.

The ciliary muscles are also immature, limiting the eyes' ability to accommodate and focus on an object for any length of time.

The infant can track and follow objects. The pupils react to light, the blink reflex is responsive to minimal stimulus, and the corneal reflex is activated by a light touch.

Tear glands usually do not begin to function until 2 to 4 weeks of age.

Newborns focus momentarily on a bright/moving object within 20 cm and in the midline of the visual field.

Visual acuity is reported to be between 20/100 and 20/400, depending on the vision measurement techniques.

Infants also demonstrate visual preferences: medium colors

(yellow, green, pink) over bright (red, orange, blue) or dim colors;

black-and-white contrasting patterns

Hearing

After the amniotic fluid has drained from the ears, infants probably have auditory acuity similar to that of adults

Infants younger than 3 days old can discriminate the mother's voice from that of other women.

The internal and middle ear is large at birth, but the external canal is small.

The mastoid process and the bony part of the external canal have not yet developed.

Consequently, the tympanic membrane and facial nerve are very close to the surface and can be easily damaged.

Smell

Newborns react to strong odors such as alcohol and vinegar by turning their heads away. Breastfed infants are able to smell breast milk and will cry for their mothers when they smell leaking milk.

Infants are also able to differentiate the breast milk of their mothers from the breast milk of other women by scent alone.

Maternal odors are believed to influence the attachment process and successful breastfeeding. Unnecessary routine washing of the breast may interfere with establishment of early breastfeeding.

Taste

The newborn has the ability to distinguish among tastes and various types of solutions elicit differing facial reflexes.

A tasteless solution elicits no facial expression; a sweet solution elicits an eager suck and a look of satisfaction; a sour solution causes puckering of the lips; and a bitter liquid produces an angry, upset expression.

Touch

At birth, infants are able to perceive tactile sensation in any part of the body, although the face (especially the mouth), hands, and soles of the feet seem to be most sensitive. Evidence shows that touch and motion are essential to normal growth and development.

Gentle patting of the back or rubbing of the abdomen usually elicits a calming response from infants. In turn, painful stimuli, such as a pinprick, elicit an upset response.

Assessment of Reflexes in the Newborn

Eyes

- Blinking or corneal:

  Infant blinks at sudden appearance of a bright light or at approach of an object toward cornea; persists throughout life.

- Pupillary

  Pupil constricts when a bright light shines toward it; persists throughout life.

- Doll's eye:

  As head is moved slowly to right or left, eyes lag behind and do not immediately adjust to new position of head; disappears as fixation develops; if persists, indicates neurologic damage.

Mouth and Throat

- Rooting: Touching or stroking the cheek alongside of mouth causes infant to turn head toward that side and begin to suck; should disappear at about 3 to 4 months old but may persist for up to 12 months.
- Extrusion: When tongue is touched or depressed, infant responds by forcing it outward; disappears by 4 months old.
- Yawn: is a spontaneous response to decreased oxygen by increasing amount of inspired air; persists throughout life.

Mass

- Moro: Sudden jarring or change in equilibrium causes sudden extension and abduction of extremities and fanning of fingers, with index finger and thumb forming a C shape followed by flexion and adduction of extremities; legs may weakly flex; infant may cry; disappears after 3 to 4 months old, usually strongest during first 2 months.
- Startle: A sudden loud noise causes abduction of the arms with flexion of elbows; hands remain clenched; disappears by 4 months old.
- Perez: While infant is prone on a firm surface, thumb is pressed along spine from sacrum to neck; infant responds by crying, flexing extremities, and elevating pelvis and head; lordosis of the spine, as well as defecation and urination, may occur;

Reflexes: Expected Behavioral Responses

Mas
s

- Tonic neck When infant's head is turned to one side, arm and leg extend on that side, and opposite arm and leg

  flex; disappears by 3 to 4 months old to be replaced by symmetric positioning of both sides of body.

- Trunk incurvation (Galant) Stroking infant's back alongside spine causes hips to move toward stimulated side; disappears by 4 weeks old.

- Dance or step If infant is held so that sole of foot touches a hard surface, there is a reciprocal flexion and extension of the leg, simulating walking; disappears after 3 to 4 weeks old to be replaced by deliberate movement.

- Crawl When placed on abdomen, infant makes crawling movements with arms and legs; disappears at about 6 weeks old.

- Placing When infant is held upright under arms and dorsal side of foot is briskly placed against hard object, such

  as table, leg lifts as if foot is stepping on table; age of disappearance varies.



A, Moro reflex. **B**, Tonic neck reflex. **C**, Dance reflex. **D**, Crawl reflex. (Courtesy of Paul Vincent Kuntz, Texas Children's Hospital, Houston, TX.)

Assessment of Reflexes in the Newborn

Reflexes	Expected Behavioral Responses
Localized	
Eyes	
Blinking or corneal	Infant blinks at sudden appearance of a bright light or at approach of an object toward cornea; persists throughout life.
Pupillary	Pupil constricts when a bright light shines toward it; persists throughout life.
Doll's eye	As head is moved slowly to right or left, eyes lag behind and do not immediately adjust to new position of head; disappears as fixation develops; if persists, indicates neurologic damage.
Nose	
Sneeze	Sneezing is a spontaneous response of nasal passages to irritation or obstruction; persists throughout life.
Glabellar	Tapping briskly on glabella (bridge of nose) causes eyes to close tightly.
Mouth and Throat	
Sucking	Infant begins strong sucking movements of circumoral area in response to stimulation; persists throughout infancy even without stimulation, such as during sleep.
Gag	Stimulation of posterior pharynx by food, suction, or passage of a tube causes infant to gag; persists throughout life.
Rooting	Touching or stroking the cheek alongside of mouth causes infant to turn head toward that side and begin to suck; should disappear at about 3 to 4 months old but may persist for up to 12 months.
Extrusion	When tongue is touched or depressed, infant responds by forcing it outward; disappears by 4 months old.
Yawn	Yawning is a spontaneous response to decreased oxygen by increasing amount of inspired air; persists throughout life.
Cough	Irritation of mucous membranes of larynx or tracheobronchial tree causes coughing; persists throughout life; usually present after first day of birth.
Extremities	
Grasp	Touching palms of hands or soles of feet near base of digits causes flexion of fingers and toes (see Fig. 7-8, *A*); palmar grasp lessens after age 3 months old to be replaced by voluntary movement; plantar grasp lessens by 8 months old.
Babinski	Stroking outer sole of foot upward from heel and across ball of foot causes toes to hyperextend and hallux to dorsiflex (see Fig. 7-8, *B*); disappears after 1 year old.
Ankle clonus	Briskly dorsiflexing foot while supporting knee in partially flexed position results in one or two oscillating movements ("beats"); eventually, no beats should be felt.
Mass	
Moro	Sudden jarring or change in equilibrium causes sudden extension and abduction of extremities and fanning of fingers, with index finger and thumb forming a **C** shape followed by flexion and adduction of extremities; legs may weakly flex; infant may cry (Fig. 7-9, *A*); disappears after 3 to 4 months old, usually strongest during first 2 months.
Startle	A sudden loud noise causes abduction of the arms with flexion of elbows; hands remain clenched; disappears by 4 months old.
Perez	While infant is prone on a firm surface, thumb is pressed along spine from sacrum to neck; infant responds by crying, flexing extremities, and elevating pelvis and head; lordosis of the spine, as well as defecation and urination, may occur; disappears by 4 to 6 months old.
Tonic neck	When infant's head is turned to one side, arm and leg extend on that side, and opposite arm and leg flex (see Fig. 7-9, *B*); disappears by 3 to 4 months old to be replaced by symmetric positioning of both sides of body.
Trunk incurvation (Galant)	Stroking infant's back alongside spine causes hips to move toward stimulated side; disappears by 4 weeks old.
Dance or step	If infant is held so that sole of foot touches a hard surface, there is a reciprocal flexion and extension of the leg, simulating walking (see Fig. 7-9, *C*); disappears after 3 to 4 weeks old to be replaced by deliberate movement.
Crawl	When placed on abdomen, infant makes crawling movements with arms and legs (see Fig. 7-9, *D*); disappears at about 6 weeks old.
Placing	When infant is held upright under arms and dorsal side of foot is briskly placed against hard object, such as table, leg lifts as if foot is stepping on table; age of disappearance varies.

Classification of High-Risk Infants

- Low birth weight (LBW) infant: An infant whose birth weight is less than 2500 g
- Very low birth weight (VLBW) infant: An infant whose birth weight is less than 1500 g
- Extremely low birth weight (ELBW) infant: An infant whose birth weight is less than 1000 g
- Intrauterine growth restriction (IUGR): Found in infants whose intrauterine growth is restricted (sometimes used as a more descriptive term for SGA infants)
- Symmetric IUGR: Growth restriction in which the weight, length, and head circumference are all affected
- Asymmetric IUGR: Growth restriction in which the head circumference remains within normal parameters while the birth weight falls below the 10th percentile
- Large-for-gestational-age (LGA) infant: An infant whose birth weight falls above the 90th percentile on intrauterine growth charts
- Preterm (premature) infant: An infant born before completion of 37 weeks of gestation
- Postterm (postmature) infant: An infant born after 42 weeks of gestational age

Nutrition

- Optimum nutrition is critical in the management of LBW and preterm infants, but there are difficulties in meeting for their nutritional needs.
- The various mechanisms for ingestion and digestion of foods are not fully developed.
- It is known that all preterm infants are at risk because of poor nutritional stores and several physical and developmental characteristics. some sucking and swallowing activities are demonstrated before birth and in preterm infants, coordination of these mechanisms does not occur until approximately 32 to 34 weeks of gestation, and they are not fully synchronized until 36 to 37 weeks.
- Nutrition can be provided by either the parenteral or the enteral route or by a combination of the two.
- Infants who are ELBW, VLBW, or critically ill often obtain the majority of their nutrients by the parenteral route because of their inability to digest and absorb enteral nutrition
- These minimal enteral (trophic gastrointestinal priming) feedings have been shown to stimulate the infant's gastrointestinal tract, preventing mucosal atrophy and subsequent enteral feeding difficulties

Preterm Infants

- Prematurity accounts for the largest number of admissions to NICUs.
- Immaturity of most organ systems places infants at risk for a variety of neonatal complications (e.g., hyperbilirubinemia, respiratory distress syndrome [RDS], intellectual and motor delays).
- According to the Centers for Disease Control and Prevention, low birth weight and prematurity were the second leading cause of infant mortality.
- Factors such as poverty, maternal infections, previous preterm delivery, multiple pregnancies, pregnancy-induced hypertension, and placental problems that interrupt the normal course of gestation before completion of fetal development are responsible for a large number of preterm births.

Etiology of Preterm Birth

- Maternal Factors

√ Socioeconomic: Malnutrition -Age - Race

√ Chronic medical conditions: Heart disease - Renal disease - Diabetes -Hypertension

√ Behavioral: Substance abuse - Smoking - Poor or absent prenatal care

- Factors Related to Pregnancy

√ Multiple pregnancy -Low body mass index (<19.8 kg/m2) - Abruptio placentae or placenta previa - Incompetent cervix - Maternal hypertension - Premature rupture of membranes or chorioamnionitis - Polyhydramnios or oligohydramnios – Infection - Trauma

- Fetal Factors

√ Chromosomal abnormalities - Congenital anomalies - Nonimmune hydrops - Erythroblastosis

Diagnostic Evaluation

- Preterm infants are very small and appear scrawny, because they have only minimal subcutaneous fat deposits and have a proportionately large head in relation to the body.
- The skin is bright pink, smooth, and shiny, with small blood vessels clearly visible underneath the thin epidermis.
- The fine lanugo hair is abundant over the body.
- The ear cartilage is soft and pliable.
- The bones of the skull and the ribs feel soft, and the eyes may be closed.

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- The skin is bright pink, smooth, and shiny, with small blood vessels clearly visible underneath the thin epidermis.
- The fine lanugo hair is abundant over the body.
- The ear cartilage is soft and pliable.
- The bones of the skull and the ribs feel soft, and the eyes may be closed.
- Male infants have few scrotal rugae, and the testes are undescended; in girls, the labia and clitoris are prominent compares the features of full-term and preterm infants.

CLINICAL EVALUATION

PRETERM

TERM

Posture—The preterm infant lies in a "relaxed attitude," limbs more extended; the body size is small, and the head may appear somewhat larger in proportion to the body size. The term infant has more subcutaneous fat tissue and rests in a more flexed attitude.





Ear—The preterm infant's ear cartilages are poorly developed, and the ear may fold easily; the hair is fine and feathery, and lanugo may cover the back and face. The mature infant's ear cartilages are well formed, and the hair is more likely to form firm, separate strands.





Sole—The sole of the foot of the preterm infant appears more turgid and may have only fine wrinkles. The mature infant's sole (foot) is well and deeply creased.





Female genitalia—The preterm female infant's clitoris is prominent, and labia majora are poorly developed and gaping. The mature female infant's labia majora are fully developed, and the clitoris is not as prominent.





Male genitalia—The preterm male infant's scrotum is undeveloped and not pendulous; minimal rugae are present, and the testes may be in the inguinal canals or in the abdominal cavity. The term male infant's scrotum is well developed, pendulous, and rugated, and the testes are well down in the scrotal sac.





Scarf sign—The preterm infant's elbow may be easily brought across the chest with little or no resistance. The mature infant's elbow may be brought to the midline of the chest, resisting attempts to bring the elbow past the midline.





NEUROLOGIC EVALUATION

PRETERM

TERM

Grasp reflex—The preterm infant's grasp is weak; the term infant's grasp is strong, allowing the infant to be lifted up from the mattress.





Heel-to-ear maneuver—The preterm infant's heel is easily brought to the ear, meeting with no resistance. This maneuver is not possible in the term infant, since there is considerable resistance at the knee.





Therapeutic Management

- At NICU, they are weighed and where IV lines, oxygen therapy, and other therapeutic interventions are initiated as needed.
- Resuscitation is conducted in the delivery area until infants can be safely transported to the NICU.
- The general care of preterm infants differs from that of full-term infants primarily in the areas of respiratory support, temperature regulation, nutrition, susceptibility to infection, activity intolerance, neurodevelopmental care, and other consequences of physical immaturity.

Postterm Infants

- Infants born beyond 42 weeks are considered to be postterm regardless of birth weight.
- This constitutes 3.5% to 15% of all pregnancies. The cause of delayed birth is unknown.
- These infants display absence of lanugo, little if any vernix caseosa, abundant scalp hair, and long fingernails.
- The skin is often cracked, parchment-like, and desquamating.
- A common finding in postterm infants is a wasted physical appearance that reflects intrauterine deprivation.
- Depletion of subcutaneous fat gives them a thin, elongated appearance.
- The little vernix caseosa that remains in the skinfolds may be stained a deep yellow or green, which is usually an indication of meconium in the amniotic fluid.
- There is increase in fetal and neonatal mortality in postterm infants compared with those born at term.
- Fetal distress associated with the decreasing efficiency of placenta, macrosomia, and meconium aspiration.
- The greatest risk occurs during the stresses of labor and delivery, particularly in an infant of a primigravida.

Questions ???

Thank You

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High Risk Related to Physiologic Factors

Hyperbilirubinemia

Refers to an excessive level of accumulated bilirubin in the blood and is characterized by jaundice, or icterus, a yellowish discoloration of the skin, sclerae, and nails.

Hyperbilirubinemia is a common finding in newborns and in most instances is relatively benign.

However, in extreme cases, it can indicate a pathologic state.

Hyperbilirubinemia may result from increased unconjugated or conjugated bilirubin.

The unconjugated form or indirect hyperbilirubinemia is the type most commonly seen in newborns.

The most common cause of hyperbilirubinemia is the relatively mild and self-limited physiologic jaundice.

Pathophysiology

Bilirubin is one of the breakdown products of the hemoglobin that results from RBC destruction.

When RBCs are destroyed, the breakdown products are released into the circulation, where the hemoglobin splits into two fractions: heme and globin. The globin (protein) portion is used by the body, and the heme portion is converted to unconjugated bilirubin, an insoluble substance bound to albumin.

In the liver, the bilirubin is detached from the albumin molecule and, in the presence of the enzyme glucuronyl transferase, is conjugated with glucuronic acid to produce a highly soluble substance, conjugated bilirubin, which is then excreted into the bile.

In the intestine, bacterial action reduces the conjugated bilirubin to urobilinogen, the pigment that gives stool its characteristic color. Most of the reduced bilirubin is excreted through the feces; a small amount is eliminated in the urine.

Pathophysiology

However, when developmental limitations or a pathologic process interferes with this balance, bilirubin accumulates in the tissues to produce jaundice.

Possible causes of hyperbilirubinemia in newborns are:

- Physiologic (developmental) factors (prematurity) • An association with breastfeeding or breast milk
- Dehydration (limited oral intake) • Excess production of bilirubin (e.g., hemolytic disease)
- Disturbed capacity of the liver to secrete conjugated bilirubin (e.g., enzyme deficiency, bile duct obstruction) and sepsis
- Some disease states (e.g., hypothyroidism, galactosemia, infant of a diabetic mother [IDM])
- Genetic predisposition to increased production or delayed metabolism (American Indians, Asians, Mediterranean)

On average, newborns produce twice as much bilirubin as adults because of higher concentrations of circulating erythrocytes and a shorter life span of RBCs (only 70 to 90 days in contrasted to 120 days in older children and adults)

In addition, the liver's ability to conjugate bilirubin is reduced because of limited production of glucuronyl transferase. Newborns also have a lower plasma-binding capacity for bilirubin because of reduced albumin concentrations compared with older children

Comparison of Major Types of Unconjugated Hyperbilirubinemia*

Physiologic Jaundice	Breastfeeding-Associated Jaundice (Early Onset)	Breast Milk Jaundice (Late Onset)	Hemolytic Disease
Cause			
Immature hepatic function plus increased bilirubin load from RBC hemolysis	Decreased milk intake related to fewer calories consumed by infant before mother's milk is well established; enterohepatic shunting	Possible factors in breast milk that prevent bilirubin conjugation Less frequent stooling	Blood antigen incompatibility causing hemolysis of large numbers of RBCs Liver's inability to conjugate and excrete excess bilirubin from hemolysis
Onset			
After 24 hours (preterm infants, prolonged)	2nd to 4th day	4th to 8th day	During first 24 hours (levels increase >5 mg/dl/day)
Peak			
3rd to 4th day	3rd to 5th day	10th to 15th day	Variable
Duration			
Declines on 5th to 7th day	Variable	May remain jaundiced for 3 to 12 weeks or more	Depends on severity and treatment
Therapy			
Increase frequency of feedings and avoid supplements. Evaluate stooling pattern. Monitor TcB or TSB level. Perform risk assessment (see Fig. 8-16, *A*). Use phototherapy if bilirubin levels increase significantly or significant hemolysis is present.	Breastfeed frequently (10 to 12 times/day); avoid supplements such as water, dextrose water, and formula. Evaluate stooling pattern; stimulate as needed. Perform risk assessment (see Fig. 8-16, *A*). Use phototherapy if bilirubin levels increase significantly or significant hemolysis is present. If phototherapy is instituted, evaluate benefits and harm of temporarily discontinuing breastfeeding; additional assessments may be required. Assist mother with maintaining milk supply; feed expressed milk as appropriate. After discharge, follow up according to hour of discharge.	Increase frequency of breastfeeding; use no supplementation, such as glucose water; cessation of breastfeeding is not recommended. Perform risk assessment (see Fig. 8-16, *A*). Consider performing additional evaluations: G6PD, direct and indirect serum bilirubin, family history, and others as necessary. May include home phototherapy with a temporary (10 to 12 hours) discontinuation of breastfeeding; a subsequent TSB may be drawn to evaluate a drop in serum levels. Assist mother with maintenance of milk supply and reassurance regarding her milk supply and therapy. Use formula supplements only at practitioner's discretion.	Monitor TcB or TSB level. Perform risk assessment (see Fig. 8-16, *A*). Postnatal: Use phototherapy; administer IV immunoglobulin per protocol; if severe, perform exchange transfusion. Prenatal: Perform transfusion (fetus). Prevent sensitization (Rh incompatibility) of Rh-negative mother with Rh_0(D) immune globulin (RhIg). If mother is breastfeeding, assist with maintenance and storage of milk; may bottle feed expressed milk as appropriate to therapy. Minimize maternal–infant separation and encourage contact as appropriate.

Physiologic jaundice

The most common cause of hyperbilirubinemia is physiologic jaundice.

Although almost all newborns experience elevated bilirubin levels, only about 50% to 60% demonstrate observable signs of jaundice.

Two phases of physiologic jaundice have been identified in full-term infants.

In the first phase, bilirubin levels gradually increase to approximately 5 to 6 mg/dl by 3 to 4 days of life and then decrease to a plateau of 2 to 3 mg/dl by the fifth day.

Bilirubin levels maintain a steady plateau state in the second phase without increasing or decreasing until approximately 12 to 14 days, at which time levels decrease to the normal value of 1 mg/dl.

This pattern varies according to racial group, method of feeding (breast vs. bottle), and gestational age.

In preterm formula-fed infants, serum bilirubin levels may peak as high as 10 to 12 mg/dl at 5 or 6 days of life and decrease slowly over a period of 2 to 4 weeks.

Breastfeeding-associated Jaundice

Breastfeeding is associated with an increased incidence of jaundice as a result of two distinct processes.

Breastfeeding-associated jaundice (early-onset jaundice) begins at 2 to 4 days of age and occurs in approximately 12% to 35% of breastfed newborns.

The jaundice is related to the process of breastfeeding and probably results from decreased caloric and fluid intake by breastfed infants before the milk supply is well established because decreased milk intake is associated with increased enterohepatic circulation of bilirubin.

Reduced fluid intake results in dehydration, which also concentrates the bilirubin in the blood.

Breast milk jaundice (late-onset jaundice) begins at age 5 to 7 days and occurs in 2% to 4% of breastfed infants. Rising levels of bilirubin peak during the second week and gradually diminish.

Despite high levels of bilirubin that may persist for 3 to 12 weeks, these infants are well.

The jaundice may be caused by factors in the breast milk (pregnanediol, fatty acids, and β-glucuronidase) that either inhibit the conjugation or decrease the excretion of bilirubin. Less frequent stooling by breastfed infants may allow for an extended time for reabsorption of bilirubin from stools.

Diagnostic Evaluation

The degree of jaundice is determined by serum bilirubin measurements. Normal values of unconjugated bilirubin are 0.2 to 1.4 mg/dl. In newborns, levels must exceed 5 mg/dl before jaundice (icterus) is observable.

The timing of the appearance of clinical jaundice; gestational age at birth; age in days since birth; family history, including maternal Rh factor; evidence of hemolysis; feeding method; infant's physiologic status; should be taken.

- The following criteria are indicators of pathologic jaundice:

Persistent jaundice over 2 weeks in a full-term formula-fed infant

Total serum bilirubin levels over 12.9 mg/dl (term infant) or over 15 mg/dl (preterm infant); the upper limit for breastfed infant is 15 mg/dl

Increase in serum bilirubin by 5 mg/dl/day

Direct bilirubin exceeding 1.5 to 2 mg/dl

Total serum bilirubin level over the 95th percentile for age (in hours) on an hour-specific nomogram

Therapeutic Management

The primary goals in the treatment of hyperbilirubinemia are to identify infants at high risk for hyperbilirubinemia; monitor serum bilirubin levels; prevent bilirubin encephalopathy; and, as in any blood group incompatibility, to reverse the hemolytic process.

The main form of treatment involves the use of phototherapy.

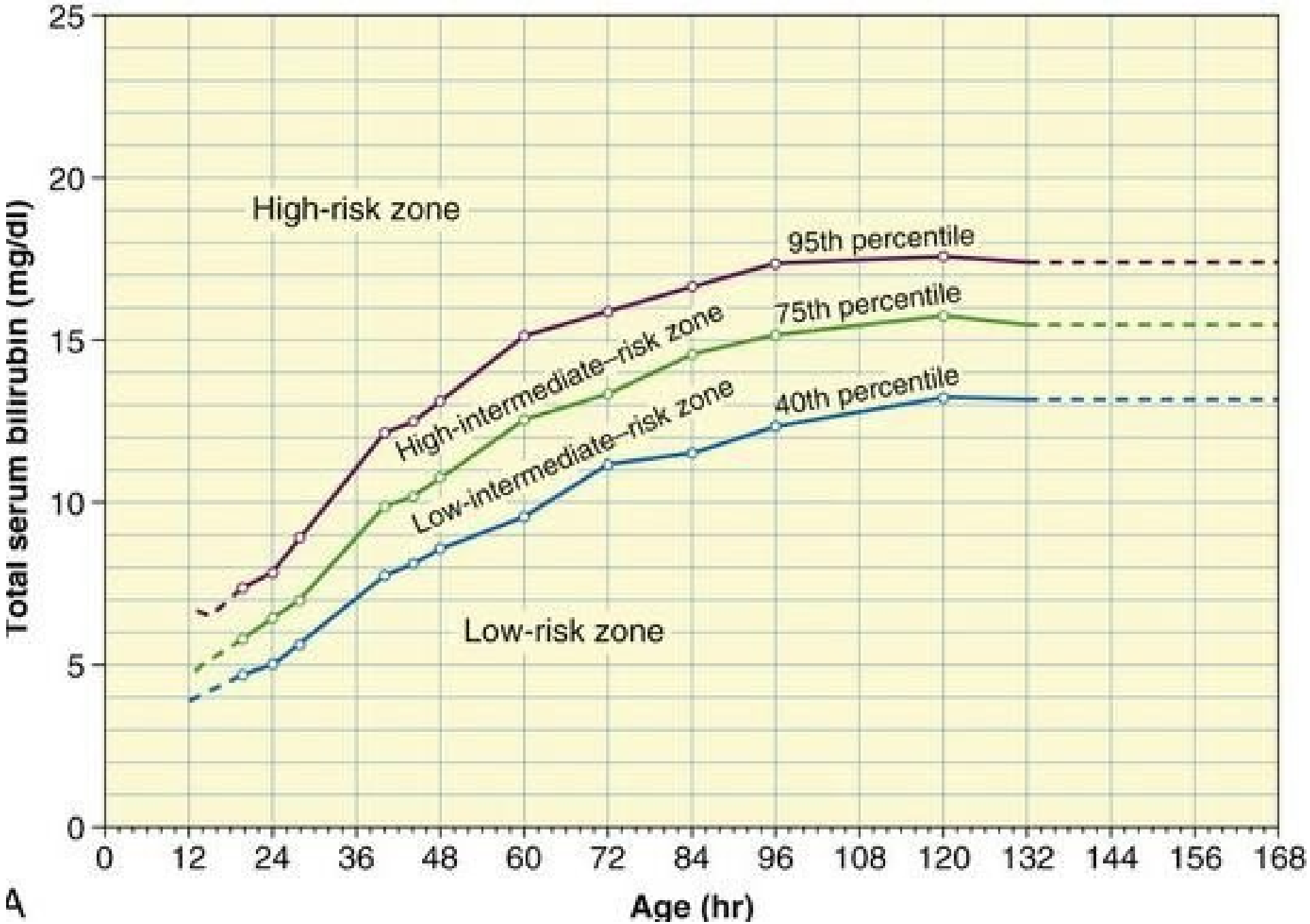
Exchange transfusion is generally used for reducing dangerously high bilirubin levels that may occur with hemolytic disease. High bilirubin levels that may occur with hemolytic disease.

Intravenous immunoglobulin (IVIG) is effective in reducing bilirubin levels in infants with Rh isoimmunization and ABO incompatibility

Phototherapy: promotes bilirubin excretion by photoisomerization, which alters the structure of bilirubin to a soluble form (lumirubin) for easier excretion,

Encouraging frequent breastfeeding, preferably every 2 hours; avoiding glucose water




Total serum bilirubin (mg/dl)
25
20
15
10
5
0
High-risk zone
95th percentile
High-intermediate–risk zone
75th percentile
Low-intermediate–risk zone
40th percentile
Low-risk zone
0
12
24
36
48
60
72
84
96
108
120
132
144
156
168
Age (hr)
A

Hemolytic Disease of the Newborn

Hyperbilirubinemia in the first 24 hours of life is most often the result of HDN, an abnormally rapidbrate of RBC destruction.

Anemia caused by this destruction stimulates the production of RBCs, which in turn provides increasing numbers of cells for hemolysis.

Major causes of increased erythrocyte destruction are isoimmunization (primarily Rh) and ABO incompatibility

- Blood Incompatibility
- The membranes of human blood cells contain a variety of antigens, also known as agglutinogens, substances capable of producing an immune response if recognized by the body as foreign.
- The reciprocal relationship between antigens on RBCs and antibodies in the plasma causes agglutination (clumping).
- In the ABO blood group system, the antibodies occur naturally. In the Rh system, the person must be exposed to the Rh antigen before significant antibody formation takes place and causes a sensitivity response known as isoimmunization.

Hemolytic Disease of the Newborn

- Rh Incompatibility (Isoimmunization)

The Rh blood group consists of several antigens (with D being the most prevalent). For simplicity, only the terms Rh positive (presence of antigen) and Rh negative (absence of antigen) are used in this discussion.

The presence or absence of the naturally occurring Rh factor determines the blood type.

Ordinarily, no problems are anticipated when the Rh blood types are the same in both the mother and the fetus or when the mother is Rh positive and the infant is Rh negative.

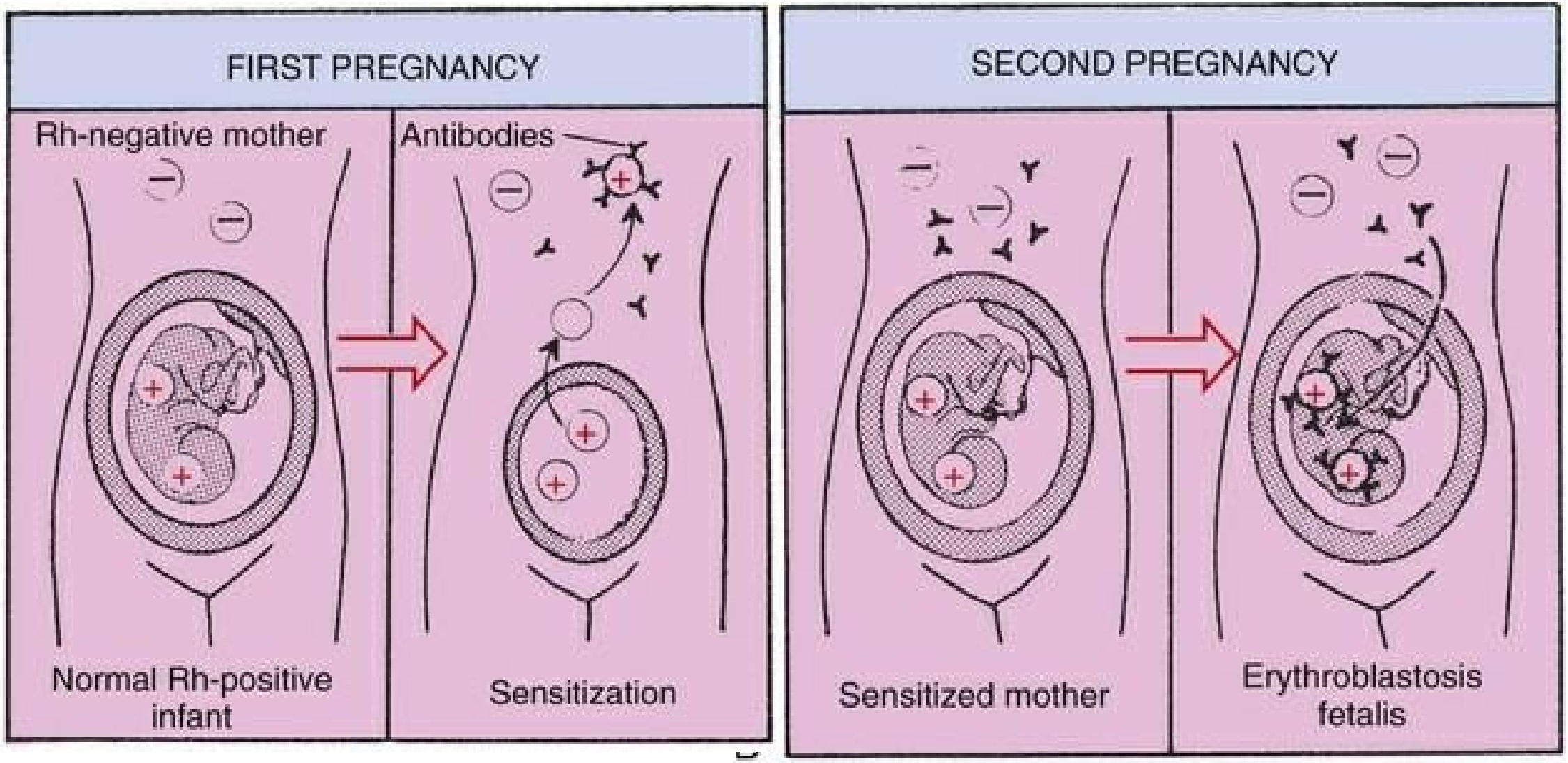
Difficulty may arise when the mother is Rh negative and the infant is Rh positive.

Although the maternal and fetal circulations are separate, there is evidence that fetal RBCs and cell-free DNA can enter the maternal circulation during pregnancy. More commonly, however, fetal RBCs enter into the maternal circulation at the time of delivery

Under normal circumstances, this process of isoimmunization has no effect during the first pregnancy with an Rh- positive fetus, because the initial sensitization to Rh antigens rarely occurs before the onset of labor. However, with the increased risk of fetal blood being transferred to the maternal circulation during placental separation, maternal antibody production is stimulated.

During a subsequent pregnancy with an Rh-positive fetus, these previously formed maternal antibodies to Rh- positive blood cells may enter the fetal circulation, where they attack and destroy fetal erythrocytes

Development of maternal sensitization to Rh antigens. A, Fetal Rh-positive erythrocytes enter the maternal system. Maternal anti-Rh antibodies are formed. B, Anti-Rh antibodies cross the placenta and attack fetal erythrocytes.



Because the condition begins in utero, the fetus attempts to compensate for the progressive hemolysis and anemia by accelerating the rate of erythropoiesis. As a result, immature RBCs (erythroblasts) appear in the fetal circulation, hence the term erythroblastosis fetalis.

Sensitization may occur during the first pregnancy if the woman had previously received an Rhpositive blood transfusion. No sensitization may occur in situations in which a strong placental barrier prevents transfer of fetal blood into the maternal circulation.

In approximately 10% to 15% of sensitized mothers, there is no hemolytic reaction in the newborn.

In addition, some Rh-negative women, even though exposed to Rh-positive fetal blood, are immunologically unable to produce antibodies to the foreign antigen.

In the most severe form of erythroblastosis fetalis, hydrops fetalis, the progressive hemolysis causes fetal hypoxia; cardiac failure; generalized edema (anasarca); and fluid effusions into the pericardial, pleural, and peritoneal spaces (hydrops).

The fetus may be delivered stillborn or in severe respiratory distress. Maternal Rh immunoglobulin (RhIg) administration, early intrauterine detection of fetal anemia by ultrasonography (serial Doppler assessment of the peak velocity in the fetal middle cerebral artery), and subsequent treatment by fetal blood transfusions or high-dose IVIG have dramatically improved the outcome of affected fetuses

ABO Incompatibility

Potential Maternal–Fetal ABO Incompatibilities

Maternal Blood Group	Incompatible Fetal Blood Group
O	A or B
B	A or AB
A	B or AB

Hemolytic disease can also occur when the major blood group antigens of the fetus are different from those of the mother.

The major blood groups are A, B, AB, and O.

46% have type O blood, 42% have type A blood, 9% have type B blood, and 3% have type AB blood.

Antibodies in the plasma of one blood group (except the AB group, which contains no antibodies) will produce agglutination (clumping) when mixed with antigens of a different blood group.

Naturally occurring antibodies in the recipient's blood cause agglutination of a donor's RBCs.

The agglutinated donor cells become trapped in peripheral blood vessels, where they hemolyze, releasing large amounts of bilirubin into the circulation.

The most common blood group incompatibility in the neonate is between a mother with O blood group and an infant with A or B blood group.

Naturally occurring anti-A or anti-B antibodies already present in the maternal circulation cross the placenta and attack the fetal RBCs, causing hemolysis.

Usually, the hemolytic reaction is less severe than in Rh incompatibility

Unlike the Rh reaction, ABO incompatibility may occur in the first pregnancy.

The risk of significant hemolysis in subsequent pregnancies is higher when the first pregnancy is complicated by significant hemolysis

- Clinical Manifestations

  Jaundice may appear shortly after birth (during the first 24 hours) in newborns affected by HDN, and serum levels of unconjugated bilirubin rise rapidly.

  Anemia results from the hemolysis of large numbers of erythrocytes, and hyperbilirubinemia and jaundice result from the liver's inability to conjugate and excrete the excess bilirubin.

  Most newborns with HDN are not jaundiced at birth.

  However, hepatosplenomegaly and varying degrees of hydrops may be evident. If the infant is severely affected, signs of anemia (notably, marked pallor) and hypovolemic shock are apparent.

  Hypoglycemia may occur as a result of pancreatic cell hyperplasia.

- Diagnostic Evaluation

  Early identification and diagnosis of RhD sensitization are important in the management and prevention of fetal complications.

  A maternal antibody titer (indirect Coombs test) should be drawn at the first prenatal visit. Genetic testing allows early identification of paternal zygosity at the RhD gene locus

  Amniocentesis can be used to test the fetal blood type of a woman whose antibody screen result is positive; the use of polymerase chain reaction may determine the fetal blood type and presence of maternal antibodies

  Alterations in the placenta, umbilical cord, and amniotic fluid volume, as well as the presence of fetal hydrops, can be detected with high-resolution ultrasonography

Therapeutic Management

Although phototherapy may control bilirubin levels in mild cases, the hemolytic process may continue, causing significant anemia between 7 and 21 days of life.

In some institutions, an IVIG is administered to decrease the formation of bilirubin in neonates with ABO incompatibility.

Prevention of Rh Isoimmunization

The administration of RhIg, a human gamma globulin concentrate of anti-D, to all unsensitized Rhnegative mothers at 28 weeks of gestation and after delivery or abortion of an Rh-positive infant or fetus prevents the development of maternal sensitization to the Rh factor. The injected anti-Rh antibodies are thought to destroy fetal RBCs passing into the maternal circulation before they can be recognized by the mother's immune system.

Because the immune response is blocked, anti-D antibodies and memory cells are not formed

The inhibition of memory cell formation is especially important because memory cells provide long-term immunity by initiating a rapid immune response after the antigen is reintroduced.

To be effective, RhIg (e.g., RhoGAM) must be administered to unsensitized mothers within 72 hours (but possibly as long as 3 to 4 weeks) after the first delivery or abortion and repeated after subsequent pregnancies or losses. The administration of RhIg at 26 to 28 weeks of gestation further reduces the risk of Rh isoimmunization. RhIg is not effective against existing Rh-positive antibodies in the maternal circulation.

Studies have demonstrated the effectiveness of IVIG at decreasing the severity of RBC destruction (hemolysis) in HDN and subsequent development of neonatal jaundice

Therapeutic Management

- ## Intrauterine Transfusion

  Infants of mothers already sensitized may be treated by intrauterine transfusion, which consists of infusing blood into the umbilical vein of the fetus.

  The need for therapy is based on the antenatal diagnosis of isoimmunization by determining the optical density of amniotic fluid (by amniocentesis) as an index of fetal hemolysis or by serial ultrasonography, which may detect the presence of fetal hydrops as early as 16 weeks of gestation. With the advance of ultrasound technology, fetal transfusion may be accomplished directly via the umbilical vein, infusing type ORh-negative packed RBCs to raise the fetal hematocrit to 40% to 50%. The frequency of intrauterine transfusions may vary according to institution and fetal hydropic status but are most often done every 2 to 3 weeks until the fetus reaches pulmonary maturity at approximately 36 weeks of gestation

- ## Exchange Transfusion

  Exchange transfusion, in which the infant's blood is removed in small amounts (usually 5 to 10 mlat a time) and replaced with compatible blood (e.g., Rh-negative blood), is a standard mode of therapy for treatment of severe hyperbilirubinemia and is the treatment of choice for hyperbilirubinemia and hydrops caused by Rh incompatibility. Exchange transfusion removes the sensitized erythrocytes, lowers the serum bilirubin level to prevent bilirubin encephalopathy, corrects the anemia, and prevents cardiac failure. Indications for exchange transfusion in full-term infants may include a rapidly increasing serum bilirubin level and hemolysis despite intensive phototherapy.

Respiratory Distress Syndrome

Respiratory Distress Syndrome

Respiratory distress is a name applied to respiratory dysfunction in neonates and is primarily a disease related to developmental delay in lung maturation.

The terms respiratory distress syndrome (RDS) and hyaline membrane disease are most often applied to this severe lung disorder, which not only is responsible for more infant deaths than any other disease but also carries the highest risk in terms of long-term respiratory and neurologic complications for a discussion of acute RDS).

It is seen almost exclusively in preterm infants. The disorder is rare in drug-exposed infants and infants who have been subjected to chronic intrauterine stress (e.g., maternal preeclampsia or hypertension).

Respiratory distress of a nonpulmonary origin in neonates may also be caused by sepsis, cardiac defects (structural or functional), exposure to cold, airway obstruction (atresia), intraventricular hemorrhage, hypoglycemia, metabolic acidosis, acute blood loss, and drugs.

Pneumonia in the neonatal period may result in respiratory distress caused by bacterial or viral agents and may occur alone or as a complication of RDS.

Pathophysiology

Preterm infants are born before the lungs are fully prepared to serve as efficient organs for gas exchange. This appears to be a critical factor in the development of RDS. The effects of lung immaturity are compounded by the presence of more cartilage in the chest wall, leading to increased compliance of the chest wall, which collapses inward in response to less compliant (stiffer) lung tissue.

There is evidence of fetal respiratory activity before birth. The lungs make feeble respiratory movements, and fluid is excreted through the alveoli. Because the final unfolding of the alveolar septa, which increases the surface area of the lungs, occurs during the last trimester of pregnancy, preterm infants are born with numerous underdeveloped and many uninflatable alveoli.

Pulmonary blood flow is limited as a result of the collapsed state of the fetal lungs, poor vascular development in general, and an immature capillary network. Because of increased pulmonary vascular resistance (PVR), the major portion of fetal blood is shunted from the lungs by way of the ductus arteriosus and foramen ovale.

At birth, infants must initiate breathing and keep the previously fluid-filled lungs inflated with air. At the same time, the pulmonary capillary blood flow increases by approximately tenfold to provide for adequate lung perfusion and to alter the intracardiac pressure that closes the fetal cardiac shunts.

. Most full-term infants successfully accomplish these adjustments, but preterm infants with respiratory distress are unable to do so. Although numerous factors are involved, a lack of stable surfactant plays a central role.

Surfactant is a surface-active phospholipid secreted by the alveolar epithelium. Acting much like a detergent, this substance reduces the surface tension of fluids that line the alveoli and respiratory passages, resulting in uniform expansion and maintenance of lung expansion at low intraalveolar pressure.

Deficient surfactant production causes unequal inflation of alveoli on inspiration and the collapse of alveoli on end expiration. Without surfactant, infants are unable to keep their lungs inflated and therefore exert a great deal of effort to reexpand the alveoli with each breath. With increasing exhaustion, infants are able to open fewer and fewer alveoli. This inability to maintain lun produces widespread atelectasis.



Following birth the oxygen concentration in the blood normally increases, the ductus arteriosus constricts and the pulmonary vessels dilate to decrease PVR. In the absence of alveolar stability (normal functional residual capacity) and with progressive atelectasis, PVR increases as resistance to blood flow into the lungs increases hypoperfusion to the lung tissue occurs. With the increase in PVR, fetal shunts (ductus arteriosus and foramen ovale) remain open allowing right-to-left shunting of blood through the persisting fetal shunts.

Inadequate pulmonary perfusion and ventilation produce hypoxemia and hypercapnia.

Pulmonary arterioles, with their thick muscular layer, constrict in response to hypoxia. Thus, a decrease in oxygen tension causes vasoconstriction in the pulmonary arterioles that is further enhanced by a decrease in blood pH. This vasoconstriction contributes to a further increase in PVR.

Prolonged hypoxemia activates anaerobic glycolysis, which produces increased amounts of lactic acid. An increase in lactic acid causes metabolic acidosis; an inability of the atelectatic lungs to blow off excess carbon dioxide produces respiratory acidosis. Acidosis causes further vasoconstriction.

With deficient pulmonary circulation and alveolar perfusion, partial pressure of oxygen in arterial

Diagnostic Evaluation

The diagnosis of RDS is made on the basis of clinical signs and chest x-ray studies.

Radiographic findings characteristic of RDS include (1) a diffuse granular pattern over both lung fields that closely resembles ground glass and represents alveolar atelectasis and (2) dark streaks, or bronchograms, within the ground glass areas that represent dilated, air-filled bronchioles. It is difficult to distinguish between RDS and pneumonia in infants with respiratory distress. The extent of respiratory compromise and acid–base status is determined by blood gas analysis.

Pulse oximetry and carbon dioxide monitoring, as well as pulmonary function studies, assist in differentiating pulmonary and extra

- Neonatal Respiratory Distress Syndrome

Room air or oxygen saturation ≥88%

Respiratory rate <60 breaths/min

Blood pH ≥7.30pulmonary illness and are used in the management of RDS.

- Clinical Manifestations of Respiratory Distress Syndrome

Tachypnea (>80 to 120 breaths/min) initially*

Dyspnea

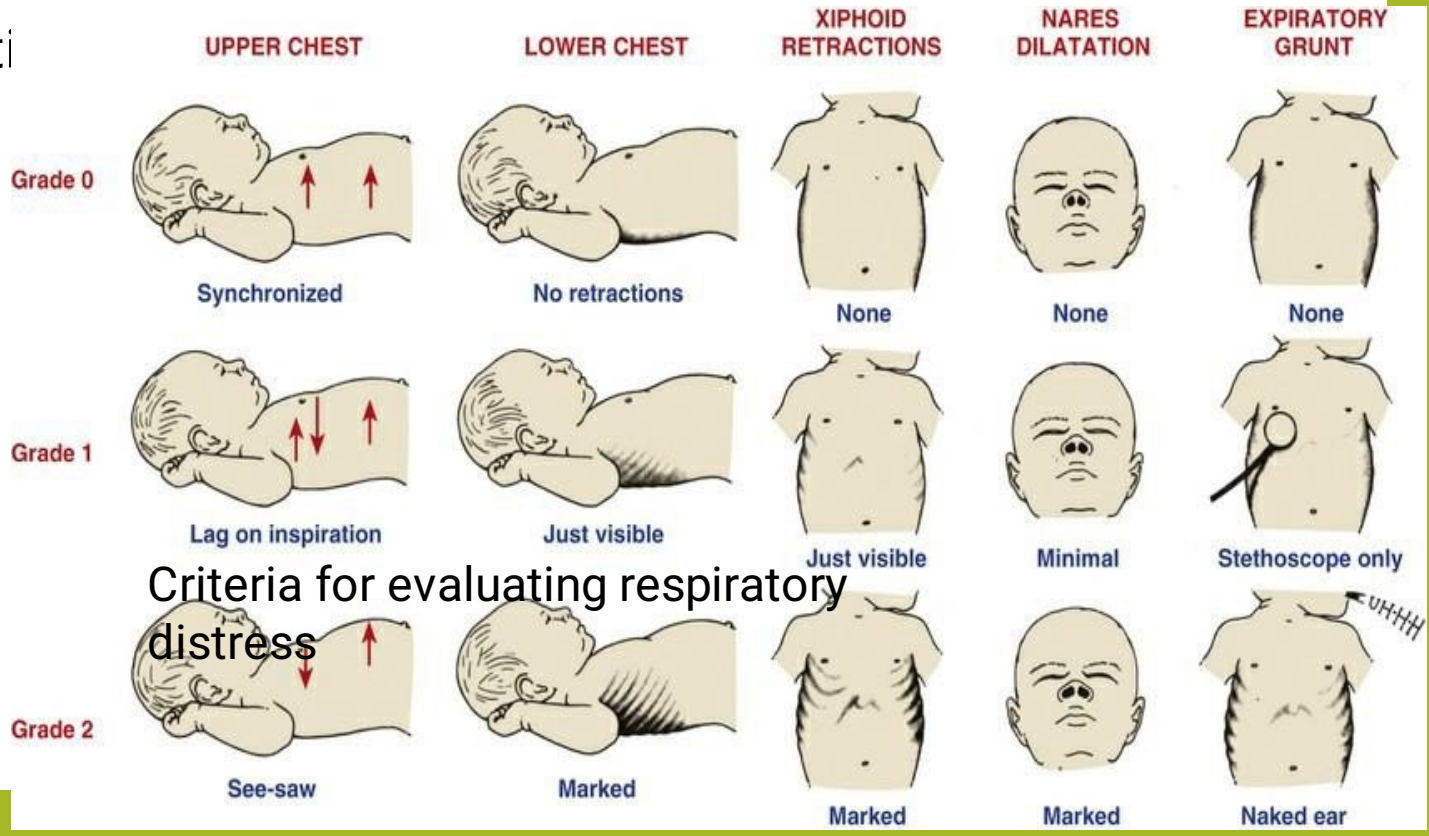
Pronounced intercostal/substernal retracti

Fine inspiratory crackles

Audible expiratory grunt

Flaring of the external nares

Cyanosis or pallor



Criteria for evaluating respiratory distress

Therapeutic Management

The treatment of RDS involves immediate establishment of adequate oxygenation and ventilation and supportive care and measures required for any preterm infant, as well as those instituted to prevent further complications associated with preterm birth. The supportive measures most crucial to a favorable outcome are to:

- Maintain adequate ventilation and oxygenation
- Maintain acid–base balance
- Maintain a neutral thermal environment
- Maintain adequate tissue perfusion and oxygenation
- Prevent hypotension
- Maintain adequate hydration and electrolyte status

Nipple feedings are contraindicated in any situation that creates a marked increase in respiratory rate because of the greater hazards of aspiration. Nutrition is provided by parenteral therapy during the acute stage of the disease, and minimal enteral feeding is provided to enhance maturation of the neonate's gastrointestinal system.

The administration of exogenous surfactant to preterm neonates with RDS has become an accepted and common therapy in most neonatal centers worldwide

Respiratory Complications

- Meconium Aspiration Syndrome
- Apnea of Prematurity
- Pneumothorax
- Persistent Pulmonary Hypertension of the Newborn
- Retinopathy of Prematurity

Positive pressure introduced by mechanical apparatus has created an increase in the incidence of ruptured alveoli and subsequent pneumothorax and bronchopulmonary dysplasia (chronic lung disease).

Retinopathy of prematurity is observed almost exclusively in preterm infants and is related primarily to prematurity and oxygen therapy.

Evidence supports the resuscitation of asphyxiated newborns with 21% oxygen rather than 100% oxygen; preliminary studies reduced mortality and neurologic morbidities in newborns resuscitated with 21% oxygen

Pulse oximetry is recommended to monitor the infant's oxygenation status during resuscitation and to prevent excessive use of oxygen in both term and preterm infants

Inhaled nitric oxide (INO) and extracorporeal membrane oxygenation (ECMO) are additional therapies used in the treatment of respiratory distress and respiratory failure in neonates.

INO is used in term and late preterm infants with conditions such as persistent pulmonary hypertension, meconium aspiration syndrome, pneumonia, sepsis, and congenital diaphragmatic hernia to decrease or reverse pulmonary hypertension, pulmonary vasoconstriction, acidosis, and

Common Methods for Assisted Ventilation in Neonatal Respiratory Distress

Method	Description	How Provided
Conventional Methods		
Continuous positive airway pressure (CPAP)	Provides constant distending pressure to airway in spontaneously breathing infant	Nasal prongs ET tube Face mask
Intermittent mandatory ventilation (IMV)*	Allows infant to breathe spontaneously at own rate but provides mechanical cycled respirations and pressure at regular preset intervals	ET intubation and ventilator
Synchronized intermittent mandatory ventilation (SIMV)	Mechanically delivered breaths are synchronized to the onset of spontaneous patient breaths; assist/control mode facilitates full inspiratory synchrony; involves signal detection of onset of spontaneous respiration from abdominal movement, thoracic impedance, and airway pressure or flow changes	Patient-triggered infant ventilator with signal detector and assist/control mode; ET tube
Volume guarantee ventilation	Delivers a predetermined volume of gas using an inspiratory pressure that varies according to the infant's lung compliance (often used in conjunction with SIMV)	Volume guarantee ventilator with flow sensor; ET tube
Alternative Methods		
High-frequency oscillation (HFO)	Application of high-frequency, low-volume, sine-wave flow oscillations to airway at rates between 480 and 1200 breaths/min	Variable-speed piston pump (or loudspeaker, fluidic oscillator); ET tube
High-frequency jet ventilation (HFJV)	Uses a separate, parallel, low-compliant circuit and injector port to deliver small pulses or jets of fresh gas deep into airway at rates between 250 and 900 breaths/min	May be used alone or with low-rate IMV; ET tube

*Also referred to as *conventional ventilation* (vs. high-frequency ventilation [HFV]).

ET, Endotracheal tube.

Prevention

The most successful approach to prevention of RDS is prevention of preterm delivery, especially in elective early delivery and cesarean section. Improved methods for assessing the maturity of the fetal lung by amniocentesis, although not a routine procedure, allow a reasonable prediction of adequate surfactant formation. Because estimation of a delivery date can be miscalculated by as much as 1 month, such tests are particularly valuable when scheduling an elective cesarean section.

The combination of maternal steroid administration before delivery and surfactant administration postnatally seems to have a synergistic effect on neonatal lungs, with the net result being a decrease in infant mortality, decreased incidence of intraventricular hemorrhage, fewer pulmonary air leaks, and fewer problems with pulmonary interstitial emphysema and RDS.

Sepsis

Sepsis, or septicemia, refers to a generalized bacterial infection in the bloodstream.

Neonates are highly susceptible to infection as a result of diminished nonspecific (inflammatory) and specific (humoral) immunity, such as impaired phagocytosis, delayed chemotactic response, minimal or absent IgA and immunoglobulin M (IgM), and decreased complement levels.

Because of infants' poor response to pathogenic agents, there is usually no local inflammatory reaction at the portal of entry to signal an infection, and the resulting symptoms tend to be vague and nonspecific.

Breastfeeding has a protective benefit against infection and should be promoted for all newborns.

Colostrum contains immunoglobulins that are effective against gram-negative bacteria.

Sepsis in the neonatal period can be acquired prenatally across the placenta from the maternal bloodstream or during labor from ingestion or aspiration of infected amniotic fluid. Prolonged rupture of the membranes always presents a risk for this type from maternal–fetal transfer of pathogenic organisms.

In utero transplacental transfer can occur with organisms and viruses such as cytomegalovirus, toxoplasmosis, and Treponema pallidum (syphilis), which cross the placental barrier during the latter half of pregnancy.

Intrapartum infection may occur via contact with an infected mother; examples of such infections include herpesvirus and human immunodeficiency virus (HIV).

Early-onset sepsis (less than 3 days after birth) is acquired in the perinatal period; infection can occur from direct contact with organisms from the maternal gastrointestinal and genitourinary tracts.

The most common infecting organism in term infants is group B streptococcus (GBS); in preterm infants, it is Escherichia coli. Despite the development of maternal screening and prophylaxis, infection rates for early- onset GBS infection remain at approximately 0.3 per 1000 live births .

E. coli, which may be present in the vagina, accounts for approximately half of all cases of sepsis caused by gram-negative organisms. GBS is an extremely virulent organism in neonates, with a high (50%) death rate in affected infants.

Other bacteria noted to cause early-onset infection include Haemophilus influenzae, Neisseria meningitidis, coagulase-negative Staphylococcus (ConS), and Streptococcus pneumoniae

Late-onset sepsis (1 to 3 weeks after birth) is primarily nosocomial, and the offending organisms are usually staphylococci, Klebsiella organisms, enterococci, E. coli, and Pseudomonas or Candida.

ConS, considered to be primarily a contaminant in older children and adults, is the most common cause of late-onset septicemia in ELBW and VLBW infants.

Bacterial invasion can occur through sites such as the umbilical stump; the skin; mucous membranes of the eye, nose, pharynx, and ear; and internal systems, such as the respiratory, nervous, urinary, and gastrointestinal systems.

Risk factors for ConS include low birth weight and early gestational age, poor hand hygiene, previous antibiotic exposure, and the presence of central IV lines.

Postnatal infection is acquired by cross-contamination from other infants, personnel, or objects in the environment. Bacteria that are commonly called “water bugs” (because they are able to grow in water) are found in water supplies, humidifying apparatus, sink drains, suction machines, and most respiratory equipment.

Organisms such as ConS, which usually colonize the skin, may infect indwelling venous and arterial catheters used for infusions, blood sampling, and monitoring of vital signs.

Neonatal sepsis is most common in infants at risk, particularly preterm infants and infants born after a difficult or traumatic labor and delivery, who are least capable of resisting such bacterial invasion.

These organisms are often transmitted by personnel from person to person or object to person by poor

Manifestations of Neonatal Sepsis

- General Signs

  Infant generally “not doing well”

  Poor temperature control—hypothermia, hyperthermia (rare in neonates)
- Circulatory System

  Pallor, cyanosis, or mottling

  Cool, clammy skin

  Hypotension

  Edema

  Irregular heartbeat—bradycardia, tachycardia
- Respiratory System

  Irregular respirations, apnea, or tachypnea

  Cyanosis

  Grunting

  Dyspnea

  Retractions

- Central Nervous System
  - Diminished activity—lethargy, hyporeflexia, coma
  - Increased activity—irritability, tremors, seizures
  - Full fontanel
  - Increased or decreased tone
  - Abnormal eye movements
- Gastrointestinal System
  - Poor feeding
  - Vomiting, Diarrhea or decreased stooling
  - Abdominal distention
  - Hepatomegaly
  - Hemoccult-positive stools
- Hematopoietic System
  - Jaundice
  - Pallor
  - Petechiae, ecchymosis
  - Splenomegaly

- ## Diagnostic Evaluation

  The definitive diagnosis is established by laboratory and radiographic examination. Cultures of blood, urine, and CSF are collected to identify the causative organism. Blood studies may show signs of anemia, leukocytosis, or leukopenia.

  Leukopenia is usually an ominous sign because of its frequent association with high mortality. An elevated number of immature neutrophils (a left shift), decreased or increased total neutrophils, and changes in neutrophil morphology also suggest an infectious process in the neonate.

  Other diagnostic data may be helpful in determination of neonatal sepsis include C-reactive protein and other acute phase reactants, such as serum amyloid A; procalcitonin; and interleukins, specifically interleukin-6

- ## Therapeutic Management

  In addition to the institution of vigorous therapeutic measures, early recognition and diagnosis are essential to increase the infant's chance for survival and reduce the likelihood of permanent neurologic damage. Antibiotic therapy is initiated before laboratory results are available for confirmation and identification of the exact organism.

  Treatment consists of circulatory support, respiratory support, aggressive administration of antibiotics, and immunotherapy.

Prevention

Several measures are important in the prevention of both early- and late-onset infection. Programs to screen pregnant women for GBS colonization (culture-based) and treatment of those women in labor have dramatically reduced the incidence of GBS infection in neonates.

Screening programs for other maternal infections, including hepatitis B and HIV, are also recommended.

In developed countries, breastfeeding by mothers infected with HIV is not recommended because the virus may be transmitted in breast milk.

Nursery procedures aimed at minimizing the risk of nosocomial infections include the practice of good hand-washing techniques, appropriate isolation precautions where indicated, and the adoption of recommended standards for spacing of infant beds.

Strategies such as the early introduction of enteral feeding aimed at reducing the indwelling time of central venous lines have been shown to reduce the risk of nosocomial infection

Necrotizing Enterocolitis

NEC is an acute inflammatory disease of the bowel with increased incidence in preterm infants. T

It appears to occur in infants whose gastrointestinal tracts have experienced vascular compromise. In

testinal ischemia of unknown etiology, immature gastrointestinal host defenses, bacterial proliferation, and feeding substrate are now believed to have a multifactorial role in the etiology of NEC. Prematurity remains the most prominent risk factor in the development of NEC .

The damage to mucosal cells lining the bowel wall may be significant. Diminished blood supply to these cells causes their death in large numbers; they stop secreting protective, lubricating mucus; and the thin, unprotected bowel wall is attacked by proteolytic enzymes. Thus, the bowel wall continues to swell and break down; it is unable to synthesize protective IgM, and the mucosa is permeable to macromolecules (e.g., exotoxins), which further hampers intestinal defenses. Gasforming bacteria invade the damaged areas to produce pneumatosis intestinalis, a radiologic finding reflecting the presence of gas in the submucosal or subserosal surfaces of the bowel.

A consistent relationship has been observed between the development of NEC and enteric feeding of hypertonic substances (e.g., formula, hyperosmolar medications). It is unclear whether this connection is a result of the formula imposing a stress on an ischemic bowel, serving as a substrate for bacterial growth, or both.

Necrotizing Enterocolitis

Clinical Manifestations of Necrotizing Enterocolitis

- Nonspecific Clinical Signs

  Lethargy

  Poor feeding

  Hypotension

  Vomiting

  Apnea

  Decreased urinary output

  Unstable temperature

  Jaundice

  Specific Signs

  Distended (often shiny) abdomen

  Blood in the stools or gastric contents

  Gastric retention (undigested formula)

  Localized abdominal wall erythema or induration

  Bilious vomitus

- Diagnostic Evaluation

Radiographic studies show a sausage-shaped dilation of the intestine that progresses to marked distention and the characteristic pneumatosis intestinalis—"soapsuds," or the bubbly appearance of thickened bowel wall and ultra lumina. There may be air in the portal circulation or free air observed in the abdomen, indicating perforation. Laboratory findings may include anemia, leukopenia, leukocytosis, metabolic acidosis, and electrolyte imbalance. In severe cases, coagulopathy (DIC) or thrombocytopenia may be evident. Gram- negative organisms are often cultured from blood, although bacteremia or septicemia may not be prominent early in the course of the disease.

- Therapeutic Management

Treatment of infants with NEC begins with prevention. Oral feedings may be withheld for at least 24 to 48 hours from infants who are believed to have experienced birth asphyxia. Breast milk is the preferred enteral nutrient because it confers some passive immunity (IgA), macrophages, and lysozymes.
Minimal enteral feedings (trophic feeding, gastrointestinal priming) have gained acceptance with no evidence of increased incidence of NEC. In particular, the use of fresh human milk has been shown to decrease the risk of NEC

Medical treatment of infants with confirmed NEC consists of discontinuation of all oral feedings; institution of abdominal decompression via nasogastric suction; administration of IV antibiotics; and correction of extravascular volume depletion, electrolyte abnormalities, acid–base imbalances, and hypoxia. Replacing oral feedings with parenteral fluids decreases the need for oxygen and circulation to the bowel. Serial abdominal radiographs (every 6 to 8 hours in the acute phase) are taken to monitor for possible progression of the disease to intestinal perforation.

Phenylketonuria

An inborn error of metabolism inherited autosomal recessive trait (PAH gene is located on chromosome 12q24), is caused by a deficiency or absence of the enzyme needed to metabolize the essential amino acid phenylalanine.

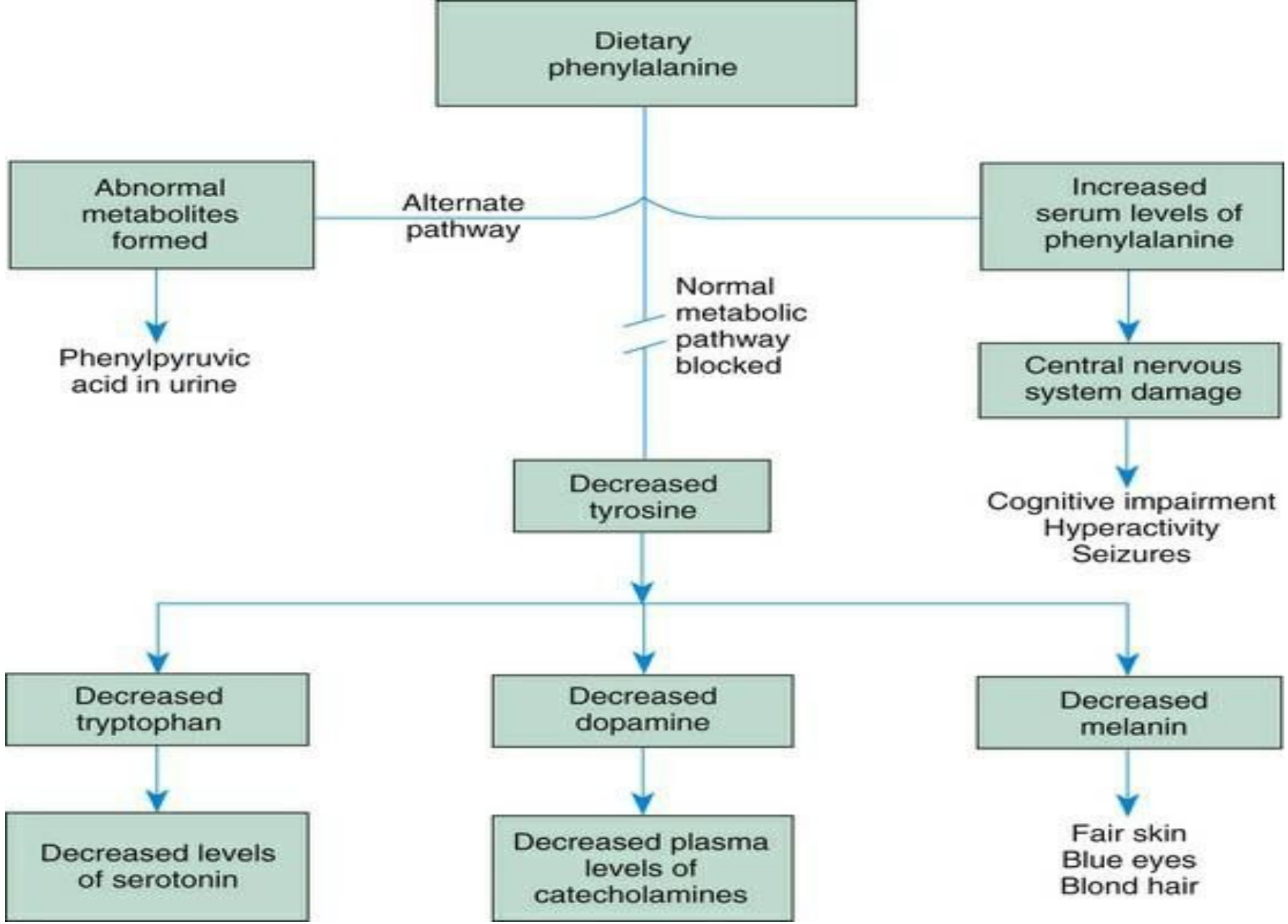
Classic PKU is at one end of a spectrum of conditions known as hyperphenylalaninemia and varying degrees of severity depending on the degree of enzyme deficiency.

In PKU, the hepatic enzyme phenylalanine hydroxylase, which normally controls the conversion of phenylalanine to tyrosine, is deficient. This results in the accumulation of phenylalanine in the bloodstream and urinary excretion of abnormal amounts of its metabolites, the phenyl acids.

One of these phenylketones, phenylacetic acid, gives urine the characteristic musty odor associated with the disease. Another is phenylpyruvic acid, which is responsible for the term phenylketonuria.

Tyrosine, the amino acid produced by the metabolism of phenylalanine, is absent in PKU.

Tyrosine is needed to form the pigment melanin and the hormones epinephrine and T4.



Metabolic error and consequences in phenylketonuria.

- Diagnosis

The objective in diagnosing and treating the disorder is to prevent cognitive impairment.

Every newborn should be screened for PKU. The most commonly used test for screening newborns is the Guthrie blood test, a bacterial inhibition assay for phenylalanine in the blood.

If performed properly, this test detects serum phenylalanine levels greater than 4 mg/dl (normal value, 1.6 mg/dl), through fresh heel blood

- Therapeutic Management

Treatment of PKU involves restricting phenylalanine in the diet.

Phenylalanine cannot be eliminated because it is an essential amino acid in tissue growth.

Therefore, dietary management must meet two criteria: (1) meet the child's nutritional need for optimum growth and (2) maintain phenylalanine levels within a safe range (2 to 6 mg/dl in neonates and children up to 12 years old, and 2 to 10 mg/dl through adolescence

Galactosemia

Galactosemia is a rare autosomal recessive disorder that results from various gene mutations leading to three distinct enzymatic deficiencies. The most common type of galactosemia (classic galactosemia) results from a deficiency of a hepatic enzyme, galactose 1-phosphate uridyltransferase (GALT), and affects approximately 1 in 50,000 births.

The other two varieties of galactosemia involve deficiencies in the enzymes galactokinase (GALK) and galactose 4'- epimerase (GALE); these are extremely rare disorders.

All three enzymes (GALT, GALK, and GALE) are involved in the conversion of galactose into glucose.

As galactose accumulates in the blood, several organs are affected. Hepatic dysfunction leads to cirrhosis, resulting in jaundice in the infant by the second week of life.

The spleen subsequently becomes enlarged as a result of portal hypertension.

Cataracts are usually recognizable by 1 or 2 months of age; cerebral damage, manifested by the symptoms of lethargy and hypotonia, is evident soon afterward.

Infants with galactosemia appear normal at birth, but within a few days of ingesting milk (which has a high lactose content), they begin to experience vomiting and diarrhea, leading to weight loss. E. coli sepsis is also a common presenting clinical sign. Death during the first month of life is frequent in untreated infants.

- Diagnostic Evaluation

  Galactosuria, increased levels of galactose in the blood

  Decreased levels of GALT activity in erythrocytes.

  The infant may display characteristics of malnutrition; hypoglycemia, jaundice, hepatosplenomegaly, sepsis, cataracts, and decreased muscle tone

- Therapeutic Management

  During infancy, treatment consists of eliminating all milk and lactose-containing formula, including breast milk. Traditionally, lactose-free formulas are used, with soy-protein formula being the feeding of choice

Failure to Thrive

Failure to thrive (FTT), or growth failure, is a sign of inadequate growth resulting from an inability to obtain or use calories required for growth. FTT has no universal definition, although one of the more common criteria is a weight (and sometimes height) that falls below the fifth percentile for the child's age.

Another definition of FTT includes a weight for age (height)

A third way to define FTT is a weight curve that crosses more than two percentile lines on a standardized growth chart after previous achievement of a stable growth pattern.

Weight for length is reported to be a better indicator of acute undernutrition.

Growth measurements alone are not used to diagnose children with FTT.

In addition to lack of consensus on the precise definition of FTT, some advocate for a change in terminology; thus, terms such as growth failure and pediatric undernutrition are used in the literature for FTT.

Approximately 5% to 10% of children in primary care in the United States have FTT with the majority presenting before 18 months old.

Some experts suggest that the previously used classifications of organic FTT and nonorganic FTT are too simplistic because most cases of growth failure have mixed causes; they suggest that FTT be

Inadequate caloric intake: Incorrect formula preparation, neglect, food fads, lack of food availability, breastfeeding problems, behavioral problems affecting eating, or central nervous system problems affecting intake

Inadequate caloric absorption: Food allergy, malabsorption, pyloric stenosis, GI atresia, inborn errors of metabolism

Excessive caloric expenditure: Hyperthyroidism, malignancy, congenital heart disease, chronic pulmonary disease or chronic immunodeficiency

The cause of FTT is often multifactorial and involves a combination of infant organic disease,dysfunctional parenting behaviors, and/or poor parent-infant bonding

- Diagnostic Evaluation

If FTT is recent, the weight, but not the height

If FTT is longstanding, both weight and height are low, indicating chronic malnutrition.

A dietary intake history, either a 24-hour food intake or a history of food consumed over a 3- to 5-day period, is also essential.

food allergies, and dietary restrictions.

An assessment of the home environment and child–parent interaction may be helpful as well. Other tests (lead toxicity, anemia, stool reducing substances, occult blood, ova and parasites, alkaline phosphatase, and zinc levels) are selected only as indicated to rule out organic problems

Therapeutic Management

The primary management of FTT is aimed at reversing the cause of the growth failure.

The goal is to provide sufficient calories to support “catch-up” growth—a rate of growth greater than the expected rate for age.

In addition to adding caloric density to feedings, the child may require multivitamin supplements and dietary supplementation with high-calorie foods and drinks.

Sudden Infant Death Syndrome

Sudden infant death syndrome (SIDS) is defined as the sudden death of an infant younger than 1 year old that remains unexplained after a complete postmortem examination.

Since 1994, the incidence of SIDS in the US has decreased due to the Safe to Sleep campaign (Back).

SIDS is the third leading cause of infant deaths and the leading cause of postneonatal deaths.

• Etiology

One hypothesis is that SIDS is related to a brainstem abnormality in the neurologic regulation of cardiorespiratory control that has maldevelopment affects arousal and physiologic responses to a lifethreatening during sleep

Abnormalities include prolonged sleep apnea, increased frequency of brief inspiratory pauses, excessive periodic breathing, and impaired arousal responsiveness to increased carbon dioxide or decreased oxygen.

A genetic :A deficiency of the complement component C4 is associated with SIDS cases

In addition, polymorphisms among interleukin genes, transforming growth factor, tumor necrosis factor, and interferon gamma are closely associated with cases of SIDS.

A brain abnormality, a critical incident in the fetal developmental period or in early neonatal life, and an environmental stressor such as prone sleep positioning

Factor	Occurrence
Incidence	55/100,000
Peak age	2 to 3 months old; 90% occur by 6 months old; preterm infants die from sudden infant death syndrome (SIDS) at mean age of 6
Gender	More among boy
Time of death	During sleep
Time of year	More in Winter
Socioeconomi	Low SE status
Birth	Higher incidence in: ● Preterm infants, especially extremely and very low birth weight ● Multiple births† ● Neonates with low Apgar scores ● Infants with CNS disturbances and respiratory disorders ● Increasing birth order (subsequent siblings Vs to firstborn child)
Sleep patterns	Prone position; use of soft bedding; overheating (thermal stress); co-sleeping with adult, co-sleeping with adult smoker
Feeding habits	Lower incidence in breastfed infants, Lower incidence in infants put to sleep with pacifier
Maternal	Young age; cigarette smoking, especially during pregnancy; poor prenatal care; substance abuse (heroin, methadone, cocaine, passive smoke

Apparent Life-Threatening Event (ALTE)

ALTE formerly referred to as SIDS, generally refers to an event that infant exhibits a combination of apnea; change in color (pallor, cyanosis, redness); change in muscle tone (usually hypotonia); and choking, gagging, or coughing.

The definition of ALTE may include apnea, but ALTE may occur without apnea.

Infants with ALTE are at increased risk for SIDS (three to five times greater in infants who experienced an ALTE).

- Risk factors:
  - Apnea and bradycardia
  - siblings of SIDS, or with ALTEs
  - preterm infants weighing less than 1750 at birth
  - born at less than 37 weeks' gestation.
- Nursing Care Management

PS support: Parents of infants on home apnea monitors report experiencing emotional distress, especially depression and hostility.

In terms of education about the equipment; education regarding observation of the infant's status; and instructions regarding immediate intervention during apneic episodes, including CPR.

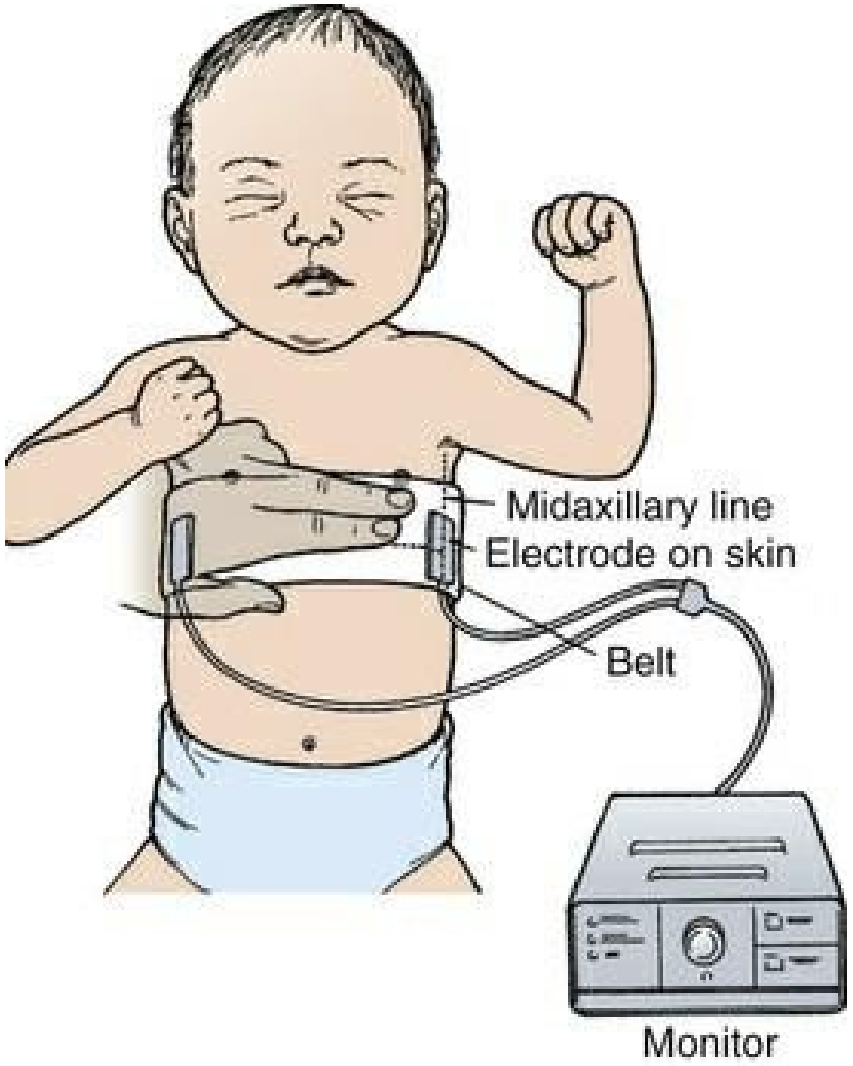
Monitors are effective to alert the caregiver to the ALTE in time to intervene.

Safety is a major concern because monitors can cause electrical burns and electrocution. The following precautions are recommended:

Remove leads from infant when not attached to the monitor.

Unplug the power cord from the electrical outlet when the cord is not plugged into the monitor.

Use safety covers on electrical outlets to discourage children from inserting objects into sockets.



Placement of electrodes or belt for apnea monitoring. In small infants, one fingerbreadth may be used.

The Child with Respiratory Dysfunction

Respiratory Infections

Infections of the respiratory tract are described according to the anatomic area of involvement. The upper respiratory tract, or upper airway, consists of the oro-nasopharynx, pharynx, larynx, and upper part of the trachea.

The lower respiratory tract consists of the lower trachea, bronchi, bronchioles, and alveoli. In this discussion, the trachea is considered with lower tract disorders, and infections of the epiglottis and larynx are categorized as croup syndromes. However, respiratory infections seldom fall into discrete anatomic areas.

Infections often spread from one structure to another because of the contiguous nature of the mucous membrane lining the entire tract. Consequently, respiratory tract infections involve several areas rather than a single structure, although the effect on one area may predominate in any given illness.

- Etiology and Characteristics

  Respiratory tract infections account for the majority of acute illnesses in children.

  The etiology and course of these infections are influenced by the age of the child, season, living conditions, and preexisting medical problems.

- Infectious Agents

  The respiratory tract is subject to a wide variety of infective organisms. Most infections are caused by viruses, particularly respiratory syncytial virus (RSV), rhinovirus, nonpolio enterovirus (coxsackievirus A and B), adenovirus, parainfluenza virus, influenza virus, and human metapneumovirus.

  Other agents involved in primary or secondary invasion include group A betahemolytic streptococci (GABHS), staphylococci, Haemophilus influenzae, Bordetella pertussis, Chlamydia trachomatis, Mycoplasma organisms, and pneumococci.

- Age

Healthy full-term infants younger than 3 months old are presumed to have a lower infection rate because of the protective function of maternal antibodies; however, infants may be susceptible to specific respiratory tract infections, namely pertussis, during this period.

The infection rate increases from 3 to 6 months old, which is the period between the disappearance of maternal antibodies and the infant's own antibody production.

The viral infection rate remains high during the toddler and preschool years.

By 5 years old, viral respiratory tract infections are less frequent, but the incidence of Mycoplasma pneumoniae and GABHS infections increases.

The amount of lymphoid tissue increases throughout middle childhood

Some viral or bacterial agents produce a mild illness in older children but severe lower respiratory tract illness or croup in infants. For example, pertussis causes a relatively harmless tracheobronchitis in childhood but is a serious disease in infancy.

- Size

  Anatomic differences influence the response to respiratory tract infections. The diameter of the airways is smaller in young children and subject to considerable narrowing from edematous mucous membranes and increased production of secretions.

- Resistance

  The ability to resist pathogens depends on several factors. Deficiencies of the immune system place the child at risk for infection. Other conditions that decrease resistance are malnutrition, anemia, and fatigue. Conditions that weaken defenses of the respiratory tract and predispose children to infection also include allergies (e.g., allergic rhinitis), preterm birth, bronchopulmonary dysplasia (BPD), asthma, history of RSV infection, cardiac anomalies that cause pulmonary congestion, and cystic fibrosis (CF).

- Seasonal Variations

  The most common respiratory pathogens appear in epidemics during the winter and spring months, but mycoplasmal infections occur more often in autumn and early winter. Whereas infection-related asthma occurs more frequently during cold weather, winter and early spring are typically RSV season.

Signs and Symptoms Associated with Respiratory Tract Infections in Infants and Small Children

- Fever
  May be absent in neonates (<28 days)
  Greatest at 6 months to 3 years reach 39.5°-40.5° C even with mild infections as first sign of infection
  May lead to listlessness and irritability, with altered activity pattern
  May precipitate febrile seizure
- Poor Feeding and Anorexia
  Common with most childhood illnesses
- Vomiting
  Common in small children with illness
  Is frequent cause of dehydration
- Diarrhea
  Usually mild, transient diarrhea but may become severe
  Frequent cause of dehydration
- Abdominal Pain
  Common complaint, Sometimes indistinguishable from pain of appendicitis
  May represent referred pain (e.g., chest pain associated with pneumonia)
  May be caused by mesenteric lymphadenitis
  May be linked to muscle spasms from vomiting, especially in nervous, tense child

- Nasal Blockage

  Small nasal passages of infants easily blocked by mucosal swelling and exudation

  Can interfere with respiration and feeding in infants

  May contribute to the development of otitis media (OM) and sinusitis
- Nasal Discharge

  Frequent occurrence: thin and watery (rhinorrhea) or thick and purulent

  Associated with itching

  May irritate upper lip and skin surrounding the nose
- Cough

  May be evident only during acute phase

  May persist several months after a disease
- Respiratory Sounds

  Sounds associated with respiratory disease: • Cough • Hoarseness • Grunting • Stridor • Wheezing

  Findings on auscultation: • Wheezing • Crackles • Absence of breath sounds (movement of air)
- Sore Throat

  Frequent complaint of older children

  Often accompanied by refusal to take oral fluids or solids

Upper Respiratory Tract Infections

- Acute Viral Nasopharyngitis

  Acute nasopharyngitis “common cold,” is caused by the rhinoviruses, RSV, adenoviruses, enteroviruses, influenza virus, and parainfluenza virus.

  Symptoms are more severe in infants and children than in adults. Symptoms typically last 4 to 14 days and symptoms peak on day 2 to 3 of illness.

  Fever is common in young children, and older children have low-grade fevers, which appear early in the course of the illness.

- Younger Children:

  Fever

  Poor feeding and decreased fluid intake

  Nasal mucus causing mouth breathing

  Irritability, restlessness

  Sneezing

  Vomiting or diarrhea

- Older Children

  Dryness and irritation of nose and throat

  Sneezing, chilling

  Cough (sometimes)

  Nasal discharge causing mouth breathing

  Muscle aches

Upper Respiratory Tract Infections

- Therapeutic Management

  Nasopharyngitis are managed at home. There is no specific treatment, and vaccines are not effective.

  Antipyretics may be indicated for fever and discomfort

  Fluids and rest are recommended.

  Cough suppressants containing dextromethorphan should be used with caution (cough is a protective way of clearing secretions) but may be prescribed every 6 to 8 hours for a dry, hacking cough, especially at night

  Pseudoephedrine and some antihistamines are not appropriate for the treatment of the common cold in infants and toddlers

  The American Academy of Pediatrics': cough and cold medications do not work for children younger than 4 years old and in some cases may pose a health risk

  Antihistamines are largely ineffective in treatment of nasopharyngitis because have a weak atropine-like effect that dries secretions, but they can cause drowsiness.

  There is no support for the usefulness of expectorants, and antibiotics are usually not indicated because most infections are viral.

- Nursing Care Management

  Most discomfort of nasopharyngitis is related to the nasal obstruction, especially in small infants.

  Elevating the head of the bed or crib mattress assists with drainage of secretions.

  Suctioning and vaporization may also provide relief.

  Saline nose drops and gentle suction with a bulb syringe before feeding and sleep time may be useful.

  Maintaining adequate fluid intake is essential.

  It is important to offer appropriate fluids to prevent dehydration.

  Because nasopharyngitis is spread from secretions, the best means for prevention is avoiding contact with affected persons.

  Family members with a cold should carefully dispose of tissues, not share towels, glasses, or eating utensils, cover the mouth and nose with tissues when coughing or sneezing, and wash hands thoroughly after nose blowing or sneezing.

  The most frequent carriers of infection are the human hands, which deposit viruses on doorknobs, faucets, and other everyday objects.

  Children should wash their hands thoroughly or use hand sanitizer and avoid touching their eyes, noses, and mouths.

Upper Respiratory Tract Infections



Pharyngitis associated with group A beta-hemolytic streptococci (GABHS) infection



Tonsillitis and pharyngitis

Acute Infectious Pharyngitis

Acute infectious pharyngitis can be caused by many bacteria or viruses.

GABHS infection is the most common causative organism for this infection.

Children with GABHS infection of the upper airway (strep throat) are at risk for rheumatic fever (RF), an inflammatory disease of the heart, joints, and CNS, and acute glomerulonephritis, an acute kidney infection.

Permanent damage can result especially RF. GABHS may also cause skin manifestations, including impetigo and pyoderma.

- Diagnostic Evaluation

Although 80% to 90% of all cases of acute pharyngitis are viral, a throat culture or rapid streptococcal antigen testing should be performed to rule out GABHS.

ASOT: antistreptolysin-O titer appear later than symptoms and are useful only for retrospective diagnosis.

A throat culture is recommended for negative test results.

- Clinical Manifestations

GABHS infection is generally a relatively brief illness that varies in severity from subclinical (no symptoms) to severe toxicity.

The onset is often abrupt and characterized by pharyngitis, headache, fever, and abdominal pain.

The tonsils and pharynx may be inflamed and covered with exudate, which usually appear by the second day of illness.

However, streptococcal infections should be suspected in children older than 2 years old

The tongue may appear edematous and red (strawberry tongue), and the child may have a fine sandpaper rash on the trunk, axillae, elbows, and groin seen in scarlet fever (caused by a strain of group A streptococcus).

The uvula is edematous and red.

Anterior cervical lymphadenopathy (in 30% to 50% of cases) usually occurs early, and the nodes are often tender.

Pain can be relatively mild to severe enough to make swallowing difficult.

Clinical manifestations usually subside in 3 to 5 days unless complicated by sinusitis or parapharyngeal, peritonsillar, or retropharyngeal abscess.

Nonsuppurative complications may appear after the onset of GABHS–AGN in about 10 days and RF in an average of 18 days. Streptococcal skin infections can occur in AGN.

Clinical Manifestations of Acute Nasopharyngitis and Pharyngitis

Nasopharyngitis

Younger Children

Fever

Irritability, restlessness

Poor feeding and decreased fluid intake

Sneezing

Nasal mucus (abundant) causing mouth breathing

Vomiting or diarrhea

Older Children

Dryness and irritation of nose and throat initially

Nasal discharge causing mouth breathing

Sneezing, chilling

Muscle aches

Cough (sometimes)

Physical Assessment Signs

Edema and vasodilation of mucosa

Pharyngitis

Younger Children

Fever

General malaise

Anorexia

Moderate sore throat

Headache

Older Children

Fever (may reach 40° C [104° F])

Headache

Anorexia

Dysphagia

Abdominal pain

Vomiting

Physical Assessment Signs

Younger Children

Mild to moderate hyperemia

Older Children

Mild to bright red, edematous pharynx

Hyperemia of tonsils and pharynx; may extend to soft palate and uvula

Often abundant follicular exudate that spreads and coalesces to form pseudomembrane on tonsils

Cervical glands enlarged and tender

- Therapeutic Management

If streptococcal sore throat infection is present, oral penicillin V or amoxicillin is prescribed for 10 days to control the acute local manifestations.

Penicillin does not prevent the development of AGN in susceptible children; however, it may prevent the spread of a nephrogenic strain of GABHS to others in the family.

Patients who have a history of RF or who remain symptomatic after a full course of antibiotics may require a follow-up throat swab.

Intramuscular (IM) benzathine penicillin G is an appropriate therapy, but it is painful and is not the first choice for children.

An oral macrolide (erythromycin, azithromycin, clarithromycin) is indicated for children who are allergic to penicillin.

Other antibiotics used to treat GABHS are oral cephalosporins, clindamycin, and amoxicillin with clavulanic acid

● Nursing Care Management

The nurse often obtains a throat swab for culture or rapid antigen testing and instructs the parents about administering oral antibiotics and analgesics as prescribed.

Cold or warm compresses to the neck may provide relief. Warm saline gargles may offer relief of throat discomfort.

Ibuprofen (for ages 6 months and older) and acetaminophen may be effective in decreasing throat pain; liquid preparations or chewable forms may be preferable because of the pain associated with swallowing.

Pain may interfere with oral intake, and children should not be forced to eat, but fluid intake is essential. Cool liquids, ice chips, or flavored ice pops may be tolerated better than solid foods or citrus juices.

If an injection of penicillin is required, it must be administered deep into a large muscle mass (e.g., vastus lateralis or ventrogluteal muscle).

To prevent pain, application of a topical anesthetic cream (4% lidocaine) or eutectic mixture of local anesthetics over the injection site before the injection is helpful

Children are considered infectious to others at the onset of symptoms and up to 24 hours after initiation of antibiotic therapy, but they should not return to school or daycare until they have been taking antibiotics for a full 24-hour period.

Nurses should remind the children to discard their toothbrushes and replace them with new ones after they have been taking antibiotics for 24 hours.

Orthodontic appliances should be washed and disinfected thoroughly because they may harbor the organisms.

Parents are cautioned to prevent other household members, especially if immunocompromised, from having close contact with the sick child and avoid sharing towels, drinking or eating items.

If the child continues to have a high fever that does not respond to antipyretics, has an extremely sore throat, refuses liquids, and appears toxic 24 to 48 hours after starting antibiotics, further evaluation by the practitioner is recommended

Tonsillitis

The tonsils are masses of lymphoid tissue located in the pharyngeal cavity.

They filter and protect the respiratory and alimentary tracts from invasion by pathogenic organisms and play a role in antibody formation.

Children generally have much larger tonsils than adolescents or adults. This difference is thought to be a protective mechanism because young children are especially susceptible to URIs.

- Pathophysiology

Several pairs of tonsils are part of a mass of lymphoid tissue encircling the nasal and oral pharynx, known as the Waldeyer tonsillar ring.

A surface of the palatine tonsils is usually visible during oral examination. The palatine tonsils are those removed during tonsillectomy.

The pharyngeal tonsils, also known as the adenoids, are located above the palatine tonsils on the posterior wall of the nasopharynx. Their proximity to the nares and eustachian tubes causes difficulties in instances of inflammation.

Tonsil

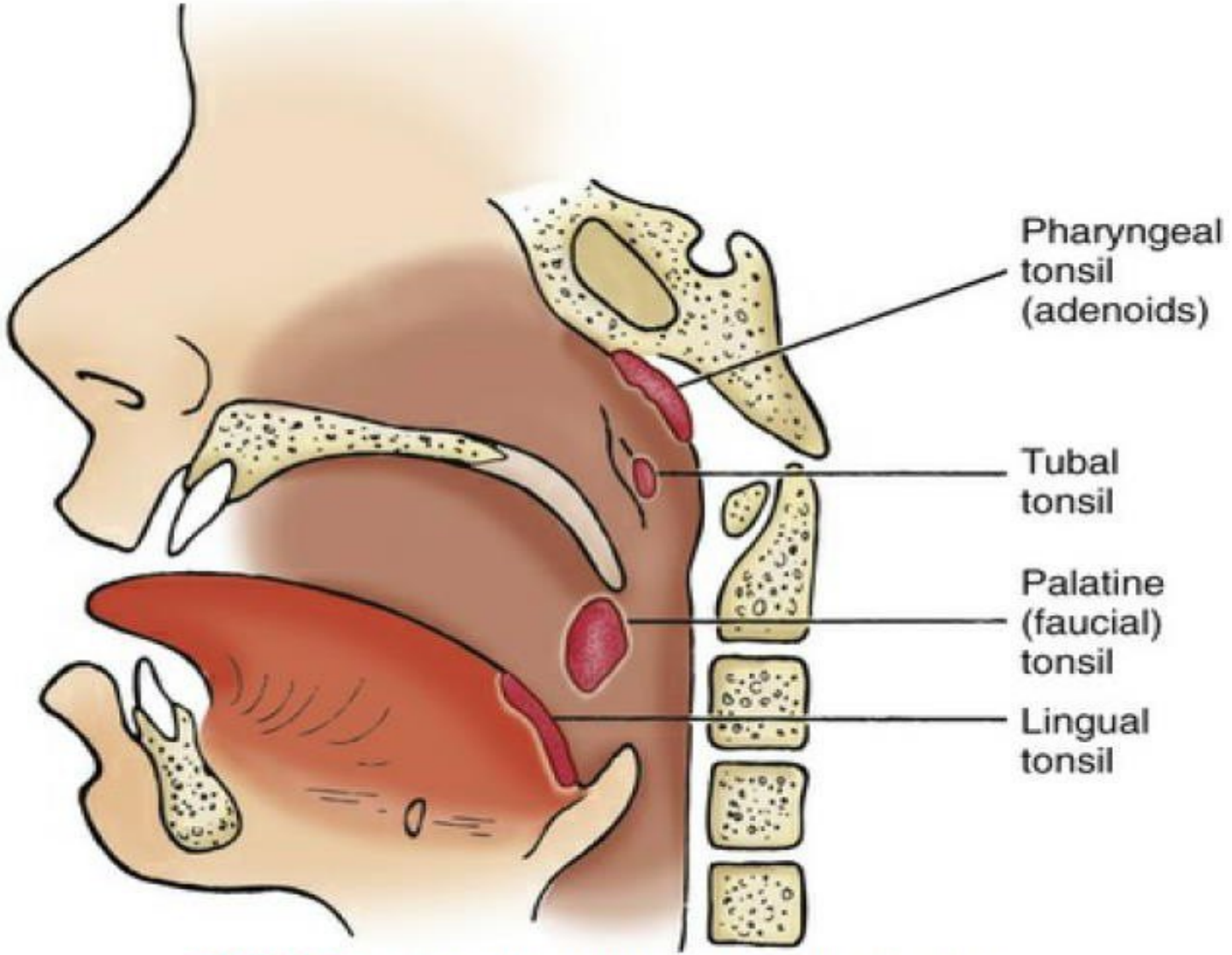


FIG 21-3 Location of various tonsillar masses.

- Etiology

  Because of the abundant lymphoid tissue and the frequency of URIs, tonsillitis is a common cause of illness in young children. The causative agent may be viral or bacterial.

- Clinical Manifestations

  Tonsils enlarge from edema (kissing tonsils), obstructing the passage of air or food.

  The child has difficulty swallowing and breathing.

  Enlarged adenoids occurs, making it difficult or impossible for air to pass from the nose to the throat. As a result, the child breathes through the mouth.

- Therapeutic Management

  Because tonsillitis is self-limiting, treatment of viral pharyngitis is symptomatic.

  Throat culture positive for GABHS infection need antibiotic treatment.

  Most infections are of viral origin, early rapid tests can eliminate unnecessary antibiotic administration.

  Tonsillectomy is the surgical removal of the palatine tonsils.

  Absolute indications for a tonsillectomy are recurrent throat infections (7 or more episodes in the preceding year, five or more episodes in each of the preceding 2 years, or three or more episodes in each of the preceding 3 years) and sleep-disordered breathing.

Adenoidectomy (the surgical removal of the adenoids) is recommended for children who have a history of four or greater episodes of recurrent purulent rhinorrhea in the previous 12 months in a child younger than 12 years old (one episode should be documented by intranasal examination or imaging)

- Other indications include

  persisting symptoms of adenoiditis after two courses of antibiotics

  sleep disturbance with nasal obstruction lasting over 3 months

  hyponasal speech,

  otitis media with effusion (OME) more than 3 months

  dental malocclusion or orofacial growth disturbance as validated by an orthodontist/dentist

  cardiopulmonary complications associated with adenoid hypertrophy

- Contraindications to either tonsillectomy or adenoidectomy are

1. cleft palate because the tonsils help minimize escape of air during speech,
2. acute infections at the time of surgery because locally inflamed tissues increase the risk of bleeding,
3. uncontrolled systemic diseases or blood dyscrasias, and (4) poor anesthetic risk.

- Nursing Care Management

Patients with sleep-disordered breathing require close monitoring of airway and breathing postoperatively.

A soft to liquid diet is preferred.

Warm saltwater gargles, warm fluids, and analgesic/antipyretic drugs (such as acetaminophen).

Often opioids are needed to reduce pain for the child to drink. Opioid medications such as oxycodone or hydrocodone (Lortab) relieve pain and should be given routinely and regularly as prescribed.

If surgery is required, the child requires the psychological and physical care as for surgical procedure

Routine suctioning is avoided, but when performed, it is done carefully to avoid trauma to the oropharynx. When alert, the child may prefer sitting up.

The child is discouraged from coughing frequently, clearing the throat, blowing the nose, and any other activity that may aggravate the operative site.

Some secretions are common, particularly dried blood from surgery.

Inspect all secretions and vomitus for evidence of fresh bleeding (some blood-tinged mucus is

The throat is sore after surgery. An ice collar may provide relief

Most children experience moderate pain after a tonsillectomy and adenoidectomy and need pain medication regularly for at least the first few days.

Analgesics may be given rectally or intravenously to avoid the oral route.

An antiemetic such as ondansetron (Zofran) or scopolamine transdermal patch (ages 12 and older) may be administered postoperatively if nausea or vomiting is present.

Food and fluids are restricted until the child is fully alert and there are no signs of hemorrhage.

Cool water, crushed ice, flavored ice pops, but fluids with a red or brown color may be avoided to distinguish fresh or old blood in emesis.

Citrus juice may cause discomfort and is usually poorly tolerated.

Soft foods, particularly gelatin, cooked fruits, soup, and mashed potatoes, are started on the first or second postoperative day.

Milk or ice-cream not given because milk products coat the mouth and throat and may cause the child to clear the throat, which can initiate bleeding.

The nurse observes the throat directly for evidence of bleeding; using a good source of light; and, if necessary, carefully inserting a tongue depressor.

Surgery may be required to ligate a bleeding vessel. Airway obstruction may also occur as a result of edema or accumulated secretions and is indicated by signs of respiratory distress, such as stridor, drooling, restlessness, agitation, increasing respiratory rate, and progressive cyanosis.

Suction equipment and oxygen should be available after tonsillectomy.

- Influenza

Influenza, or the “flu,” is classified into three groups of orthomyxoviruses, which are antigenically distinct: types A and B, which cause epidemic disease, and type C, which is antigenically stable and causes milder disease.

Influenza is spread from one individual to another by direct contact (largedroplet infection) or by articles recently contaminated by nasopharyngeal secretions.

Attack rates are highest in young children who have had no previous contact with a strain.

Influenza is frequently most severe in infants.

The disease is more common during the winter months and has a 1- to 3-day incubation period.

Affected persons are most infectious for 24 hours before and after the onset of symptoms.

The virus has a peculiar affinity for epithelial cells of the respiratory tract mucosa, where it destroys ciliated epithelium with metaplastic hyperplasia of the tracheal and bronchial epithelium with associated edema.

The alveoli may also become distended with a hyaline-like material.

The viruses can be isolated from nasopharyngeal secretions early after the onset of infection, and serologic tests identify the type by complement fixation or the subgroups by hemagglutination inhibition.

According to the more than 200,000 people in the United States are hospitalized each year for respiratory and heart conditions illnesses associated with seasonal influenza virus infections.

- Clinical Manifestations

  The manifestations of influenza may be subclinical, mild, moderate, or severe.

  Most patients have a dry throat and nasal mucosa, a dry cough, and a tendency toward hoarseness.

  A flushed face, photophobia, myalgia, hyperesthesia, and sometimes exhaustion and lack of energy accompany a sudden onset of fever and chills. Subglottal croup can occur, especially in infants.

  The symptoms of influenza last for 4 or 5 days. Complications include severe viral pneumonia (often hemorrhagic); encephalitis; and secondary bacterial infections such as otitis media, sinusitis, or pneumonia.

- Therapeutic Management

  Uncomplicated influenza in children usually requires only symptomatic treatment, including acetaminophen or ibuprofen for fever and sufficient fluids to maintain hydration.

  There are four influenza antiviral drugs but only oseltamivir (Tamiflu) and zanamivir (Relenza) are recommended because of widespread resistance to amantadine (Symmetrel) and rimantadine (Flumadine)

- Prevention

  The influenza vaccine is now recommended annually for children over 6 months old.

  Influenza vaccine (trivalent inactivated influenza vaccine [TIV]) may be given to healthy children 6 months old and older via IM injection.

Otitis Media

OM is the presence of fluid in the middle ear along with acute signs of illness and symptoms of middle ear inflammation.

OM is one of the most prevalent diseases of early childhood. Its incidence is highest in the winter months.

Many cases of bacterial OM are preceded by a viral respiratory infection.

The two viruses most likely to precipitate OM are RSV and influenza.

Most episodes of acute otitis media (AOM) occur in the first 24 months of life, but the incidence decreases with age except for a small increase at 5 or 6 years old when children enter school.

OM occurs infrequently in children older than 7 years old.

Preschool-age boys are affected more frequently than preschool-age girls.

Passive smoking increases the risk of persistent middle ear effusion by enhancing attachment of the pathogens that cause otitis to the respiratory epithelium in the middle ear space, by prolonging the inflammatory response, and by impeding drainage through the eustachian tube

- Standard Terminology for Otitis Media
- Otitis media (OM): An inflammation of the middle ear without reference to etiology or pathogenesis
- Acute otitis media (AOM): An inflammation of the middle ear space with a rapid onset of the signs and symptoms of acute infection—namely, fever and otalgia (ear pain)
- Otitis media with effusion (OME): Fluid in the middle ear space without symptoms of acute infection

- Etiology

  Streptococcus pneumoniae, H. influenzae, and Moraxella catarrhalis are three most common bacteria causing AOM.

  OM may occur because of blocked eustachian tubes, which results in negative ear pressure. Fluid is pulled from the mucosal lining, which accumulates and becomes colonized by infectious organisms.

  Predisposing factors include URIs, allergic rhinitis, Down syndrome, cleft palate, daycare attendance, exposure to secondhand smoke, and bottle propping during feeding.

  Infants fed breast milk have a lower incidence of OM than formula-fed infants because it may protect infants against respiratory viruses and allergy because breast milk contains secretory immunoglobulin A

  Reflux of milk up the eustachian tubes is less likely in breastfed infants because of the semivertical positioning during breastfeeding compared with bottle feeding.

- Diagnostic Evaluation

  Careful assessment of tympanic membrane mobility with a pneumatic otoscope to differentiate AOM from OME

  A diagnosis of AOM is made if visual inspection of the tympanic membrane reveals a purulent discolored effusion and a bulging or full, opacified, or reddened immobile membrane.

  Some practitioners also consider the presence of acute onset of less than 48 hours of ear pain as criteria in AOM.

  An immobile tympanic membrane or an orange, discolored membrane indicates OME.

  In AOM, symptoms such as acute onset of ear pain, fever, and a bulging yellow or red tympanic membrane are usually present.

- Clinical Manifestations of Otitis Media
- Acute Otitis Media
- Follows an upper respiratory tract infection
- Otalgia (earache)
- Fever—may or may not be present
- Purulent discharge (otorrhea)—may or may not be present
- Infants and Very Young Children
- Crying, fussiness, restlessness, irritability, especially on lying down
- Tendency to rub, hold, or pull affected ear
- Rolling head from side to side
- Difficulty comforting child
- Loss of appetite, refusal to feed
- Older Children
- Crying or verbalizing feelings of discomfort
- Irritability
- Lethargy
- Loss of appetite
- Chronic Otitis Media
- Hearing loss
- Difficulty communicating
- Feeling of fullness, tinnitus, or vertigo may be present

• Therapeutic Management

Treatment for AOM is one of the most common reasons for antibiotic use.

Recently, however, concerns about drug-resistant S. pneumoniae and other drug resistances have led infectious disease authorities to recommend careful use of antibiotics for the treatment of this illness.

Current literature indicates that waiting up to 72 hours for spontaneous resolution is safe and appropriate management of AOM without severe signs and symptoms in healthy infants older than 6 months of age.

Furthermore, some reviews of the treatment of AOM reveal no clear evidence that antibiotics improve outcomes in children younger than 2 years old with uncomplicated AOM.

In addition, all cases of AOM in infants younger than 6 months old should be treated with antibiotics because of their immature immune systems and the potential for infection with bacteria.

Oral amoxicillin in high doses (80 to 90 mg/kg/day divided twice daily) is the treatment of choice for initial episodes of AOM in children who have not received antibiotics within the past month

Second-line antibiotics used to treat OM include amoxicillin/clavulanate and cephalosporins(such as cefdinir, cefuroxime). IM ceftriaxone is used if the causative organism is a highly resistant pneumococcus

Supportive care of AOM includes treating the fever and pain. Topic pain relief is recommended by external application of heat or cold, or the practitioner may prescribe topical pain relief drops such

Myringotomy, a surgical incision of the eardrum, necessary to alleviate the severe pain of AOM or OME.

A myringotomy is also performed to drain infected middle ear fluid in the presence of complications (mastoiditis, labyrinthitis, or facial paralysis) or to allow purulent middle ear fluid to drain into the ear canal for culture.

A minimally invasive laser-assisted myringotomy procedure may be performed in outpatient settings.

Tympanostomy tube placement and adenoidectomy are surgical procedures that may be done to treat recurrent chronic OM

Placement of tympanostomy tubes is recommended after a total of 4 to 6 months of bilateral effusion with a bilateral hearing deficit

OME is frequently associated with mild to moderate impairment of hearing; therefore, a hearing test should also be performed if OME persists for 3 months or more or if there is evidence of language or learning delays.

- Prevention

Routine immunization with the pneumococcal conjugate vaccine PCV7 (Prevnar 7) has reduced the incidence of AOM in some infants and children. The vaccine is administered as a four-dose series beginning at 2 months old.

- Nursing Care Management

Nursing objectives for children with AOM include (1) relieving pain, (2) facilitating drainage (3) preventing complications or recurrence, (4) educating the family in care of the child, and (5) providing emotional support to the child and family.

Analgesic such as acetaminophen (all ages) and ibuprofen (6 months of age and older) are used to mild pain.

If the ear is draining, the external canal may be cleaned with sterile cotton swabs or pledgets coupled with topical antibiotic treatment.

Tympanostomy tubes may allow water to enter the middle ear, but recommendations for earplugs are inconsistent.

Parents also need anticipatory guidance regarding methods to reduce the risks of OM, especially in children younger than 2 years old. Reducing the chances of OM is possible with measures such as sitting or holding an infant upright for feedings, maintaining routine childhood immunizations, and exclusively breastfeeding until at least 6 months old.

Early detection of middle ear effusion is essential to prevent complications. Infants and preschool children should be screened for effusion, and all schoolchildren, especially those with learning disabilities, should be tested for hearing deficits related to a middle ear effusion.

Infectious Mononucleosis

Infectious mononucleosis is an acute, self-limiting infectious disease that is common among young people under 25 years old.

Symptoms include fever, exudative pharyngitis, lymphadenopathy, hepatosplenomegaly, and an increase in atypical lymphocytes.

- Etiology and Pathophysiology

The herpes-like Epstein-Barr virus (EBV) is the cause of infectious mononucleosis.

It appears in both sporadic and epidemic forms, but the sporadic cases are more common.

The virus is believed to be transmitted in saliva by direct intimate contact, blood transfusion, or transplantation.

The incubation period after exposure is approximately 30 to 50 days

- Diagnostic Tests

The onset of symptoms may be acute or insidious and may appear anywhere from 10 days to 6 weeks after exposure. The presenting symptoms vary greatly in type, severity, and duration the convalescent phase is shorter in younger children than in older children and young adults.

The monospot test, a form of the heterophile antibody test, is a rapid test for infectious mononucleosis due to Epstein–Barr virus (EBV). It is an improvement on the Paul–Bunnell test.

Clinical Manifestations of Infectious Mononucleosis

Early Signs

Headache

Epistaxis

Malaise

Fatigue

Chills

Low-grade fever

Loss of appetite

Puffy eyes

Acute Disease

Cardinal Features

Fever

Sore throat

Cervical adenopathy

Common Features

Splenomegaly (may persist for several months)

Palatine petechiae

Macular eruption (especially on trunk)

Exudative pharyngitis or tonsillitis

Hepatic involvement to some degree, often associated with jaundice

Therapeutic Management

No specific treatment exists for infectious mononucleosis.

A mild analgesic is often sufficient to relieve the headache, fever, and malaise.

Rest is encouraged for fatigue.

Contact sports are discouraged in the presence of splenomegaly.

Antibiotics are contraindicated unless beta-hemolytic streptococci are present (amoxicillin or ampicillin can cause a rash in patients with EBV infection).

If sore throat is severe, effective therapies include gargles, warm drinks, anesthetic troches, or analgesics, including opioids.

Corticosteroids have been used to treat respiratory distress from significant tonsillar inflammation, myocarditis, hemolytic anemia, thrombocytopenia, and neurologic complications; however, routine use of steroids is not recommended

- Prognosis

The course of this disease is usually self-limiting and uncomplicated.

Acute symptoms often disappear within 7 to 10 days, and persistent fatigue subsides within 2 to 4 weeks. Some adolescents may need to restrict their activities for 2 to 3 months, but the disease rarely extends for longer periods.

The child is encouraged to maintain limited exercise to prevent deconditioning.

- Croup Syndromes

Croup is a general term applied to a symptom complex characterized by hoarseness, a barking" or "brassy cough " (croupy), inspiratory stridor, and varying degrees of respiratory distress resulting from swelling or obstruction in the region of the larynx and subglottic airway.

Acute infections of the larynx are important in infants and small children because of their increased incidence in these age groups and because the small diameter of the airway in infants and children places them at risk for significant narrowing with inflammation.

Croup syndromes can affect the larynx, trachea, and bronchi. However, laryngeal involvement often dominates the clinical picture because of the severe effects on the voice and breathing.

Croup syndromes are described according to the primary anatomic area affected (i.e., epiglottitis [or supraglottitis], laryngitis, laryngotracheobronchitis [LTB], and tracheitis). In general, LTB occurs in very young children, and epiglottitis is more common in older children.

Comparison of Croup Syndromes

	Acute Epiglottitis	Acute Laryngotracheobronchitis	Acute Spasmodic Laryngitis	Acute Tracheitis
Age group affected	2 to 5 years old but varies	Infant or child younger than 5 years old	1 to 3 years old	1 month old to 6 years old
Etiologic agent	Bacterial	Viral	Viral with allergic component	Viral or bacterial with allergic component
Onset	Rapidly progressive	Slowly progressive	Sudden; at night	Moderately progressive
Major symptoms	Dysphagia Stridor aggravated when supine Drooling High fever Toxic appearance Rapid pulse and respirations	URI Stridor Brassy cough Hoarseness Dyspnea Restlessness Irritability Low-grade fever Nontoxic appearance	URI Croupy cough Stridor Hoarseness Dyspnea Restlessness Symptoms awakening child but disappearing during day Tendency to recur	URI Croupy cough Purulent secretions High fever No response to LTB therapy
Treatment	Airway protection Corticosteroids Fluids Antibiotics Reassurance	Humidified oxygen if needed Corticosteroids Fluids Reassurance	Cool mist Reassurance	Antibiotics Fluids

LTB

Acute Epiglottitis

Acute epiglottitis, or acute supraglottitis, is a medical emergency.
It is a serious obstructive inflammatory process that occurs predominantly in children 2 to 5 years old but can occur from infancy to adulthood.
The obstruction is supraglottic as opposed to the subglottic obstruction of laryngitis. The responsible organism is usually H. influenzae. LTB and epiglottitis do not occur together.

- Clinical Manifestations

The onset of epiglottitis is abrupt, and it can rapidly progress to severe respiratory distress. The child usually goes to bed asymptomatic to awaken later, complaining of sore throat and pain on swallowing.

The child has a fever; appears sicker than clinical findings suggest; and insists on sitting upright and leaning forward (tripod position) with the chin thrust out, mouth open, and tongue protruding. Drooling of saliva is common because of the difficulty or pain on swallowing and excessive secretions.

The child is irritable, extremely restless, and has an anxious, apprehensive, and frightened expression. The voice is thick and muffled, with a froglike croaking sound on inspiration, but the child is not hoarse. Suprasternal and substernal retractions may be evident.

Therapeutic Management

Progressive obstruction leads to hypoxia, hypercapnia, and acidosis followed by decreased muscle tone;
reduced level of consciousness; and, when obstruction becomes more or less complete, a rather sudden death.

The child who is suspected of having epiglottitis should be examined in a setting where emergency airway equipment is readily available.

Examination of the throat with a tongue depressor is contraindicated until experienced personnel and equipment are available to proceed with immediate intubation or tracheostomy in the event that the examination precipitates further or complete obstruction lateral neck radiograph of the soft tissues is indicated for diagnosis.

Nasotracheal intubation or on occasion, tracheostomy, is considered for the child with epiglottitis with severe respiratory distress

Humidified oxygen is administered as necessary either via mask in older children or blow-by in younger children to avoid further agitation

The epiglottal swelling usually decreases after 24 hours of antibiotic therapy (ceftriaxone sodium), and the epiglottis is near normal by the third day. Intubated children are generally extubated at this time.

The use of corticosteroids for reducing edema may be beneficial during the early treatment phase.

Continuous monitoring of respiratory status, including pulse oximetry

Infections of the Lower Airways

The reactive portion of the lower respiratory tract includes the bronchi and bronchioles in children.

The smooth muscle in these structures represents a major factor in the constriction of the airway, particularly in the bronchioles, the portion that extends from the bronchi to the alveoli.

Comparison of Conditions Affecting the Bronchi

	Asthma*	Bronchitis	Bronchiolitis
Description	Exaggerated response of bronchi to a trigger such as URI, animal dander, cold air, exercise Bronchospasm, exudation, and edema of bronchi Inflammatory response	Usually occurs in association with URI Seldom an isolated entity	Most common infectious disease of lower airways. Maximum obstructive impact at bronchiolar level
Age group affected	Infancy to adolescence	First 4 years of life	Usually children 2 to 12 months old; rare after 2 years old Peak incidence, approximately 6 months old
Etiologic agents	Most often viruses such as RSV in infants but may be any of a variety of URI pathogens	Usually viral Other agents (e.g., bacteria, fungi, allergic disorders, airborne irritants) can trigger symptoms	Viruses, predominantly RSV; also adenoviruses, parainfluenza viruses, human metapneumovirus, and *Mycoplasma pneumoniae*
Predominant characteristics	Wheezing, cough, labored respirations	Persistent dry, hacking cough (worse at night) becoming productive in 2 to 3 days	Labored respirations, poor feeding, cough, tachypnea, retractions and flaring nares, emphysema, increased nasal mucus, wheezing, may have fever
Treatment	Inhaled corticosteroids, bronchodilators, leukotriene modifiers, allergen and "triggers" control, long-term antiinflammatory medications	Cough suppressants if needed	Supplemental oxygen if saturations ≤90%; bronchodilators (optional) Suctioning nasopharynx Ensure adequate fluid intake Maintain adequate oxygenation

Bronchitis

Bronchitis (sometimes referred to as tracheobronchitis) is inflammation of the large airways (trachea and bronchi), which is frequently associated with URIs. Viral agents are the primary cause of the disease, although M. pneumoniae is a common cause in children older than 6 years of age.

A dry, hacking, nonproductive cough that worsens at night and becomes productive in 2 or 3 days characterizes this condition.

Bronchitis is a mild, self-limiting disease that requires only symptomatic treatment, including analgesics, antipyretics, and humidity.

Cough suppressants may be useful to allow rest but can interfere with clearance of secretions.

Most patients recover uneventfully in 5 to 10 days. It can be associated with other underlying conditions (such as CF and bronchiectasis) and can become chronic in nature (cough >3 months).

Adolescents with chronic bronchitis (>3 months) should be screened for tobacco or marijuana use.

Respiratory Syncytial Virus and Bronchiolitis

Bronchiolitis is a common, acute viral infection with upper respiratory symptoms and lower respiratory infection of the bronchioles due to inflammation. The infection occurs primarily in winter and early spring.

RSV infection is the most frequent cause of hospitalization in children younger than 1 year old. .

Severe RSV infections in the first year of life represent a significant risk factor for the development of asthma up to 13 years old.

RSV infection may also occur in children older than 1 year of age who have a chronic or serious disabling illness.

Althoughmost cases of bronchiolitis are caused by RSV, adenoviruses and parainfluenza viruses are also

implicated; human metapneumovirus has also been associated with bronchiolitis in children.

It can also rarely be caused by M. pneumoniae.

RSV is transmitted from exposure to contaminated secretions. RSV can live on fomites for several hours and on hands for 30 minutes

- ## Pathophysiology

RSV affects the epithelial cells of the respiratory tract. The ciliated cells swell, protrude into the lumen, and lose their cilia. The walls of the bronchi and bronchioles are infiltrated with inflammatory cells, and varying degrees of intraluminal obstruction lead to hyperinflation, obstructive emphysema resulting from partial obstruction, and patchy areas of atelectasis.

Dilation of bronchial passages on inspiration allows sufficient space for intake of air, but narrowing of the passages on expiration prevents air from leaving the lungs. Thus, air is trapped distal to the obstruction and

Signs and Symptoms of Respiratory Syncytial Virus Infection

Initial

Rhinorrhea

Pharyngitis

Coughing, sneezing

Wheezing

Possible ear or eye drainage

Intermittent fever

With Progression of Illness

Increased coughing and wheezing

Tachypnea and retractions

Cyanosis

Severe Illness

Tachypnea, >70 breaths/min

Listlessness

Apneic spells

Poor air exchange; poor breath sounds

- ## Diagnostic Evaluation

  Identification has been simplified by the development of tests done on nasopharyngeal secretions, using either a rapid immunofluorescent antibody/direct fluorescent antibody (DFA) staining or an enzyme-linked immunosorbent assay (ELISA) for RSV antigen detection .

  Hyperinflation of the lungs is generally seen on the chest radiograph.

- ## Therapeutic Management

  Children with bronchiolitis are cared for home if they are maintaining hydration, do not have respiratory distress, and do not need oxygen therapy. Hospitalization is recommended for children with respiratory distress or those who cannot maintain adequate hydration. Other reasons for hospitalization include complicating conditions, such as underlying lung or heart disease,

  In-patients are treated symptomatically with humidified oxygen, adequate fluid intake, airway maintenance, and medications.

  In general, the illness peaks in 5 to 7 days but the cough can persist for 2 to 3 weeks.

  Children with thickened secretions may benefit from extra humidity blended with oxygen administration and continuous positive airway pressure (CPAP) via a high flow nasal cannula (HFNC)

  Routine chest percussion and postural drainage

  Fluids by mouth may be contraindicated because of tachypnea, weakness, and fatigue; therefore, IV fluids may be used until the acute stage of the disease has passed. Nasogastric (NG) fluids may be required if the infant is unable to tolerate oral fluids and a peripheral IV is difficult to establish

  no evidence to support the use of antibiotics for bronchiolitis.

  Ribavirin, an antiviral agent, is the only specific therapy approvedfor hospitalized children



FIG 21-4 High flow nasal cannula (HFNC).

Pneumonia

Pneumonia, inflammation of the pulmonary parenchyma, is common in childhood but occurs more frequently in early childhood. Clinically, pneumonia may occur either as a primary disease or as a complication of another illness. The causative agent is either inhaled into the lungs directly or comes from the bloodstream.

The most useful classification of pneumonia is based on the etiologic agent (e.g., viral, bacterial, mycoplasmal, or aspiration of foreign substances) . Viral infections of the respiratory tract render the affected child more susceptible to secondary bacterial invasion, especially when there is denuded bronchial mucosa.

Many organisms can cause pneumonia, and these vary according to the child's age

Neonates: Group B streptococci, gram-negative enteric bacteria, cytomegalovirus, Ureaplasma urealyticum, Listeria monocytogenes, C. trachomatis

Infants: RSV, parainfluenza virus, influenza virus, adenovirus, metapneumovirus, S. pneumoniae, H. influenzae, M. pneumoniae, Mycobacterium tuberculosis

Preschool children: RSV, parainfluenza virus, influenza virus, adenovirus, metapneumovirus, S. pneumoniae, H. influenzae, M. pneumoniae, M. tuberculosis

School-age children: M. pneumoniae, Chlamydia pneumoniae, M. tuberculosis, and respiratory viruses

- Viral Pneumonia

Viral pneumonias, which occur more frequently than bacterial pneumonias, are seen in children of all ages and are often associated with viral URIs. Viruses that cause pneumonia include RSV in infants and parainfluenza, influenza, human metapneumovirus, enterovirus, and adenovirus in older children.

Differentiation among viruses is usually made by clinical features, such as child's age, medical history, season of the year, and radiographic and laboratory examination

Viral infections of the respiratory tract render the affected child more susceptible to secondary bacterial invasion, especially when there is denuded bronchial mucosa. Treatment is symptomatic and includes measures to promote oxygenation and comfort, such as oxygen administration with cool mist, postural drainage, antipyretics for fever management, monitoring fluid intake, and family support.

Antimicrobial therapy is usually reserved for children in whom a bacterial infection is demonstrated by appropriate cultures.

- General Signs of Pneumonia

  Fever: Usually high

  Respiratory

  - Cough: Unproductive to productive with whitish sputum
  - Tachypnea
  - Breath sounds: Crackles, decreased breath sounds
  - Dullness with percussion
  - Chest pain
  - Retractions
  - Nasal flaring
  - Pallor to cyanosis (depends on severity)

  Chest radiography: Diffuse or patchy infiltration with peribronchial distribution

  Behavior: Irritability, restlessness, malaise, lethargy

  Gastrointestinal: Anorexia, vomiting, diarrhea, abdominal pain

Primary Atypical Pneumonia

Atypical pneumonia refers to pneumonia that is caused by pathogens other than the traditionally most common and readily cultured bacteria (e.g., S. pneumoniae).

In the category of atypical pneumonias, M. pneumoniae is the most common cause of community-acquired pneumonia in children 5 to 15 years old.

It occurs primarily in the fall and winter months and is more prevalent in crowded living conditions. Most affected persons recover from acute illness at home in 7 to 10 days with symptomatic treatment followed by 1 week of convalescence.

The incubation period is 2 to 3 weeks, but the cough may last several weeks.

Chlamydial pneumonia, caused by C. trachomatis, can occur in infants and generally appears between 2 and 19 weeks after delivery. The infant contracts this from the infected genital tract of the mother at birth. Chlamydial pneumonia is characterized by a persistent cough, tachypnea, and sometimes rales.

Oral azithromycin is the treatment of choice; alternatively, erythromycin or ethylsuccinate can be given

- Bacterial Pneumonia

S. pneumoniae is the most common bacterial pathogen responsible for community-acquired pneumonia in both children and adults.

Other bacteria that cause pneumonia in children are group A streptococcus, S. aureus, M. catarrhalis, M. pneumonia, and C. pneumoniae.

Symptoms include fever, malaise, rapid and shallow respirations, cough, and chest pain. The associated cough may persist for several weeks or months. The pain of pneumonia may be referred to the abdomen in young children. Chills and meningeal symptoms (meningism) without meningitis are common.

Most older children with pneumonia can be treated at home if the condition is recognized and treatment is initiated early.

Antibiotic therapy, rest, liberal oral intake of fluid, and administration of an antipyretic for fever are the principal therapeutic measures. Chest percussion and postural drainage may be indicated, but there is a lack of evidence to show that they have benefit to children with pneumonia.

- Nursing Care Management
- Nursing care of the child with pneumonia is primarily supportive and symptomatic but necessitates thorough respiratory assessment and administration of supplemental oxygen (as required), fluids, and antibiotics.
- The child's respiratory rate, rhythm and depth, oxygenation, general disposition, and level of activity are frequently assessed.
- To prevent dehydration, fluids may be needed intravenously during the acute phase.
- Nursing care of the child with a chest tube and drainage device used are monitored for proper function (i.e., drainage is not impeded, vacuum setting is correct, tubing is free of kinks, dressing covering chest tube insertion site is intact, water seal is maintained [if used], and chest tube remains in place).
- Movement in bed and ambulation with a chest tube are encouraged according to the child's respiratory status, but children require frequent doses of analgesia.
- Supplemental oxygen may be required in the acute phase of the illness and may be administered by nasal cannula, face mask, blow-by, or face tent.
- Children are usually more comfortable in a semierect position but should be allowed to determine the position of comfort.
- Lying on the affected side if the pneumonia is unilateral ("good lung up") splints the chest on that side and reduces the pleural rubbing that often causes discomfort.
- Fever is controlled by cooling the environment and administrating antipyretic drugs.
- Chest percussion, postural drainage, and nebulized bronchodilator treatments may be prescribed depending on the child's condition.



Child placed in semierect position is often more comfortable, and this position enhances diaphragmatic expansion.

Other Infections of the Respiratory Tract
Pertussis (Whooping Cough)

Pertussis, or whooping cough, is an acute respiratory tract infection caused by B. pertussis, which in the past primarily occurred in children younger than 4 years old who were not immunized. It is highly contagious and is particularly threatening in young infants, who have a higher morbidity and mortality rate.

Complications in adolescents can include syncope, rib fractures, and pneumonia; whereas in younger children, seizures, pneumonia, intracranial bleeding, conjunctival bleeding, and death can occur Infants younger than 6 months old may not come in to the practitioner with the typical cough; in this age group, apnea is a common presenting manifestation.

Likewise, older children are known to manifest the disease with a persistent cough.

Most children with pertussis can be managed at home; care is supportive in nature, including encouraging adequate hydration and administering antipyretics Admission to the hospital occurs if respiratory symptoms are severe or if apnea occurs.

Diagnosis is obtained via culture or B. pertussis polymerase chain reaction (PCR) test Patients are considered infectious until at least 5 days of antibiotics have been completed or for 3 weeks if no antibiotics have been administered.

Tuberculosis

Tuberculosis (TB) along with human immunodeficiency virus is the leading cause of death from a single infectious disease.

Children with human immunodeficiency virus (HIV) infection have an increased incidence of TB disease, and all children with TB should be tested for HIV.

TB is caused by M. tuberculosis, an acid-fast bacillus. Children are susceptible to the human (M. tuberculosis) and the bovine(Mycobacterium bovis) organisms.

In parts of the world where TB in cattle is not controlled or milk is not pasteurized, the bovine type is a common source of infection from the milk or is spread via airborne transmission.

The source of TB infection in children is usually an infected member of the household or a frequent visitor to the home, such as a babysitter or domestic worker. The airway is the usual portal of entry for the organism

Diagnostic Evaluation

Diagnosis is based on information derived from physical examination, history, tuberculin skin testing, radiographic examinations, and cultures of the organism.

The clinical manifestations of the disease are extremely variable

Clinical Manifestations of Tuberculosis

- May be asymptomatic or produce a broad range of symptoms:
  Fever
  Malaise
  Anorexia
  Weight loss
  Cough (may or may not be present; progresses slowly over weeks to months)
  Aching pain and tightness in the chest
  Hemoptysis (rare)
- With progression:
  Increasing respiratory rate
  Poor expansion of lung on the affected side
  Diminished breath sounds and crackles
  Dullness to percussion
  Persistent fever
  Generalized symptoms
  Pallor, anemia, weakness, and weight loss

The tuberculin skin test (TST) is the most important indicator of whether a child has been infected with the tubercle bacillus.

- Tuberculin Skin Test Recommendations for Infants, Children, and Adolescents:
- Children for Whom Immediate Tuberculin Skin Test Is Indicated

  Contacts of persons with confirmed or suspected contagious tuberculosis (TB; contact investigation)

  Children with radiographic or clinical findings suggesting TB disease

  Children immigrating from endemic countries (e.g., Asia, Middle East, Africa, Latin America)

  Children with travel histories to endemic countries with indigenous persons from such countries
- Children Who Should Have Annual Tuberculin Skin Test

  Children infected with human immunodeficiency virus (HIV)

  Incarcerated adolescents

- The tuberculin skin test (TST)

1-ml syringe intradermally into the volar aspect of the forearm.

The tuberculin is injected intradermally with the bevel of the needle pointing upward. A wheal 6 to 10 mm in diameter should form between the layers of the skin when the solution is injected properly.

If the wheal is not formed, the procedure is repeated.

The reaction to the skin test is determined in 48 to 72 hours by a health care professional. Reactions occurring after 72 hours should be measured and considered the result.

The size of the transverse diameter of induration, not the erythema, is measured.

A positive reaction indicates that the individual has been infected and has developed sensitivity to the protein of the tubercle bacillus.

The test is usually positive 2 to 10 weeks after initial infection with the organism.

Any negative reaction does not exclude active disease because false negatives can immunosuppression or certain medications.

Induration ≥5 mm



Induration and erythema resulting from a positive tuberculosis (TB) skin test.

The term latent tuberculosis infection (LTBI) is used to indicate infection in a person who has a positive TST, no physical findings of disease, and normal chest radiograph findings

The term TB disease or clinically active TB is used when a child has clinical symptoms or radiographic manifestations caused by the M. tuberculosis organism.

A diagnosis of TB disease represents recent transmission of the M. tuberculosis organism and is a sentinel event for public health.

Prompt evaluation, treatment, and identification and treatment of contacts are key components to managing TB.

Sputum specimens are difficult or impossible to obtain from infants and young children, because they swallow any mucus coughed from the lower respiratory tract.

- Therapeutic Management

Medical management of TB disease in children consists of adequate nutrition, pharmacotherapy, general supportive measures, and sometimes surgical procedures.

Family members and other contacts should also be assessed for symptoms by public health and treated accordingly.

Ethambutol, isoniazid, pyrazinamide (PZA), and rifampin are common medications used to treat TB in children.

Prevention

The only definite means to prevent TB is to avoid contact with the tubercle bacillus. Maintaining an optimal state of health with adequate nutrition and avoiding debilitating infections promote natural resistance but do not prevent infection.

Pasteurization and routine testing of milk and elimination of diseased cattle have reduced the incidence of bovine TB.

Limited immunity can be produced by administration of bacillus Calmette-Guérin (BCG), a live vaccine containing bovine bacilli with reduced virulence (attenuated). In most instances, positive tuberculin reactions develop after inoculation with BCG

Pulmonary Dysfunction Caused by Noninfectious Irritants

Foreign Body Aspiration

Small children characteristically explore matter with their mouths and are prone to aspirate foreign bodies (FBs). Small children also place objects such as beads, toys, paper clips, small magnets, or food items in the nose, which can easily be aspirated into the trachea.

FB aspiration can occur at any age but is most common in children 1 to 3 years old. Severity is determined by the location, type of object aspirated, and extent of obstruction.

For example, dry vegetable matter, such as a seed, nut, or piece of carrot or popcorn, that does not dissolve and that may swell when wet creates a particularly difficult problem.

The high fat content of potato chips and peanuts may cause the added risk of lipoid pneumonia. “Fun foods” such as hard candy and hot dogs are the worst offenders in terms of potential for choking.

- Diagnostic Evaluation

  The diagnosis of FB aspiration is suspected on the basis of the history and physical signs. Initially, an FB in the air passages can cause choking, gagging, wheezing, or coughing.

  Laryngotracheal obstruction most commonly causes dyspnea, cough, stridor, and hoarseness because of decreased air entry. Up to half of all children with FB ingestion may be asymptomatic. Cyanosis may occur if the obstruction becomes worse.

- Therapeutic Management

  FB aspiration may result in life-threatening airway obstruction, especially in infants because of the small diameters of their airways. Current recommendations for the emergency treatment of the choking child include the use of abdominal thrusts for children older than 1 year of age and back blows and chest thrusts for children younger than 1 year old.

  An FB is rarely coughed up spontaneously. Most frequently, it must be removed instrumentally by bronchoscopy.

Long-Term Respiratory Dysfunction
Asthma

Asthma is a chronic inflammatory disorder of the airways characterized by recurring symptoms, airway obstruction, bronchial hyperresponsiveness, and an underlying inflammation process.

In susceptible children, inflammation causes recurrent episodes of wheezing, breathlessness, chest tightness, and cough, especially at night or in the early morning. The airflow limitation or obstruction is reversible either spontaneously or with treatment.

Inflammation causes an increase in bronchial hyperresponsiveness to a variety of stimuli.

Recognition of the key role of inflammation has made the use of antiinflammatory agents, especially inhaled steroids, a major component in the treatment of asthma.

Asthma is classified into four categories based on the symptom indicators of disease severity. These categories are intermittent, mild, moderate, and severe.

These categories provide a stepwise approach to the pharmacologic management, environmental control, and educational interventions needed for each category These categories emphasize the multifaceted aspect of the disease for consideration of effects on present quality of life and functional capacity and the future risk of adverse events.

- Asthma Severity Classification in Children
- Step 5 or 6: Severe Asthma

  Continual symptoms throughout the day

  Frequent nighttime symptoms (>1 time/week ages 0 to 4 and 7 nights/week, ages 5 and older)

  Pulmonary expiratory flow (PEF): <60%

  Forced expiratory volume in 1 second (FEV1): <75% of predicted value

  Interference with normal activity: Extremely limited

  Use of short-acting β-agonist for symptom control: Several times a day
- Step 3 or 4: Moderate Asthma

  Daily symptoms

  Nighttime symptoms: Three to four times a month (0 to 4 years old), >1/week but not nightly (5 to 11 years old)

  PEF: 60% to 80% of predicted value (ages 5 and older)

  FEV1: 75% to 80% (ages 5 and older)

  PEF variability: >30%

  Interference with normal activity: Some limitation

  Use of short-acting β-agonist for symptom control: Daily

- Step 2: Mild Asthma

  Symptoms >2 times/week but <1 time/day

  Nighttime symptoms: One to two times a month (0 to 4 years old), three or four times a month (5 to 11 years old)

  PEF or FEV1: ≥80% of predicted value

  PEF variability: 20% to 30%

  Interference with normal activity: Minor limitation

  Use of short-acting β-agonist for symptom control: >2 days/wk but not daily
- Step 1: Intermittent Asthma

  Symptoms ≤2 days/wk

  Nighttime symptoms (awakenings): None (0 to 4 years old); ≤2 nights per month (5 to 11 years old)

  PEF or FEV1: ≥80% of predicted value

  PEF variability: <20%

  Interference with normal activity: None

  Use of short-acting β-agonist for symptom control: <2 days/wk

Asthma is the most common chronic disease of childhood, the primary cause of school absences, and the third leading cause of hospitalizations in children younger than 15 years old

Although the onset of asthma may occur at any age, 80% to 90% of children have their first symptoms before 4 or 5 years old.

Boys are affected more frequently than girls until adolescence

Morbidity and mortality increases may result from worsening air pollution, more premature infants with chronic lung disease, poor access to medical care, under diagnosis, and under treatment.

- Etiology

In fact, atopy, or the genetic predisposition for the development of an immunoglobulin E (IgE)–mediated response to common aeroallergens, is the strongest identifiable predisposing factor for developing asthma

Evidence shows that viral respiratory infections, including RSV infection, may also have a significant role in the development and expression of asthma

- Triggers Tending to Precipitate or Aggravate Asthma Exacerbations

  Allergens

  Outdoor: Trees, shrubs, weeds, grasses, molds, pollens, air pollution, spores

  Indoor: Dust or dust mites, mold, cockroach antigen

  Irritants: Tobacco smoke, wood smoke, odors, sprays

  Exposure to occupational chemicals

  Exercise

  Cold air

  Changes in weather or temperature

  Environmental change: Moving to new home, starting new school, and so on

  Colds and infections

  Animals: Cats, dogs, rodents, horses

  Medications: Aspirin, NSAIDs, antibiotics, beta-blockers

  Strong emotions: Fear, anger, laughing, crying

  Conditions: Gastroesophageal reflux, tracheoesophageal fistula

  Food additives: Sulfite preservatives

  Foods: Nuts, milk or other dairy products

  Endocrine factors: Menses, pregnancy, thyroid disease

  NSAID, Nonsteroidal antiinflammatory drug.

Pathophysiology

It appears that asthma results from complex interactions among inflammatory cells, mediators, and the cells and tissues present in the airways

The mechanisms responsible for the obstructive symptoms in asthma include:

1. inflammatory response to stimuli;
2. airway edema and accumulation and secretion of mucus;
3. spasm of the smooth muscle of the bronchi and bronchioles, which decreases the caliber of the bronchioles;
4. airway remodeling, which causes permanent cellular changes

Airflow is determined by the size of the airway lumen, degree of bronchial wall edema, mucus production, smooth muscle contraction, and muscle hypertrophy.

As the severity of obstruction increases, there is a reduced alveolar ventilation with carbon dioxide retention; hypoxemia; respiratory acidosis; and, eventually, respiratory failure.

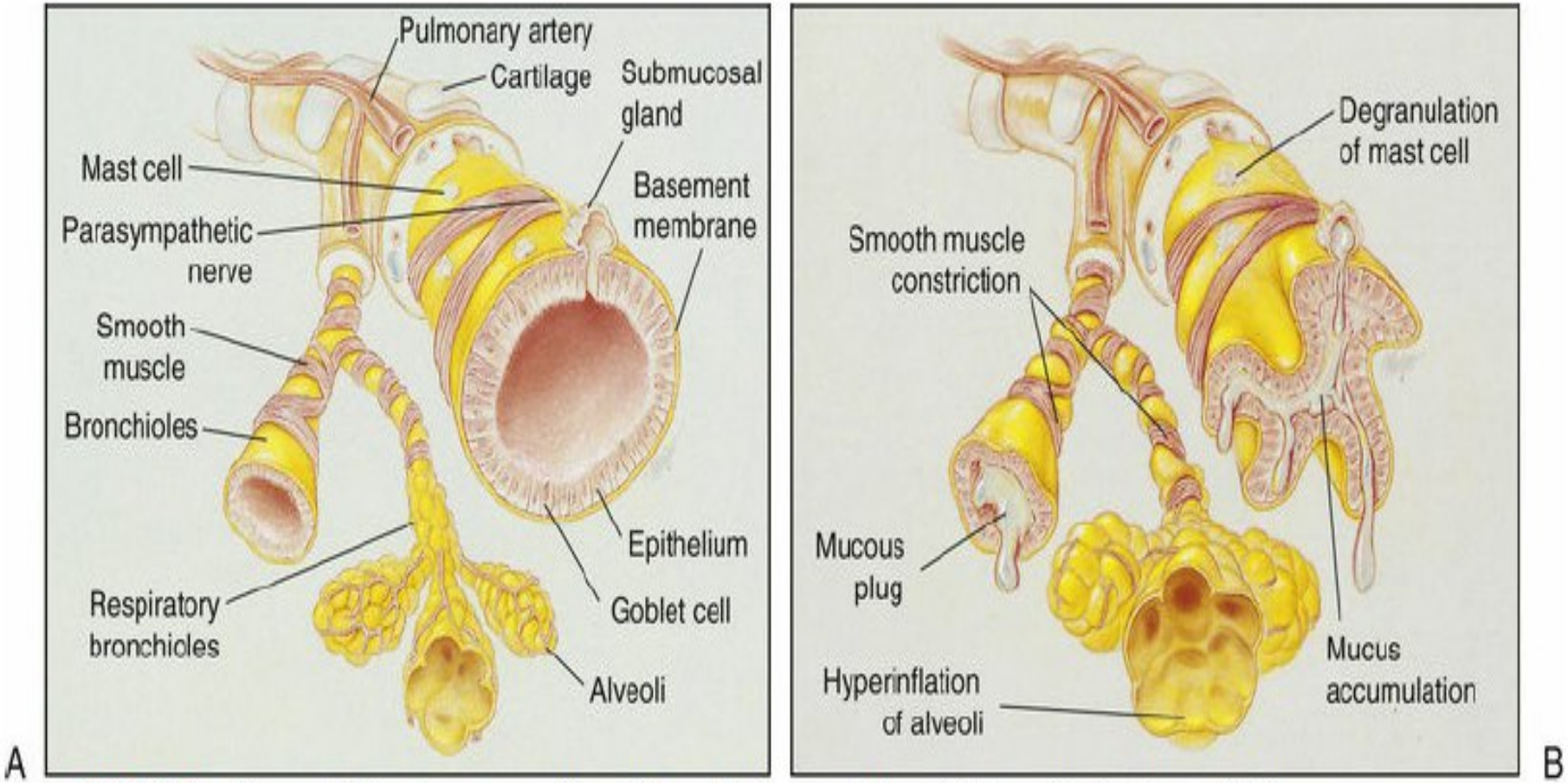


FIG 21-7 Airway obstruction caused by asthma. **A,** A normal lung. **B,** Bronchial asthma: Thick mucus, mucosal edema, and smooth muscle spasm causing obstruction of small airways; breathing becomes labored, and expiration is difficult. (Modified from Des Jardins T, Burton GG: *Clinical manifestations and assessment of respiratory disease,* ed 3, St Louis, 1995, Mosby.)

- Diagnostic Evaluation

The classic manifestations of asthma are dyspnea, wheezing, and coughing.

An attack may develop gradually or appear abruptly and may be preceded by a URI.

The age of the child is often a significant factor because the first attack frequently occurs before 5 years old, with some children manifesting clinical signs and symptoms in infancy.

In infancy, an attack usually follows a respiratory infection.

Some children may experience a prodromal itching at the front of the neck or over the upper part of the back just before an attack, especially if the attack is related to allergies

The diagnosis is determined primarily on the basis of clinical manifestations, history, physical examination, laboratory tests.

Generally, chronic cough in the absence of infection or diffuse wheezing during the expiratory phase of respiration is sufficient to establish a diagnosis.

Clinical Manifestations of Asthma

- Cough

  Hacking, paroxysmal, irritative, and nonproductive

  Becomes rattling and productive of frothy, clear, gelatinous sputum

- Respiratory-Related Signs

  Shortness of breath

  Prolonged expiratory phase

  Audible wheeze

  May have a malar flush and red ears

  Lips deep, dark red color

  May progress to cyanosis of nail beds or circumoral cyanosis

  Restlessness

  Apprehension

  Prominent sweating as the attack progresses

  Older children sitting upright with shoulders in a hunched-over position, hands on the bed or chair, and arms braced (tripod)

  Speaking with short, panting, broken phrases

Hyperresonance on percussion

Coarse, loud breath sounds

Wheezes throughout the lung fields

Prolonged expiration

Crackles

Generalized inspiratory and expiratory wheezing; increasingly high pitched

- With Repeated Episodes

Barrel chest

Elevated shoulders

Use of accessory muscles of respiration

Facial appearance—flattened malar bones, dark circles beneath the eyes, narrow nose, prominent upper teeth

Diagnosis of Asthma

Pulmonary function tests (PFTs) provide an objective method of evaluating the presence and degree of lung disease, as well as the response to therapy. Spirometry can generally be performed reliably on children by 5 or 6 years old.

spirometry testing be done at the time of initial assessment of asthma, after treatment is initiated and symptoms have stabilized, and at least every 1 to 2 years to assess the maintenance of airway function

Nursing Care Guidelines

Interpreting Peak Expiratory Flow Rates*

- **Green (80% to 100% of personal best)** signals all clear. Asthma is under reasonably good control. No symptoms are present, and the routine treatment plan for maintaining control can be followed.
- **Yellow (50% to 79% of personal best)** signals caution. Asthma is not well controlled. An acute exacerbation may be present. Maintenance therapy may need to be increased. Call the practitioner if the child stays in this zone.
- **Red (<50% of personal best)** signals a medical alert. Severe airway narrowing may be occurring. A short-acting bronchodilator should be administered. Notify the practitioner if the peak expiratory flow rate (PEFR) does not return immediately and stay in yellow or green zones.



Another measurement to consider is the peak expiratory flow rate (PEFR), which measures the maximum flow of air (in liters per minute) that can be forcefully exhaled in 1 second using a peak expiratory flow meter (PEFM).

The reliability of the PEFM is controversial, because it relies on the child's ability to use the PEFM and willingness to participate. Each child needs to establish his or her personal best value during a 2- to 3-week period when the child's asthma is stable.

Diagnosis

Bronchoprovocation testing, direct exposure of the mucous membranes to a suspected antigen in increasing concentrations helps to identify inhaled allergens. Exposure to methacholine (methacholine challenge), histamine, or cold or dry air may be performed to assess airway responsiveness or reactivity. Exercise challenges may be used to identify children with exercise induced bronchospasm (EIB). These tests should be done under close observation in a qualified laboratory or clinic.

Skin prick testing (SPT) and serological testing (with quantification of sIgE) for allergen-specific immunoglobulin E (sIgE) may be used to identify environmental allergens that trigger asthma.

It is recommended that all patients with year-round asthma symptoms be tested with skin tests or laboratory blood analysis to determine sensitization to perennial allergens (e.g., house dust mites, cats, dogs, cockroaches, molds, and fungus).

laboratory tests (complete blood count with differential), presence of eosinophilia of greater than 500/mm3 suggests the presence of an allergic or inflammatory disorder.

chest radiographs: Frontal and lateral radiographs may show infiltrates and hyperexpansion of the airways, with the anteroposterior diameter on physical examination indicating an increased diameter (suggestive of barrel chest).

Therapeutic Management

The overall goals of asthma management are to maintain normal activity levels, maintain normal pulmonary function, prevent chronic symptoms and recurrent exacerbations, provide optimum drug therapy with minimum or no adverse effects.

- Several treatment principles need to be followed

  Regular visits to the health care provider is necessary to evaluate therapeutic response and plan of care.

  Prevention of exacerbations includes avoiding triggers, avoiding allergens, and using medications as needed.

  Therapy includes efforts to reduce underlying inflammation and relieve/prevent symptomatic airway narrowing.

  Therapy includes education, environmental control, pharmacologic management.

- Allergen Control

  Nonpharmacologic therapy is aimed at the prevention and reduction of exposure to airborne allergens and irritants. House dust mites and other components of house dust are frequent agents identified in children who are allergic to inhalants.

  The cockroach, mouse, cat and dog are important allergen in many locations.

  Tobacco smoke, wood-burning stoves, cleaning products, pesticides, mold spores, nitrogen dioxide, and sulfur dioxide; these are believed to contribute to asthma

  Living in homes close to busy roads, damp homes with mold, and exposure to tobacco smoke are significant contributing factors in the development of asthma in infants and small children removing carpeting from the home of a child sensitive to mold and dust particles will decrease the frequency of asthma episodes.

- ## Drug Therapy

Pharmacologic therapy is used to prevent and control asthma symptoms, reduce the frequency and severity of asthma exacerbations, and reverse airflow obstruction.

Inflammation is considered an early/persistent feature of asthma, therapy is directed toward long-term suppression of inflammation.

Asthma control has two domains:

1. Reducing impairment (associated with the frequency and intensity of symptoms and functional limitations)
2. Reducing risk (preventing attacks, decline in lung function, as well as watching for medication side effects)

Asthma medications are categorized into two general classes:

- long-term control medications (preventive medications) to achieve and maintain control of inflammation,
- and quick-relief medications (rescue medications) to treat symptoms and exacerbations.

Quick-relief and long-term medications are often used in combination.

Inhaled corticosteroids, cromolyn sodium, long-acting β2-agonists (LABAs), methylxanthines, and leukotriene modifiers are used as long-term control medications.

Short-acting β2-agonists, anticholinergics, and systemic corticosteroids are used as quick-relief medications.

√ Drug Therapy

√ Corticosteroids are antiinflammatory drugs used to treat reversible airflow obstruction, control symptoms, and reduce bronchial hyperresponsiveness in chronic asthma. Inhaled corticosteroids are used as first-line therapy in children older than 5 years of age.

√ β-Adrenergic agonists (short acting) (primarily albuterol, levalbutero) are used for treatment of acute exacerbations and for the prevention of exercise-induced bronchoconstriction (EIB).

√ Salmeterol (Serevent) is a (LABA bronchodilator) that is used twice a day. This drug is anti-inflammatory therapy and used for long-term prevention of symptoms, especially nighttime symptoms, and EIB.

√ Theophylline is a methylxanthine drug used for decades to relieve symptoms and prevent asthma attacks

√ Cromolyn sodium is a medication used in maintenance therapy for asthma in children older than 2 years old. It stabilizes mast cell membranes; inhibits activation and release of mediators from eosinophil and epithelial cells; and inhibits the acute airway narrowing after exposure to exercise, cold dry air, and sulfur dioxide.

√ Anticholinergics (atropine and ipratropium [Atrovent]) help relieve acute bronchospasm. However, have

Status Asthmaticus

Status asthmaticus is a medical emergency that can result in respiratory failure and death if untreated.

The condition may develop gradually or rapidly, often coincident with complicating conditions, such as pneumonia or a respiratory virus, that can influence the duration and treatment of the exacerbation.

Therapy for status asthmaticus is aimed at improving ventilation, decreasing airway resistance, relieving bronchospasm, correcting dehydration and acidosis, allaying child and parent anxiety related to the severity of the event, and treating any concurrent infection.

Humidified oxygen is recommended and should be given to maintain SaO2 greater than 90%.

Inhaled aerosolized short acting β2-agonists are recommended for all patients.

Three treatments of β2-agonists spaced 20 to 30 minutes apart are usually given as initial therapy, and continuous administration of β2-agonists via nebulizer may be initiated.

A systemic corticosteroid (oral, IV, or IM) is given to decrease the effects of inflammation.

An anticholinergic agent may be added to the aerosolized solution of the β2-agonist. Anticholinergics have been shown to result in additional bronchodilation in patients with severe airflow obstruction.

An IV infusion is often initiated to provide a means for hydration and to administer medications.

Correction of dehydration, acidosis, hypoxia, and electrolyte disturbance is guided by frequent determination of arterial pH, blood gases, and serum electrolytes.

In acute asthma attacks, IV magnesium sulfate, a potent muscle relaxant that decreases inflammation and improves pulmonary function and peak flow rate among patients with moderate to severe asthma at ED or ICU.

Heliox (mixture of Helium in Oxygen) may be given to decrease airway resistance that decrease the work of breathing

- Nursing Care Management
- Acute Asthma Care

Pulse oximetry is monitored along with rate and depth of breathing, auscultation of air movement, and any signs of respiratory distress (e.g., nasal flaring, tachypnea, retractions).

The child in status asthmaticus should be placed on continuous cardiorespiratory (including blood pressure) and pulse oximetry monitoring.

Many asthma scoring tools assesses the child's respiratory rate, oxygen requirements, auscultation findings, retractions, and degree of dyspnea. Nurses and other members of the health care team can use this tool to evaluate how the child is responding to the medications and other therapies.

- Provide Long-Term Asthma Care

Nurses who are involved with children in the home, hospital, school, outpatient clinic, or practitioner's office play an important role in helping children and their families learn to live with the condition.

The disease can be managed so that it does not require hospitalization or interfere with family life, physical activity, or school attendance.

Parents are taught how to prevent exacerbations, to recognize and respond to symptoms of bronchospasm, to maintain health and prevent complications, and to promote normal activities. The nurse should determine any cultural or ethnic beliefs or practices that influence self-management and



FIG 21-8 Children with asthma may take a nebulized aerosol treatment with **(A)** a mask or **(B)** mouthpiece. (Courtesy of Texas Children's Hospital, Houston, TX.)



FIG 21-9 Child using metered-dose inhaler (MDI) with spacer and face mask.

A spacer or AeroChamber device should be used with MDI inhalers. These devices allow the parent or child to deliver the medication from the MDI and slowly inhale it. Spacers also help prevent yeast infections in the mouth when corticosteroids are inhaled via an MDI.

- Cystic Fibrosis

CF is a life-shortening disease, inherited as an autosomal recessive trait. The affected child inherits the defective gene from both parents, with an overall risk of one in four if both parents carry the gene.

The condition has a frequency of 1 in 3500 live births among Caucasians.

The mutated gene responsible for CF is located on the long arm of chromosome 7.

This gene codes protein of 1480 amino acids called cystic fibrosis transmembrane conductance regulator CFTR.

- Pathophysiology

CF is characterized by several clinical features, which are increased viscosity of mucous gland secretions, a striking elevation of sweat electrolytes, an increase in several organic and enzymatic constituents of saliva, and abnormalities in autonomic nervous system function.

Although both sodium and chloride are affected, the defect appears to be primarily a result of abnormal chloride movement; the CFTR appears to function as a chloride channel. Children with CF demonstrate an increase in sodium and chloride in both saliva and sweat.

This characteristic is the basis for the sweat chloride diagnostic test.

The primary factor, and the one that is responsible for many of the clinical manifestations of the disease, is mechanical obstruction caused by the increased viscosity of mucous gland secretions.

Small passages in organs (such as the pancreas and bronchioles) become obstructed as secretions precipitate or coagulate to form concretions in glands and ducts.

The earliest postnatal manifestation of CF is often meconium ileus in the newborn, in which the small

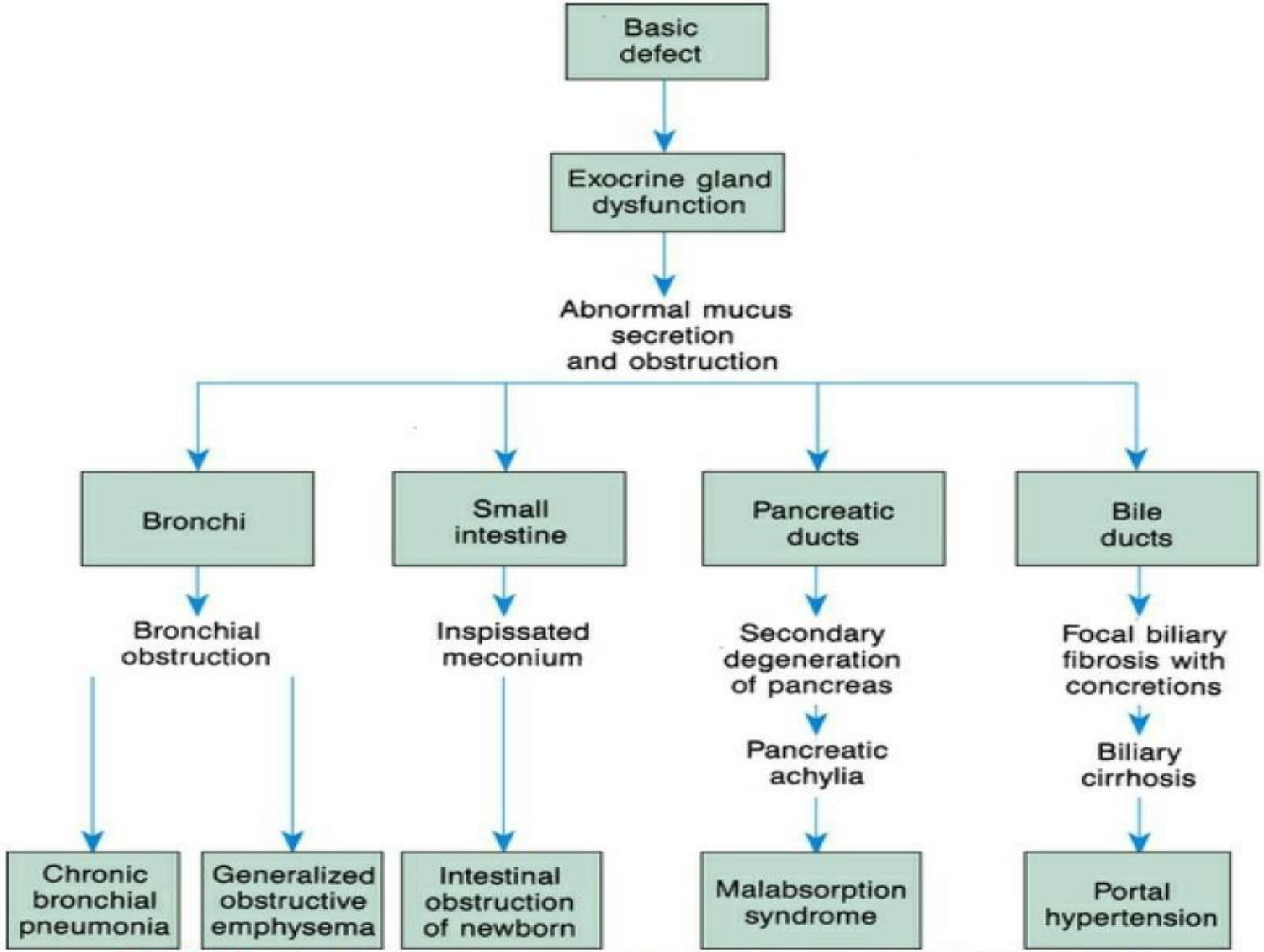


FIG 21-10 Various effects of exocrine gland dysfunction in cystic fibrosis (CF).

In the pancreas, the thick secretions block the ducts, eventually causing pancreatic fibrosis. This blockage prevents essential pancreatic enzymes from reaching the duodenum, which causes marked impairment in the digestion and absorption of nutrients.

The disturbed function is reflected in bulky stools that are frothy from undigested fat (steatorrhea) and foul smelling from putrefied protein (azotorrhea).

The incidence of diabetes mellitus (cystic fibrosis–related diabetes [CFRD]) is greater in CF children than in the general population, which may be caused by changes in pancreatic architecture and diminished blood supply over time.

CFRD is reported to be the most common complication associated with CF; by age 30 years, approximately 50% of people with CF will develop diabetes, which is associated with increased morbidity (sixfold) and mortality and poor lung function.

A common gastrointestinal complication associated with CF is prolapse of the rectum, which occurs in infancy and childhood and is related to large, bulky stools; malnutrition; and increased intra-abdominal pressure secondary to paroxysmal cough.

Pulmonary complications are present in almost all children with CF, with eventual bacterial colonization leading to destruction of lung tissue.

The abnormally viscous and tenacious secretions are difficult to expectorate and gradually obstruct the bronchi and bronchioles

The reproductive systems of both males and females with CF are affected. Fertility can be inhibited by highly viscous cervical secretions, which act as a plug, blocking sperm entry. Women with CF who become pregnant have an increased incidence of premature labor and delivery and infant low birth weight

Box 21-17

Clinical Manifestations of Cystic Fibrosis

Meconium Ileus*

Abdominal distention

Vomiting

Failure to pass stools

Rapid development of dehydration

Gastrointestinal Manifestations

Large, bulky, loose, frothy, extremely foul-smelling stools

Voracious appetite (early in disease)

Loss of appetite (later in disease)

Weight loss

Marked tissue wasting

Failure to grow

Distended abdomen

Thin extremities

Sallow skin

Evidence of deficiency of fat-soluble vitamins A, D, E, and K

Anemia

Pulmonary Manifestations

Initial signs:

Wheezy respirations

Dry, nonproductive cough

Eventually:

Increased dyspnea

Paroxysmal cough

Evidence of obstructive emphysema and patchy areas of atelectasis

Progressive involvement:

Overinflated, barrel-shaped chest

Cyanosis

Clubbing of fingers and toes

Repeated episodes of bronchitis and bronchopneumonia

- Diagnostic Evaluation

Traditionally, the diagnosis of CF was based on the presence of one or more characteristic features (chronic sinopulmonary disease, gastrointestinal or nutritional abnormalities, salt loss syndromes, genital abnormalities in males), a history of CF in a sibling or a positive newborn screen plus laboratory confirmation of an abnormality in the CFTR gene or protein.

An in utero diagnosis of CF is also possible based on detection of two CF mutations in the fetus.

The consistent finding of abnormally high sodium and chloride concentrations in the sweat is a unique characteristic of CF.

Parents may report that their infant tastes “salty” when they kiss him or her. The quantitative sweat chloride test (pilocarpine iontophoresis) remains the best diagnostic tool for CF and involves stimulating the production of sweat with a special device

The quantitative analysis requires a sufficient volume of sweat (>75 mg).

Two separate samples are collected to ensure the reliability of the test for any individual.

Normally, sweat chloride content is less than 40 mEq/L, with a mean of 18 mEq/L.

A chloride concentration greater than 60 mEq/L in a child 6 months old or older is diagnostic of CF, a concentration between 40 and 59 mmol/L is indeterminate and a repeat test should be performed in 1 to 2 months

- Management of Pulmonary Problems

Management of pulmonary problems is directed toward prevention and treatment of pulmonary infection by improving ventilation, removing mucopurulent secretions, and administering antimicrobial agents.

Many children develop respiratory symptoms by 3 years old. The large amounts and viscosity of respiratory secretions in children with CF contribute to the likelihood of respiratory tract infections.

Recurrent pulmonary infections in children with CF result in greater damage to the airways; small airways are destroyed, causing bronchiectasis.

Airway clearance therapies (ACTs) are an essential part of CF management and include percussion and postural drainage, positive expiratory pressure (PEP), active-cycle-of-breathing technique, autogenic drainage, oscillatory PEP, high-frequency chest compressions (HFCCs), and exercise.

Studies have demonstrated that no particular ACT has any advantage over the other in relation to outcomes of sputum production; however, it is recommended that individualized assessment occur to determine the best ACT for each patient.

The Flutter mucus clearance device is a small handheld plastic pipe with a stainless-steel ball on the inside that facilitates removal of mucus

Bronchodilator medication delivered in an aerosol opens bronchi for easier expectoration and is administered before percussion and postural drainage

Nebulized hypertonic saline (7%) has been shown to be effective in improving airway hydration

and increases mucus clearance in patients with CF

IV antibiotics may be administered

Pancreatic enzymes should be taken before a meal or snack or within 30 minutes of eating.



Questions ???

Thank You

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The Child with Gastrointestinal Dysfunction

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Distribution of Body Fluids

The distribution of body fluids, or total body water (TBW), involves the presence of intracellular fluid (ICF) and extracellular fluid (ECF). Water is the major constituent of body tissues, and the TBW in an individual ranges from 75% (in term newborns) to 45% (in late adolescence) of total body weight.

The ICF refers to the fluid contained within the cells, whereas the ECF is the fluid outside the cells. The ECF is further broken down into several components: intravascular (contained within the blood vessels), interstitial (surrounding the cell; the location of most ECF), and transcellular (contained within specialized body cavities, such as cerebrospinal, synovial, and pleural fluid). In the newborn about 50% of the body fluid is contained within the ECF, whereas 30% of a toddler's body fluid is contained within the ECF.

Under normal conditions, the amount of water ingested closely approximates the amount of urine excreted in a 24-hour period

- Nurses should be alert for altered fluid requirements in various conditions:
- Increased requirements:

  Fever (add 12% per rise of 1° C)

  Vomiting, diarrhea

  High-output kidney failure

  Diabetes insipidus

  Diabetic ketoacidosis

  Burns

  Shock

  Tachypnea

  Radiant warmer (preterm infant)

  Phototherapy (infants)

  Postoperative bowel surgery (e.g., gastroschisis)

- Decreased requirements:

  Heart failure

  Syndrome of inappropriate antidiuretic hormone (SIADH)

  Mechanical ventilation

  After surgery

  Oliguric renal failure

  Increased intracranial pressure

- <u>Daily Maintenance Fluid Requirements:</u>

1. (1- 10 kg) …. 100ml/kg
2. (11-20 kg) …1000 ml plus 50ml/kg for each kg > 10 kg
3. (more than 20kg) … 1500ml plus 20ml/kg for each kg > 20 kg

Water Excess		
Water intake in excess of output: • Excessive oral intake • Hypotonic fluid overload • Plain water enemas Failure to excrete water in presence of normal intake: • Kidney disease • Syndrome of inappropriate syndrome of inappropriate anti-diuretic hormone • Heart failure • Malnutrition	Edema: • Generalized • Pulmonary (moist rales or crackles) • Intracutaneous (noted especially in loose areolar tissue) Elevated central venous pressure Hepatomegaly Slow, bounding pulse Weight gain Lethargy Increased spinal fluid pressure CNS manifestations (seizures, coma) Laboratory findings: • Low urine specific gravity • Decreased serum electrolytes • Decreased hematocrit • Variable urine volume	Limit fluid intake. Administer diuretics. Monitor vital signs. Monitor neurologic signs as necessary. Determine and treat cause of water excess. Analyze serum electrolyte measurements. Implement seizure precautions.

TABLE 22-2

Disturbances of Select Fluid and Electrolyte Balance

Mechanisms and Situations	Manifestations	Management and Nursing Care
Water Depletion		
Failure to absorb or reabsorb water Complete or sudden cessation of intake or prolonged diminished intake: • Neglect of intake by self or caregiver—confused, psychotic, unconscious, or helpless • Loss from GI tract—vomiting, diarrhea, NG suction, fistula Disturbed body fluid chemistry: Inappropriate ADH secretion Excessive renal excretion: Glycosuria (diabetes) Loss through skin or lungs: • Excessive perspiration or evaporation—febrile states, hyperventilation, increased ambient temperature, increased activity (BMR) • Impaired skin integrity—transudate from injuries • Hemorrhage Iatrogenic: • Overzealous use of diuretics • Improper perioperative fluid replacement • Use of radiant warmer or phototherapy	General symptoms depend to some extent on proportion of electrolytes lost with water Thirst Variable temperature—increased (infection) Dry skin and mucous membranes Poor skin turgor Poor perfusion (decreased pulse, prolonged capillary refill time) Weight loss Fatigue Diminished urinary output Irritability and lethargy Tachycardia Tachypnea Altered level of consciousness, disorientation Laboratory findings: • High urine specific gravity • Increased hematocrit • Variable serum electrolytes • Low serum bicarbonate (HCO_3) • Variable urine volume • Increased BUN • Increased serum osmolality	Provide replacement of fluid losses commensurate with volume depletion. Provide maintenance fluids and electrolytes. Determine and correct cause of water depletion. Measure fluid intake and output. Monitor vital signs. Monitor urine specific gravity. Monitor body weight. Monitor serum electrolytes.

Sodium Depletion (Hyponatremia)		
Prolonged low-sodium diet Decreased sodium intake Fever Excess sweating Increased water intake without electrolytes Tachypnea (infants) Cystic fibrosis Burns and wounds Vomiting, diarrhea, NG suction, fistulas Adrenal insufficiency Renal disease DKA Malnutrition	Associated with water loss: • Same as with water loss—dehydration, weakness, dizziness, nausea, abdominal cramps, apprehension • Mild—apathy, weakness, nausea, weak pulse • Moderate—decreased blood pressure, lethargy Laboratory findings: • Sodium concentration <130 mEq/L (may be normal if volume loss) • Urine specific gravity depends on water deficit or excess	Determine and treat cause of sodium deficit. Administer IV fluids with appropriate saline concentration. Monitor fluid intake and output.
Sodium Excess (Hypernatremia)		
High salt intake—enteral or IV Renal disease Fever Insufficient breast milk intake in neonate (dehydration hypernatremia) High IWL: • Increased temperature • Increased humidity • Hyperventilation • Diabetes insipidus • Hyperglycemia	Intense thirst Dry, sticky mucous membranes Flushed skin Temperature possibly increased Hoarseness Oliguria Nausea and vomiting Possible progression to disorientation, seizures, muscle twitching, nuchal rigidity, lethargy at rest, hyperirritability when aroused Laboratory findings: • Serum sodium concentration ≥150 mEq/L • High plasma volume • Alkalosis	Determine and treat cause of sodium excess. Administer IV fluids as prescribed. Measure fluid intake and output. Monitor laboratory data. Monitor neurologic status. Ensure adequate intake of breast milk and provide lactation assistance with new mother/baby pair before hospital discharge.

Potassium Depletion (Hypokalemia)		
Starvation Clinical conditions associated with poor food intake Malabsorption IV fluid without added potassium GI losses—diarrhea, vomiting, fistulas, NG suction Diuresis Administration of diuretics Administration of corticosteroids Diuretic phase of nephrotic syndrome Healing stage of burns Potassium-losing nephritis Hyperglycemic diuresis (e.g., diabetes mellitus) Familial periodic paralysis IV administration of insulin in DKA Alkalosis	Muscle weakness, cramping, stiffness, paralysis, hyporeflexia Hypotension Cardiac arrhythmias, gallop rhythm Tachycardia or bradycardia Ileus Apathy, drowsiness Irritability Fatigue Laboratory findings: • Decreased serum potassium concentration ≤3.5 mEq/L • Abnormal ECG—notched or flattened T waves, decreased ST segment, premature ventricular contractions	Determine and treat cause of potassium deficit. Monitor vital signs, and ECG. Administer supplemental potassium. Assess for adequate renal output before administration. For IV replacement, administer potassium slowly. Always monitor ECG for IV bolus potassium replacement. For oral intake, offer high-potassium fluids and foods. Evaluate acid-base status.
Potassium Excess (Hyperkalemia)		
Renal disease Renal failure Adrenal insufficiency (Addison disease) Associated with metabolic acidosis Too rapid administration of IV potassium chloride Transfusion with old donor blood Severe dehydration Crushing injuries Burns Hemolysis Dehydration Potassium-sparing diuretics Increased intake of potassium (e.g., salt substitutes)	Muscle weakness, flaccid paralysis Twitching Hyperreflexia Bradycardia Ventricular fibrillation and cardiac arrest Oliguria Apnea—respiratory arrest Laboratory findings: • High serum potassium concentration ≥5.5 mEq/L • Variable urine volume • Flat P wave on ECG, peaked T waves, widened QRS complex, increased PR interval	Determine and treat cause of potassium excess. Monitor vital signs, including ECG. Administer exchange resin, if prescribed. Administer IV fluids as prescribed. Administer IV insulin (if ordered) to facilitate movement of potassium into cells. Monitor potassium levels. Evaluate acid-base status.

- Water Intoxication

Water intoxication, or water overload, is observed less often than dehydration. However, it is important that nurses and others who care for children be alert to this possibility in certain situations.

Children who ingest excessive amounts of electrolyte-free water develop a concurrent decrease in serum sodium accompanied by central nervous system (CNS) symptoms. There is a large urinary output, and because water moves into the brain more rapidly than sodium moves out, the child may also exhibit irritability, somnolence, headache, vomiting, diarrhea, or generalized seizures. The affected child usually appears well hydrated but may be edematous or even dehydrated.

Fluid intoxication can occur during acute intravenous (IV) fluid replacement, too rapid dialysis, tap water enemas, feeding of incorrectly mixed formula, or excess water ingestion

- Dehydration

Water constitutes 60% to 70% of an infant's weight.

Dehydration is a common body fluid disturbance in the nursing care of infants and children; it occurs whenever the total output of fluid exceeds the total intake, regardless of the underlying cause.

Although dehydration can result from impaired oral intake, more often it is a result of abnormal losses, such as those that occur in vomiting or diarrhea, when oral intake only partially compensates for the abnormal losses. Other significant causes of dehydration include diabetic ketoacidosis and extensive burns.

Dehydration is often described according to plasma sodium concentrations (i.e., isonatremic, hyponatremic, or hypernatremic).

1. Isotonic (isosmotic or isonatremic) dehydration occurs in conditions in which electrolyte and water deficits are present in approximately balanced proportions. This is the primary form of dehydration occurring in children.
2. Hypotonic (hyposmotic or hyponatremic) dehydration occurs when the electrolyte deficit exceeds the water deficit.
3. Hypertonic (hyperosmotic or hypernatremic) dehydration results from water loss in excess of electrolyte loss and is usually caused by a proportionately larger loss of water or a larger intake of electrolytes. This type of dehydration is the most dangerous and requires more specific fluid therapy.

The degree of dehydration has been described as a percentage of body weight dehydrated:

mild—less than 3% in older children or less than 5% in infants;

moderate—5% to 10% in infants and 3% to 6% in older children

Severe—more than 10% in infants and more than 6% in older children

Clinical Manifestations of Dehydration

Manifestation	Isotonic (Loss of Water and Sodium)	Hypotonic (Loss of Sodium in Excess of Water)	Hypertonic (Loss of Water in Excess of Sodium)
Skin			
Color	Gray	Gray	Gray
Temperature	Cold	Cold	Cold or hot
Turgor	Poor	Very poor	Fair
Texture	Dry	Clammy	Thickened, doughy
Mucous membranes	Dry	Slightly moist	Parched
Tearing and salivation	Absent	Absent	Absent
Eyeball	Sunken	Sunken	Sunken
Fontanel	Sunken	Sunken	Sunken
Body temperature	Subnormal or elevated	Subnormal or elevated	Subnormal or elevated
Pulse	Rapid	Very rapid	Moderately rapid
Respirations	Rapid	Rapid	Rapid
Behavior	Irritable to lethargic	Lethargic or comatose; seizures	Marked lethargy with extreme hyperirritability on stimulation

- Diagnostic Evaluation

- To initiate a therapeutic plan, several factors must be determined:

√ The degree of dehydration based on physical assessment

√ The type of dehydration based on the pathophysiology of the specific illness responsible for the dehydrated state

√ Specific physical signs other than general signs

√ Initial plasma sodium concentrations

√ Serum bicarbonate concentration (HCO_3)

√ Any associated electrolyte (especially serum potassium) and acid-base imbalances

● Therapeutic Management

Medical management is directed at correcting the fluid imbalance and treating the underlying cause.

When the child is alert, awake, and not in danger, correction of dehydration may be attempted with oral fluid administration.

Mild cases of dehydration can be managed at home by this method.

Oral rehydration management consists of replacement of fluid loss over 4 to 6 hours, replacement of continuing losses, and provision for maintenance fluid requirements.

In general, a mildly dehydrated child may be given 50 ml/kg of oral rehydration solution (ORS), whereas the child with moderate dehydration may be given 100 ml/kg of ORS. The child with fluid losses from diarrhea or vomiting may be given an additional 10 ml/kg for each stool or vomitus

Amounts and rates are determined from body weight and the severity of dehydration

Gastrointestinal Dysfunction

The primary function of the GI tract is the digestion and absorption of nutrients.
The extensive surface area of the GI tract and its digestive function represent the major means of exchange between the human organism and the environment. Thus any dysfunction of the GI tract can cause significant problems with the exchange of fluids, electrolytes, and nutrients.

- Disorders of Motility

- Diarrhea

Diarrhea is a symptom that results from disorders involving digestive, absorptive, and secretory functions. Diarrhea is caused by abnormal intestinal water and electrolyte transport.

Diarrheal disturbances involve the stomach and intestines (gastroenteritis), the small intestine (enteritis), the colon (colitis), or the colon and intestines (enterocolitis).

The incidence and morbidity of diarrhea are more prominent in low-income countries, such as areasof Asia and Africa

Diarrhea is classified as acute or chronic.

- Acute diarrhea is defined as a sudden increase in frequency and a change in consistency of stools, often caused by an infectious agent in the GI tract. It may be associated with upper respiratory or urinary tract infections, antibiotic therapy, or laxative use.

  Acute infectious diarrhea (infectious gastroenteritis) is caused by a variety of viral (Rotavirus), bacterial (Escherichia coli, Salmonella, Shigella, Clostridium difficile), and parasitic pathogens
- Chronic diarrhea is an increase in stool frequency and increased water content with a duration of more than 14 days.

  It is often caused by chronic conditions, such as malabsorption syndromes, inflammatory bowel disease (IBD), immunodeficiency, food allergy, lactose intolerance, or chronic nonspecific diarrhea (CNSD), or as a result of inadequate management of acute diarrhea.
- Intractable diarrhea of infancy is a syndrome that occurs in the first few months of life, persists for longer than 2 weeks with no recognized pathogens, and is refractory to treatment.

  The most common cause is acute infectious diarrhea that was not managed adequately.
- Chronic nonspecific diarrhea (CNSD), also known as irritable colon of childhood and toddlers' diarrhea, is a common cause of chronic diarrhea in children 6 to 54 months old. These children have loose stools, often with undigested food particles, and diarrhea lasting longer than 2 weeks' duration.

  Children with CNSD grow normally and have no evidence of malnutrition, no blood in their stool, and no enteric infection. Poor dietary habits and food sensitivities have been linked to chronic diarrhea.

  The excessive intake of juices and artificial sweeteners such as sorbitol, which is a substance found in many commercially prepared beverages and foods, may be a factor.

- Etiology

Most pathogens that cause diarrhea are spread by the fecal–oral route through contaminated food or water or are spread from person to person where there is close contact (e.g., daycare centers).

Lack of clean water, crowding, poor hygiene, nutritional deficiency, and poor sanitation are major risk factors, for bacterial or parasitic pathogens.

Infants are often more susceptible to frequent and severe bouts of diarrhea because their immune system has not been exposed to many pathogens and has not acquired protective antibodies.

Worldwide, the most common causes of acute gastroenteritis are infectious agents, viruses, bacteria, and parasites.

Rotavirus is the most important cause of serious gastroenteritis among children, with 28% of all cases causing fatality

The virus is spread through the fecal–oral route or by person-to-person contact, and almost all children are infected with rotavirus at least once by 5 years old

Rotavirus is the most common cause of diarrhea associated hospitalization in children younger than 5 years old.

Salmonella, Shigella, and Campylobacter organisms are most frequently isolated bacterial pathogens (negative bacteria) and can be contracted through raw or undercooked food, contaminated food or water, or through the fecal–oral route.

Among children younger than 5 years old, Salmonella; Campylobacter and Shigella

Antibiotic administration is frequently associated with diarrhea because antibiotics alter the normal intestinal flora, resulting in an overgrowth of other bacteria.

Clostridium difficile is the most common bacterial overgrowth and accounts for approximately 20% of all antibiotic- associated diarrhea

- Pathophysiology

  Invasion of the GI tract by pathogens results in increased intestinal secretion as a result of enterotoxins, cytotoxic mediators, or decreased intestinal absorption secondary to intestinal damage or inflammation.

  The most serious and immediate physiologic disturbances associated with severe diarrheal disease are dehydration, acid-base imbalance with acidosis, and shock that occurs when dehydration progresses to the point that circulatory status is seriously impaired.

Treatment of Acute Diarrhea

Degree of Dehydration	Signs and Symptoms	Rehydration Therapy*	Replacement of Stool Losses	Maintenance Therapy
Mild (5% to 6%)	Increased thirst Slightly dry buccal mucous membranes	ORS, 50 ml/kg within 4 hours	ORS, 10 ml/kg (for infants) or 150 to 250 ml at a time (for older children) for each diarrheal stool	Breastfeeding, if established, should continue; give regular infant formula if tolerated. If lactose intolerance suspected, give undiluted lactose-free formula (or half-strength lactose-containing formula for brief period only); infants and children who receive solid food should continue their usual diet.
Moderate (7% to 9%)	Loss of skin turgor, dry buccal mucous membranes, sunken eyes, sunken fontanel	ORS, 100 ml/kg within 4 hours	Same as above	
Severe (>9%)	Signs of moderate dehydration plus one of following: rapid, thready pulse; cyanosis; rapid breathing; lethargy; coma	IV fluids (Ringer lactate), 40 ml/kg until pulse and state of consciousness return to normal; then 50 to 100 ml/kg or ORS	Same as above	

*If no signs of dehydration are present, rehydration therapy is not necessary. Proceed with maintenance therapy and replacement of stool losses.

- Prevention

  Personal hygiene, protection of the water supply from contamination, and careful food preparation.

- Constipation

Constipation is an alteration in the frequency, consistency, or ease of passing stool. It is defined as a decrease in bowel movement frequency or increased stool hardness for more than 2 weeks

Constipation is often associated with painful bowel movements, blood-streaked or retained stool, abdominal pain, lack of appetite, and stool incontinence (i.e., soiling)

Constipation with fecal soiling is encopresis

Constipation may arise secondary to a variety of organic disorders or in association with a wide range of systemic disorders.

Structural disorders of the intestine (such as strictures, ectopic anus, and Hirschsprung disease, may be associated with constipation.

Systemic disorders associated with constipation include hypothyroidism, hypercalcemia resulting from hyperparathyroidism or vitamin D excess, and chronic lead poisoning. Constipation is also associated with use of drugs, such as antacids, diuretics, antiepileptics, antihistamines, opioids, and iron supplementation

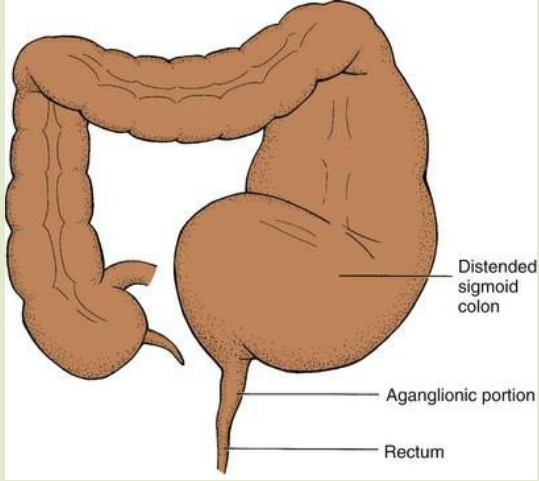
The majority of children have idiopathic or functional constipation because no underlying cause can be identified

- Therapeutic Management

The management of simple constipation consists of a plan to promote regular bowel movements.

Changing the diet to provide more fiber and fluids, eliminating foods known to be constipating, and establishing a bowel routine that allows for regular passage of stool.

An increase in dietary fiber is recommended as a treatment for constipation in the healthy child.



- Hirschsprung Disease (Congenital Aganglionic Megacolon)

Hirschsprung disease is a congenital anomaly that results in mechanical obstruction from inadequate motility of part of the intestine.

The incidence is 1 in 5000 live births. It is four times more common in males than in females

Mutations in the RET protooncogene confirmed association between RET polymorphisms and Hirschsprung dz

- Pathophysiology

The pathology of Hirschsprung disease relates to the absence of ganglion cells in the affected areas of the intestine, resulting in a loss of the rectosphincteric reflex and an abnormal microenvironment of the cells of the affected intestine.

The term congenital aganglionic megacolon describes the primary defect, which is the absence of ganglion cells in the myenteric plexus of Auerbach and the submucosal plexus of Meissner

The absence of ganglion cells in the affected bowel results in a lack

of enteric nervous system stimulation, which decreases the internal

sphincter's ability to relax.

A neonate usually is seen with distended abdomen, feeding intolerance with bilious vomiting, and delay in the passage of meconium

The majority of children with Hirschsprung disease require surgery rather than medicalfrequent enemas

- Diagnostic Evaluation

  Most children with Hirschsprung disease are diagnosed in the first few months of life. Clinical manifestations vary according to the age when symptoms are recognized and the presence of complications, such as enterocolitis

- Clinical Manifestations of Hirschsprung Disease

  - Newborn Period

    Failure to pass meconium within 24 to 48 hours after birth, Refusal to feed, Bilious vomiting, Abdominal distention
  - Infancy

    Failure to thrive, Constipation, Abdominal distention, Episodes of diarrhea and vomiting, Signs of enterocolitis, Explosive, watery diarrhea, Fever, Appears significantly ill
  - Childhood

    Constipation, foul-smelling stools, Abdominal distention, Visible peristalsis, Easily palpable fecal mass, Undernourished, anemic appearance

- Therapeutic Management

The majority of children with Hirschsprung disease require surgery rather than medical therapy with frequent enemas. After the child is stabilized with fluid and electrolyte replacement, if needed, surgery is performed, with a high rate of success.

Surgical management consists primarily of the removal of the a ganglionic portion of the bowel to relieve obstruction, restore normal motility, and preserve the function of the external anal sphincter.

- Vomiting

Vomiting is the forceful ejection of gastric contents through the mouth. It is a well-defined, complex, coordinated process that is under CNS control and is often accompanied by nausea and retching. Vomiting has many causes including acute infectious diseases, increased intracranial pressure, toxic ingestions, food intolerances and allergies, mechanical obstruction of the GI trace, metabolic disorders, nephrologic disease, and psychogenic problems

Vomiting is common in childhood, is usually self-limiting, and requires no specific treatment. However, complications may occur, including acute fluid volume loss (dehydration) and electrolyte disturbances, malnutrition, aspiration, and Mallory-Weiss syndrome (small tears in the distal esophageal mucosa).

● Therapeutic Management

Management is directed toward detection and treatment of the cause of the vomiting and prevention of complications, such as dehydration and malnutrition. If vomiting leads to dehydration, oral rehydration or parenteral fluids may be required. Antiemetic drugs may be indicated. Adverse effects with earlier-generation antiemetics (such as promethazine and metoclopramide) include somnolence, nervousness, irritability, and dystonic reactions and should not be routinely administered to children

- Esophageal Atresia and Tracheoesophageal Fistula

Congenital esophageal atresia (EA) and tracheoesophageal fistula (TEF) are rare malformations that represent a failure of the esophagus to develop as a continuous passage and a failure of the trachea and esophagus to separate into distinct structures.

These defects may occur as separate entities or in combination (Fig. 22-5); and without early diagnosis and treatment, they pose a serious threat to the infant's well-being.

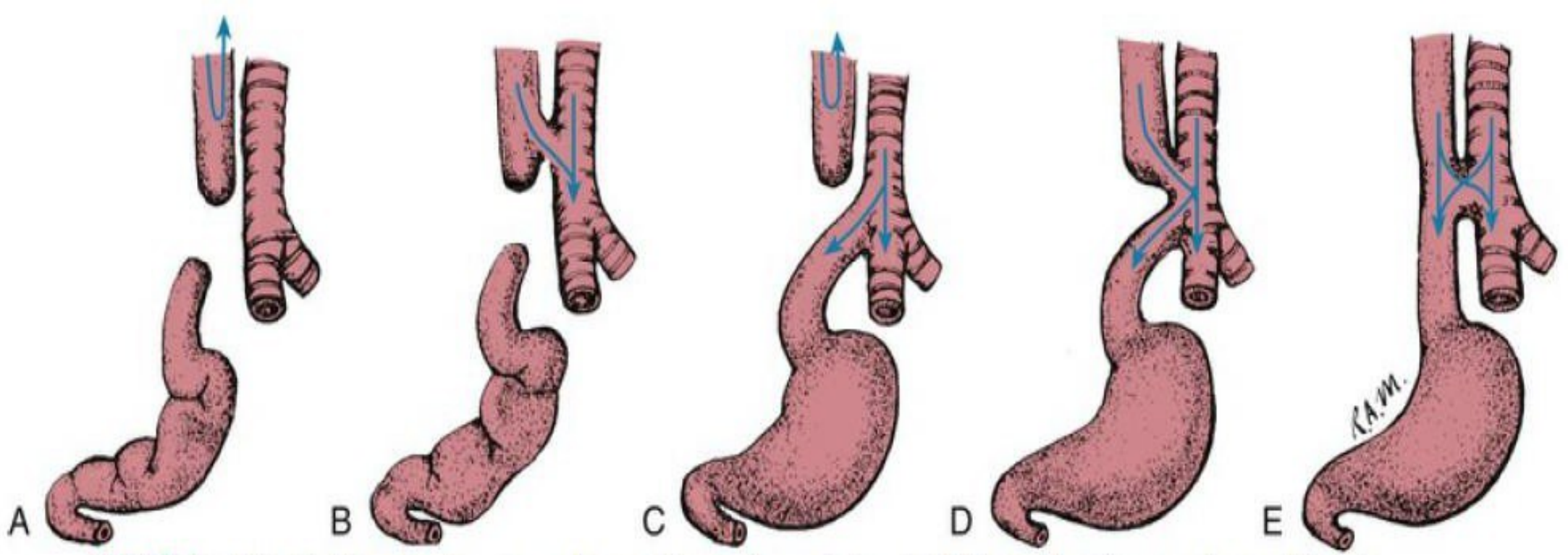


FIG 22-5 **A** to **E**, Five most common types of esophageal atresia (EA) and tracheoesophageal fistula (TEF)

Anatomical classification of EA

Type	Description	Frequency (%)
I	EA alone	7
II	EA and proximal fistula	1
III	EA and distal fistula	87
IV	EA &proximal & distal fistula	1
V	"H" type fistula. No EA	4

Box 22-7

Clinical Manifestations of Tracheoesophageal Fistula

Excessive frothy mucus from nose and mouth

Three Cs of tracheoesophageal fistula (TEF):

Coughing

Choking

Cyanosis

Apnea

Increased respiratory distress during feeding

Abdominal distention

Therapeutic Management

The treatment of patients with EA and TEF includes maintenance of a patent airway, prevention of pneumonia, gastric or blind pouch decompression, supportive therapy, and surgical repair of the anomaly.

- Hypertrophic Pyloric Stenosis

Hypertrophic pyloric stenosis (HPS) occurs when the circumferential muscle of the pyloric sphincter becomes thickened, resulting in elongation and narrowing of the pyloric channel.

This produces an outlet obstruction and compensatory dilation, hypertrophy, and hyperperistalsis of the stomach.

This condition usually develops in the first few weeks of life, causing nonbilious vomiting, occurs after a feeding.

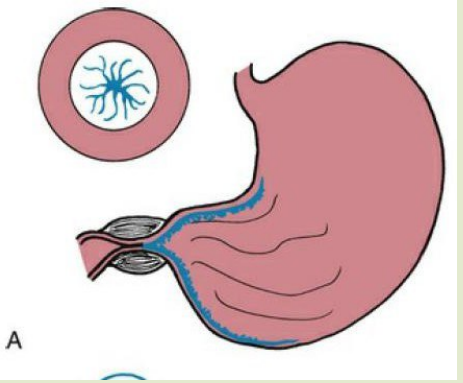
If the condition is not diagnosed early, dehydration, metabolic alkalosis, and failure to thrive may occur.

The precise etiology of HPS is unknown. Boys are affected four to six times more frequently than girls

The olive-like mass is easily palpated when the stomach is empty, the infant is quiet, and the abdominal muscles are relaxed.

Vomiting usually occurs 30 to 60 minutes after feeding and becomes projectile as the obstruction progresses. Emesis is nonbilious in the early stages

Surgical relief of the pyloric obstruction by pyloromyotomy is the standard therapy for this disorder.



● Celiac Disease (Gluten-Sensitive Enteropathy)

Celiac disease, also known as gluten-induced enteropathy, gluten-sensitive enteropathy, and celiac sprue, is a permanent intestinal intolerance to dietary gluten, a protein present in wheat, barley, rye, and oats that causes damage to the villi in the small intestine

- Pathophysiology

Celiac disease is characterized by villous atrophy in the small intestine in response to the protein gluten.

When individuals are unable to digest the gliadin component of gluten (an accumulation of a toxic substance that is damaging to the mucosal cells), damage to the mucosa of the small intestine leads to villous atrophy, hyperplasia of the crypts, and infiltration of the epithelial cells with lymphocytes.

Villous atrophy leads to malabsorption caused by the reduced absorptive surface area

Intestinal symptoms are common in children diagnosed within the first 2 years of life. Other symptoms include failure to thrive, chronic diarrhea, abdominal distention and pain, muscle wasting, aphthous ulcers, and fatigue.

The first step is a serological blood test for tissue transglutaminase and antiendomysial antibodies in children 18 months old or older

Positive serological markers should be followed by an upper GI endoscopy with biopsy.

Although the diet is called “gluten free,” it is actually low in gluten because it is impossible to remove every source of this protein.

Specific nutritional deficiencies, such as iron, folic acid, and fat-soluble vitamin deficiencies, are treated with appropriate supplements.

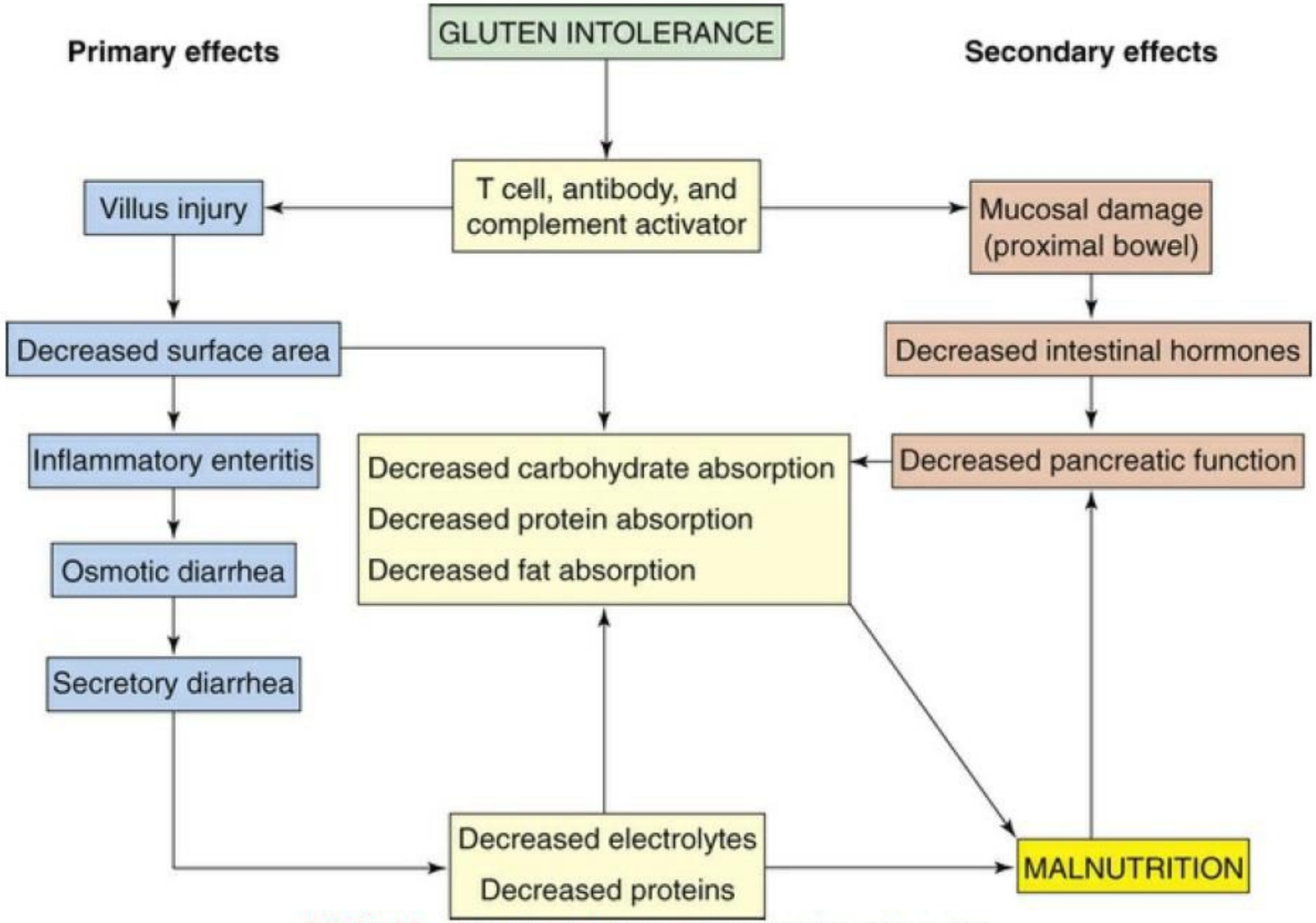


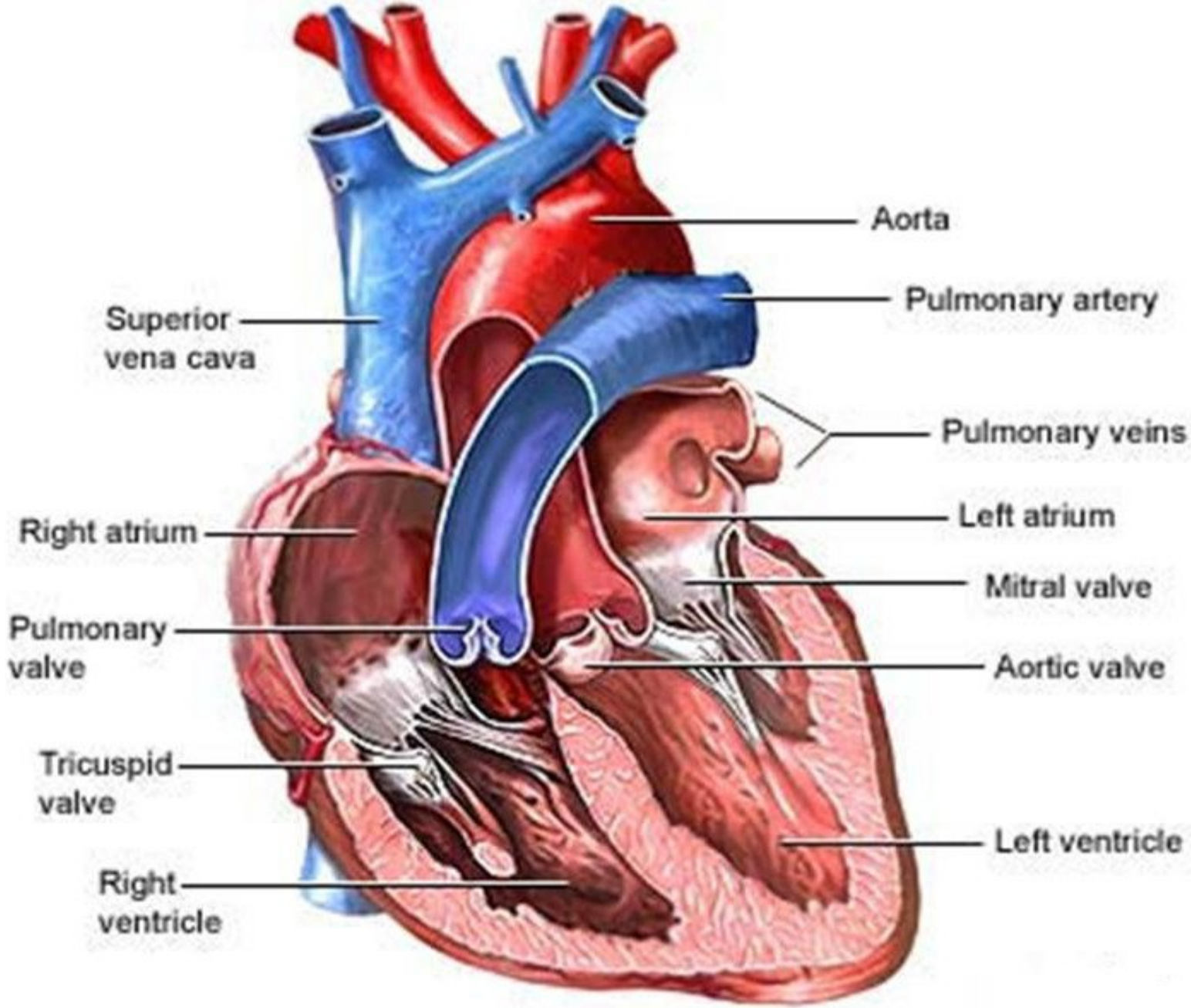
FIG 22-10 Pathophysiology of gluten-sensitive enteropathy.

The Child with Problems Related to the Production and Circulation of Blood

The Child with Cardiovascular Dysfunction

Amy Delaney, Annette L. Baker, Heather Bastardi, Patricia O'Brien




Aorta
Pulmonary artery
Superior vena cava
Pulmonary veins
Left atrium
Right atrium
Mitral valve
Pulmonary valve
Aortic valve
Tricuspid valve
Left ventricle
Right ventricle

Cardiovascular Function

- Cardiovascular = Heart, Arteries, Veins, Blood

- Function
  - Transportation
  - Blood = transport vehicle
  - Carries oxygen, nutrients, wastes, and hormones
  - Movement provided by pumping of heart

Cardiac Tissues

- Outermost = Pericardium & Epicardium
  - Pericardium is a membrane anchoring heart to diaphragm and sternum
  - Pericardium secretes lubricant (serous fluid)
  - Epicardium is outermost muscle tissue
- Middle = Myocardium
  - Contains contractile muscle fibers
- Innermost = Endocardium
  - Lines Cardiac Chambers

Cardiovascular Dysfunction

Cardiovascular disorders in children are divided into two major groups, congenital heart disease and acquired heart disorders. Congenital heart disease (CHD) includes primarily anatomic abnormalities present at birth that result in abnormal cardiac function. The clinical consequences of congenital heart defects fall into two broad categories, heart failure (HF) and hypoxemia.

Acquired cardiac disorders are disease processes or abnormalities that occur after birth and can be seen in the normal heart or in the presence of congenital heart defects. They result from various factors, including infection, autoimmune responses, environmental factors, and familial tendencies

Evaluation of an Infant with suspected CHD

- History and Physical Examination

  Taking an accurate health history is an important first step in assessing an infant or child for possible heart disease. Parents may have specific concerns, such as an infant with poor feeding or fast breathing. Others may not realize that their child has a medical problem because their baby has always been pale and fussy.

- Inspection

  Nutritional state: Failure to thrive or poor weight gain is associated with heart disease.

  Color: Cyanosis is a common feature of CHD, and pallor is associated with poor perfusion.

  Chest deformities: An enlarged heart sometimes distorts the chest configuration.

  Unusual pulsations: Visible pulsations of the neck veins are seen in some patients.

  Respiratory excursion: This refers to the ease or difficulty of respiration (e.g., tachypnea, dyspnea, expiratory grunt).

  Clubbing of fingers: This is associated with cyanosis.

- Palpation and Percussion

  Chest: These maneuvers help discern heart size and other characteristics (e.g., thrills) associated with heart disease.

  Abdomen: Hepatomegaly or splenomegaly may be evident.

  Peripheral pulses: Rate, regularity, and amplitude (strength) may reveal discrepancies.

- Auscultation

  Heart rate and rhythm: Listen for fast heart rates (tachycardia), slow heart rates (bradycardia), and irregular rhythms.

  Character of heart sounds: Listen for distinct or muffled sounds, murmurs, and additional heart sounds.

Procedures for Cardiac Diagnosis

Procedure	Description
Chest radiography (x-ray)	Provides information on heart size and pulmonary blood flow patterns
ECG	Graphic measure of electrical activity of heart
Holter monitor	24-hour continuous ECG recording used to assess dysrhythmias
Echocardiography	Use of high-frequency sound waves obtained by a transducer to produce an image of cardiac structures
Transthoracic	Done with transducer on chest
M-mode	One-dimensional graphic view used to estimate ventricular size and function
Two-dimensional	Real-time, cross-sectional views of heart used to identify cardiac structures and cardiac anatomy
Doppler	Identifies blood flow patterns and pressure gradients across structures
Fetal	Imaging fetal heart in utero
TEE	Transducer placed in esophagus behind heart to obtain images of posterior heart structures or in patients with poor images from chest approach
Cardiac catheterization	Imaging study using radiopaque catheters placed in a peripheral blood vessel and advanced into heart to measure pressures and oxygen levels in heart chambers and visualize heart structures and blood flow patterns
Hemodynamics	Measures pressures and oxygen saturations in heart chambers
Angiography	Use of contrast material to illuminate heart structures and blood flow patterns
Biopsy	Use of special catheter to remove tiny samples of heart muscle for microscopic evaluation; used in assessing infection, inflammation, or muscle dysfunction disorders; also to evaluate for rejection after heart transplant
EPS	Special catheters with electrodes employed to record electrical activity from within heart; used to diagnose rhythm disturbances
Exercise stress test	Monitoring of heart rate, BP, ECG, and oxygen consumption at rest and during progressive exercise on a treadmill or bicycle
Cardiac MRI	Noninvasive imaging technique; used in evaluation of vascular anatomy outside of heart (e.g., COA, vascular rings), estimates of ventricular mass and volume; uses for MRI are expanding

- Electrocardiogram

Electrocardiography (ECG or EKG) measures the electrical activity of the heart, provides a graphic display and supplies information on heart rate and rhythm, abnormal rhythms or conduction, ischemic changes, and other information.

A standard ECG uses 12 leads to get different views of the heart. An ECG takes about 15 minutes to perform, infants and young children may be fussy with lead placement.

Bedside cardiac monitoring with a single lead of the ECG is commonly used in pediatrics, especially in the care of children with heart disease. An alarm can be set with parameters for individual patient requirements and will sound if the heart rate is above or below the set parameters.

- Echocardiography “Echo”

Echocardiography involves the use of ultra-high-frequency sound waves to produce an image of the heart's structure.

A transducer placed directly on the chest wall delivers repetitive pulses of ultrasound and processes the returned signals (echoes). It is the most frequently used test for describing cardiac anatomy and detecting cardiac dysfunction in children.

In many cases, a prenatal diagnosis of CHD can be made by fetal echocardiography. Although the test is noninvasive, painless, and associated with no known side effects, it can be stressful for children.

- Diagnostic catheterizations

These studies are used to diagnose congenital cardiac defects, particularly in symptomatic infants and before surgical repair.

Congenital Heart Disease
Etiology & Incidence

CHD is one of the most common forms of congenital anomalies
It involves the chambers, valves, and vessels arising from the heart
In most cases, the exact cause of CHD is not known

CHD is the major cause of death (other than prematurity) in the first year of life
CHD affects 8-12 of every 1,000 neonates.
90% of cases are unknown
The heart begins as a single cell and develops into four chambered pumping system during the third to eight weeks of gestation.

Most are thought to be a result of multiple factors, including a complex interaction of genetic and environmental influences. Some risk factors are known to be associated with increased incidence of congenital heart defects. Maternal risk factors include chronic illnesses (such as diabetes or poorly controlled phenylketonuria), alcohol consumption, and exposure to environmental toxins and infections.

- Congenital heart anomalies are often associated with chromosomal abnormalities, specific syndromes, or congenital defects in other body systems.

Marfan's syndrome:- mitral valve prolapse (MVP), dilated aortic root.

Turner's syndrome:- aortic valve stenosis (AVS), coarctation of the aorta (COA)

Noonan's syndrome:- dysplastic pulmonary valve

William's syndrome:- supravalvular pulmonary stenosis (PS)

DiGeorge syndrome: a syndrome characterized by deletion of part of chromosome 22q11

Holt-Oram syndrome: (upper limb anomalies and atrial septal defect [ASD]).


Symptoms & Signs of DiGeorge Syndrome
Cleft palate
Cyanosis
Abnormal facial features
Learning difficulties
© www.medindia.net
source:wikipedia.org

Classification of Defects

Common Heart Malformation

- Obstructive lesion: (obstruction to blood flow)

  AS- valvular, subvalvular

  CoA

  PS- valvular, subvalvular

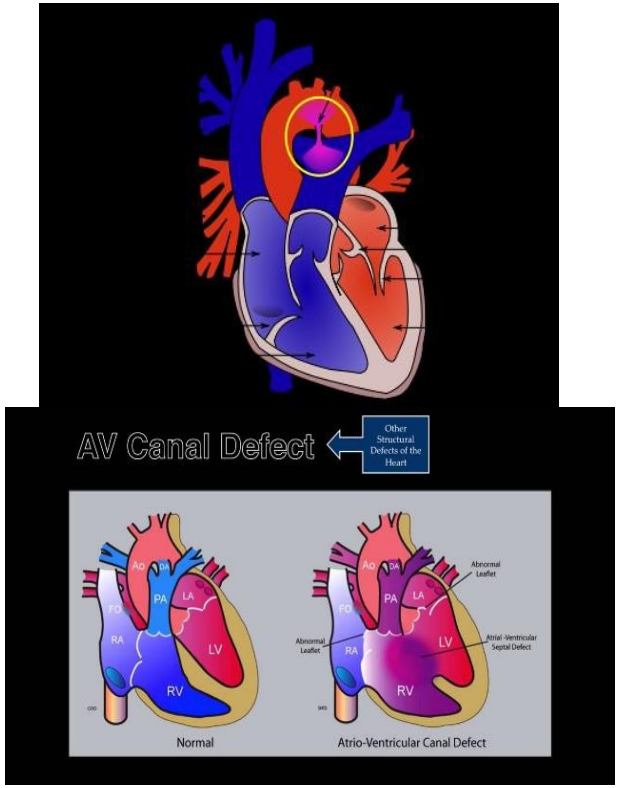
- Increase Pulmonary blood flow (acyanotic)

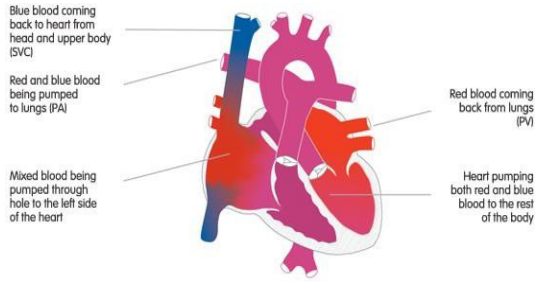
  Patent Ductus Arteriosus (PDA)

  Atrial Septal Defect (ASD)

  Ventricular Septal Defect (VSD)

  AV canal





- Decrease Pulmonary blood flow (cyanotic)

  Tetralogy of Fallot (TOF)

  Tricuspid Atresia (TA)

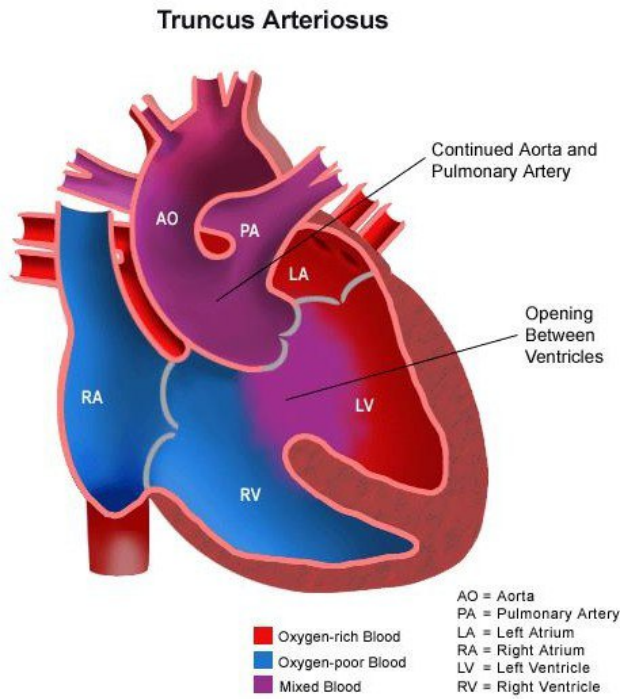
  Total Anomalous Pulmonary Venous Return (TAPVR)

  Truncus arteriosus

  Hypoplastic Left Heart Syndrome (HLHS)

  Double outlet right ventricle (DORV)





Congenital heart disease: Relative Frequency of Congenita

Heart Lesions.	%of all lesion
1-Ventricular septal defect	25-30%
2-A trial septal defect	6-8%
3-Patent ductus arteriosus	6-8%
4-Coarctation of aorta	5-7%
5-Tetralogy of Fallot	5-7%
6- Pulomnary Valve Sterosis	5-7%
7- Aortic Valve Stenosis	4-7%
8-Transposition of great arteries	3-5%
9- Hypo plastic left ventricle	1-3%

Atrial Septal Defect

- Abnormal opening between the atria, allowing blood from the higher pressure left atrium to flow into the lower pressure right atrium.
- Clinical manifestations: Patients may be asymptomatic. They may develop HF. There is a characteristic systolic murmur with a fixed split second heart sound. There may also be a diastolic murmur. Patients are at risk for atrial dysrhythmias
- Surgical treatment: Surgical patch closure (pericardial patch or Dacron patch) is done for moderate to large defects. Open repair with cardiopulmonary bypass is usually performed before school age.
- Nonsurgical treatment: ASD 2 closure with a device during cardiac catheterization is becoming commonplace and can be done as an outpatient procedure
- Prognosis: Operative mortality is very low (<0.5%).

Ventricular Septal Defect

- Abnormal opening between the right and left ventricles. May be classified according to location: membranous (accounting for 80%) or muscular. May vary in size from a small pinhole to absence of the septum, which results in a common ventricle. VSDs are frequently associated with other defects, such as pulmonary stenosis, transposition of the great vessels, PDA, atrial defects, and COA.
- Clinical manifestations: HF is common. There is a characteristic loud holosystolic murmur heard best at the left sternal border. Patients are at risk for BE and pulmonary vascular obstructive disease.
- Surgical treatment: Complete repair (procedure of choice): Small defects are repaired with sutures.

Patent Ductus Arteriosus

- Failure of the fetal ductus arteriosus (artery connecting the aorta and pulmonary artery) to close within the first weeks of life. The continued patency of this vessel allows blood to flow from the higher pressure aorta to the lower pressure pulmonary artery, which causes a left to- right shunt.
- Clinical manifestations: Patients may be asymptomatic or show signs of HF. There is a characteristic machinery-like murmur.
- Medical management: Administration of indomethacin (a prostaglandin inhibitor) has proved successful in closing a PDA in preterm infants and some newborns.
- Surgical treatment: Surgical division or ligation of the patent vessel is performed via a left thoracotomy.

Obstructive Defects
Coarctation of the Aorta

- Localized narrowing near the insertion of the ductus arteriosus, which results in increased pressure proximal to the defect (head and upper extremities) and decreased pressure distal to the obstruction (body and lower extremities).
- Clinical manifestations: The patient may have high BP and bounding pulses in the arms, weak or absent femoral pulses, and cool lower extremities with lower BP. There are signs of HF in infants. In infants with critical coarctation, the hemodynamic condition may deteriorate rapidly with severe acidosis and hypotension.
- Surgical treatment: Surgical repair is the treatment of choice for infants younger than 6 months old and for patients with long-segment stenosis or complex anatomy; it may be performed for all patients with coarctation.
- Nonsurgical treatment: Balloon angioplasty is being performed as a primary intervention for COA in older infants and children, Balloon angioplasty has a higher associated rate of re coarctation than surgical repair and the rate of complication, particularly femoral artery injury is high during infancy

Co-arctation of Aorta

Narrowing or hypo-plastic segment of the aortic arch

It account 8-10% oh CHDs

Most common in male

increase the workload of Lt ventricle (S pressure)

Asymptomatic until PDA begin to close

Tachypnea, acidosis, absent femoral & pedal pulse

Nose bleeding, headache, leg cramp

Diagnosis

Auscultation: murmur

Chest x ray: cardiomegaly & pulmonary edema

Echocardiogram with Doppler (color)

Invasive study (catheterization)

Cardiac MRI

Management

Stabilize with prostaglandin E1 (PGE1)

Inotropic support as needed

Intubation & ventilation as needed

Infective endocarditis prophylaxis (lifelong)

Catheterization (balloon)

Surgical (subclavian flap repair, end to end anastomosis)

Beta-adrenergic blockers

Complications

Systematic hypertension

CHF

Cerebral hemorrhage

Lt ventricular failure

Aortic aneurysm

Aortic stenosis (AS)

May be caused by bicuspid aortic valve with fused commissures that does not open completely

Blood flows at an increase velocity across the obstructive valve or stenosis area

During systole, Lt ventricular pressure rise dramatically to overcome increased resistance.

Myocardial ischemia may occur RT O2 required

Lt side heart failure

Clinical manifestation (AS)

Severe congestive heart failure (CHF)

Metabolic acidosis

Tachypnea, shortness of breathing

Poor capillary refill, cool skin

Poor feeding

Chest pain

Palpitation

Sudden death

Diagnosis (AS)

Auscultation (murmur RT upper sternal)

Palpation (thrill may be palpable in RT upper sternal)

ECG

X ray (increase cardiac silhouette)

Echocardiogram (visualize the anatomy and gradient across the valve)

Management (AS)

Stabilize with prostaglandin (maintain cardiac output)

Inotropic support as needed

Intubation and ventilation as needed

Infective endocarditis prophylaxis

Cardiac catheter (aortic balloon valvuloplasty or angioplasty)

Surgical valvotamy

Restrict strenuous exercise

Pulmonic Stenosis

Narrowing at the entrance to the pulmonary artery. Resistance to blood flow causes right ventricular hypertrophy and decreased pulmonary blood flow. Pulmonary atresia is the extreme form of PS in that there is total fusion of the commissures and no blood flows to the lungs.

Clinical manifestations: Patients may be asymptomatic; some have mild cyanosis or HF. Progressive narrowing causes increased symptoms. Newborns with severe narrowing are cyanotic. A loud systolic ejection murmur at the upper left sternal border may be present. Cardiomegaly is evident on chest radiography

Defects with Decreased Pulmonary Blood Flow
Tetralogy of Fallot

The classic form includes four defects: (1) VSD, (2) PS, (3) overriding aorta, and (4) right ventricular hypertrophy. Tetralogy of Fallot occurs in 5% to 10% of all CHDs and is the most common cyanotic lesion.

Clinical manifestations: Some infants may be acutely cyanotic at birth; others have mild cyanosis that progresses over the first year of life as the PS worsens. There is a characteristic systolic murmur that is often moderate in intensity. Anoxic spells occur when the infant's oxygen requirements exceed the blood supply, usually during crying or after feeding

Tricuspid Atresia

The tricuspid valve fails to develop; consequently there is no communication from the right atrium to the right ventricle. Blood flows through an ASD or a patent foramen ovale to the left side of the heart and through a VSD to the right ventricle and out to the lungs. The condition is often associated with PS and TGA.

Clinical manifestations: Cyanosis is usually seen in the newborn period. There may be tachycardia and dyspnea. Older children have signs of chronic hypoxemia with clubbing.

Modified Fontan procedure: Systemic venous return is directed to the lungs without a ventricular pump through surgical connections between the right atrium and the pulmonary artery.

Mixed Defects

Many complex cardiac anomalies are classified together in the mixed category, because survival in the postnatal period depends on mixing of blood from the pulmonary and systemic circulations within the heart chambers.

Clinically, these patients have a variable picture that combines some degree of desaturation (although cyanosis is not always visible) and signs of HF. Some defects, such as transposition of the great arteries, cause severe cyanosis in the first days of life and later cause HF. Others, such as truncus arteriosus, cause severe HF in the first weeks of life and mild desaturation.

Transposition of the Great Arteries, or Transposition of the Great Vessels

The pulmonary artery leaves the left ventricle, and the aorta exits from the right ventricle with no communication between the systemic and pulmonary circulations.

Clinical manifestations: These depend on the type and size of the associated defects. Newborns with minimum communication are severely cyanotic and have depressed function at birth. Those with large septal defects or a PDA may be less cyanotic but have symptoms of HF. Heart sounds vary according to the type of defect present. Cardiomegaly is usually evident a few weeks after birth.

Therapeutic management (to provide intracardiac mixing): The administration of IV prostaglandin E1 may be initiated to keep the ductus arteriosus open to temporarily increase blood mixing and provide an oxygen saturation of 75% or to maintain cardiac output.

Hypoplastic Left Heart Syndrome

Underdevelopment of the left side of the heart, resulting in a hypoplastic left ventricle and aortic atresia. Most blood from the left atrium flows across the patent foramen ovale to the right atrium, to the right ventricle, and out the pulmonary artery. The descending aorta receives blood from the PDA supplying systemic blood flow.

Clinical manifestations: The patient has mild cyanosis and signs of HF until the PDA closes and then progressive deterioration with cyanosis and decreased cardiac output, leading to cardiovascular collapse. The condition is usually fatal in the first months of life without intervention.

Therapeutic management: Neonates require stabilization with mechanical ventilation and inotropic support preoperatively. A prostaglandin E1 infusion is needed to maintain ductal patency and ensure adequate systemic blood flow.

Pediatric Oncology

Cancer in Children

Few situations in nursing exceed the challenges of caring for a child with cancer. Despite the dramatic improvements in survival rates for these children, the family's needs are tremendous as they cope with a serious physical illness and the fear that the child will not be cured. Nurses should base support of patients and their families on the premise that communication promotes understanding and clarity.

Cancer is the second leading cause of death in children ages 1 to 14

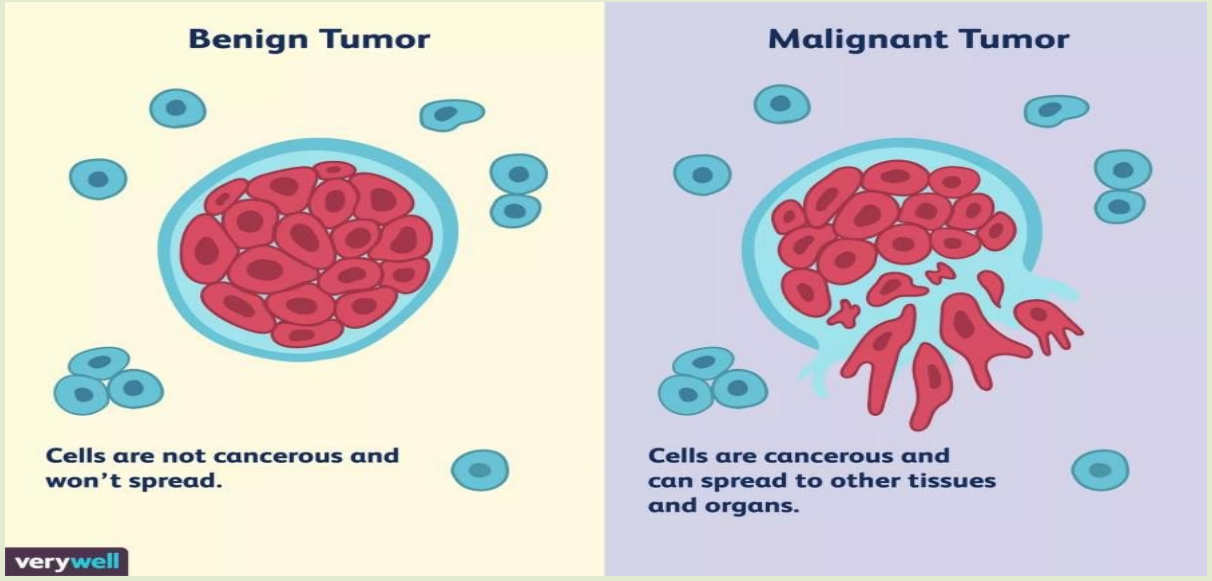
Normal cells grow and divide, then eventually die.

Cancer occurs when the body is unable to regulate cell growth, leading to an overgrowth of abnormal cells.

Benign v. Malignant

Benign: overgrowth of cells that are NOT cancerous

Malignant: overgrowth of cells that ARE cancerous



Benign Vs Malignant

- Characteristics of Benign Tumors

  Cells tend not to spread

  Most grow slowly

  Do not invade nearby tissue

  Do not metastasize (spread) to other parts of the body

  Tend to have clear boundaries

  Under a pathologist's microscope, shape, chromosomes, and DNA of cells appear normal

  Do not secrete hormones or other substances (an exception: pheochromocytomas of the adrenal gland)

  May not require treatment if not health-threatening

  Unlikely to recur if removed or require further treatment such as radiation or chemotherapy

- Characteristics of Malignant Tumors

  Cells can spread

  Usually grow fairly rapidly

  Often invade basal membrane that surrounds nearby healthy tissue

  Can spread via bloodstream or lymphatic system, or by sending "fingers" into nearby tissue

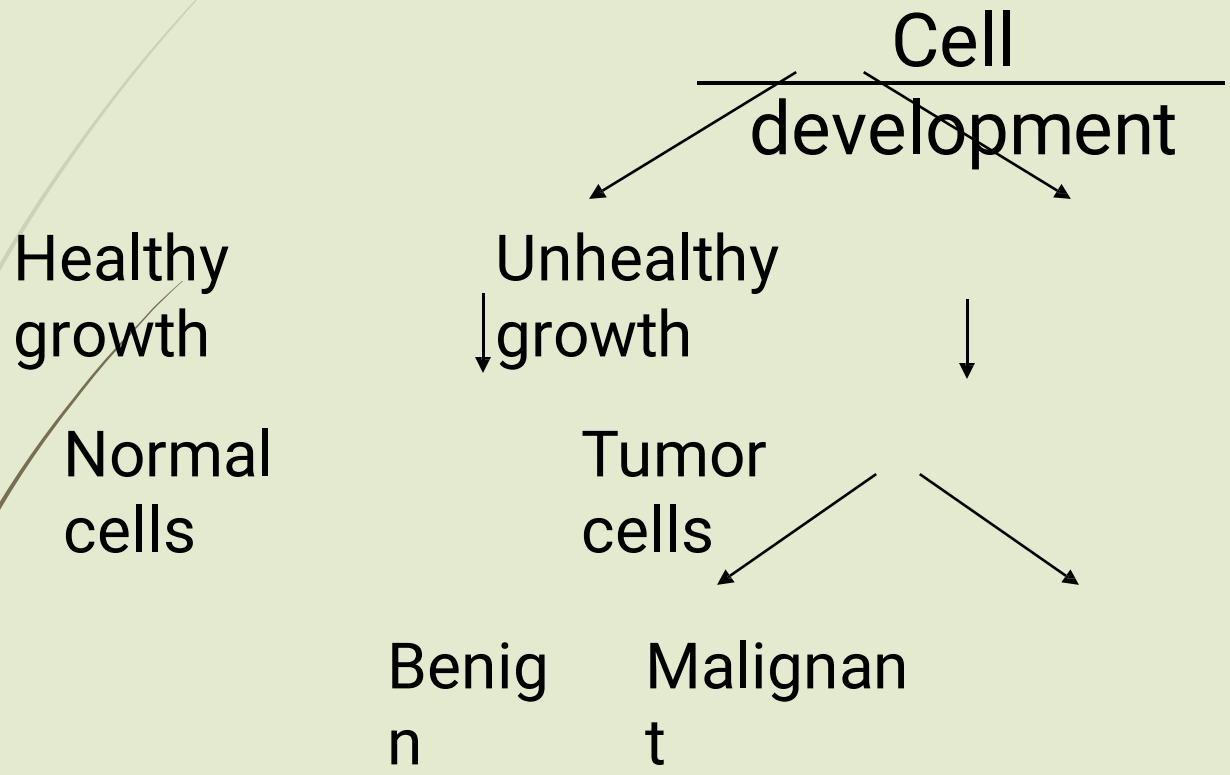
  May recur after removal, sometimes in areas other the original site

  Cells have abnormal chromosomes and DNA characterized by large, dark nuclei; have abnormal shape

  Can secrete substances that cause fatigue and weight loss (paraneoplastic syndrome)

  May require aggressive treatment, including surgery, radiation, chemotherapy, and immunotherapy medications

What is cancer?



Grade Vs stage

The grade of a tumor is an indication of how quickly it is likely to grow and spread, the grade refers to the appearance of the cancer cells under the microscope.

The stage of a solid tumor refers to the extent of the tumor, lymph node involvement, and whether or not it has spread to other organs and tissues. The actual number used to describe a tumor stage – 0, 1, 2, 3, or 4

Cardinal Symptoms of Cancer in Children

1. Unusual mass or swelling
2. Unexplained paleness and loss of energy
3. Sudden tendency to bruise
4. Persistent, localized pain or limping
5. Prolonged, unexplained fever or illness
6. Frequent headaches, often with vomiting
7. Sudden eye or vision changes
8. Excessive, rapid weight loss

Treatment Modalities

The use of multimodal therapy consisting of surgery, chemotherapy, and radiotherapy; enrollment of large numbers of children in cooperative group clinical trials or protocols; and improvements in supportive care have greatly increased the survival of children with cancer. Eighty percent of these patients are now expected to be cured of their disease.

Current efforts are aimed at increasing the survival of patients with high-risk tumors, decreasing the acute and long-term side effects of treatment, and studying the biology of the diseases to better identify patients who are at different risk levels for disease recurrence and can therefore benefit from risk-adapted therapies.

Early Side Effects of Radiotherapy

Site	Effects	Nursing Interventions
Gastrointestinal tract	Nausea and vomiting	Give antiemetic on schedule around the clock. Measure amount of emesis to assess for dehydration.
	Anorexia	Encourage fluids and foods best tolerated, usually light, soft diet and small, frequent meals. Monitor weight.
	Mucosal ulceration	Use frequent mouth rinses and oral hygiene to prevent mucositis.
	Diarrhea	Control with antispasmodics and kaolin pectin preparations. Observe for signs of dehydration.
Skin	Alopecia (within 2 weeks; hair may regrow by 3 to 6 months)	Introduce idea of wig. Stress necessity of scalp hygiene and need for head covering in sun and cold weather.
	Dry or moist desquamation	Do not refer to skin change as a "burn" (implies use of too much radiation). Avoid lotions and other creams to skin. Wash daily, using soap (e.g., Dove) sparingly. Do not remove skin marking for radiation fields. Avoid exposure to sun. For desquamation, consult practitioner for skin hygiene and care.
Head	Nausea and vomiting (from stimulation of vomiting center in brain)	Same as for gastrointestinal tract.
	Alopecia	Same as for skin.
	Mucositis	Encourage regular dental care, fluoride treatments.
	Potential effects: • Parotitis • Sore throat • Loss of taste	Provide analgesics as needed to relieve discomfort.
	Xerostomia (dry mouth)	Combat severe dryness of mouth with oral hygiene and liquid diet.
Urinary bladder	Rarely cystitis	Encourage liberal fluid intake and frequent voiding. Evaluate for hematuria.
Bone marrow	Myelosuppression	Observe for fever (temperature >101° F [38.3° C]). Initiate workup for sepsis as ordered. Administer antibiotics as prescribed. Avoid use of suppositories, rectal temperatures. Institute bleeding precautions. Observe for signs of anemia.

Pediatric Oncology Facts

- Most common childhood cancers:
  - Leukemia (blood)
  - Brain and nervous system
  - The lymphatic system (lymphoma)
  - Kidneys (Wilm's tumor)
  - Bones (osteosarcoma & Ewing's sarcoma)
  - Muscles (rhabdomyosarcoma)



Cancers of Blood and Lymph Systems
Leukemias

Leukemia is a broad term given to a group of malignant diseases of the bone marrow and lymphatic system. It is a complex disease of varying heterogeneity. Consequently, classification has become increasingly complex, sophisticated, and essential because identification of the subtype of leukemia has therapeutic and prognostic implications.

In children, two forms are generally recognized: ALL, Acute lymphoblastic leukemia (ALL) Acute non lymphoblastic leukemia (ANLL) and acute myelogenous leukemia (AML).

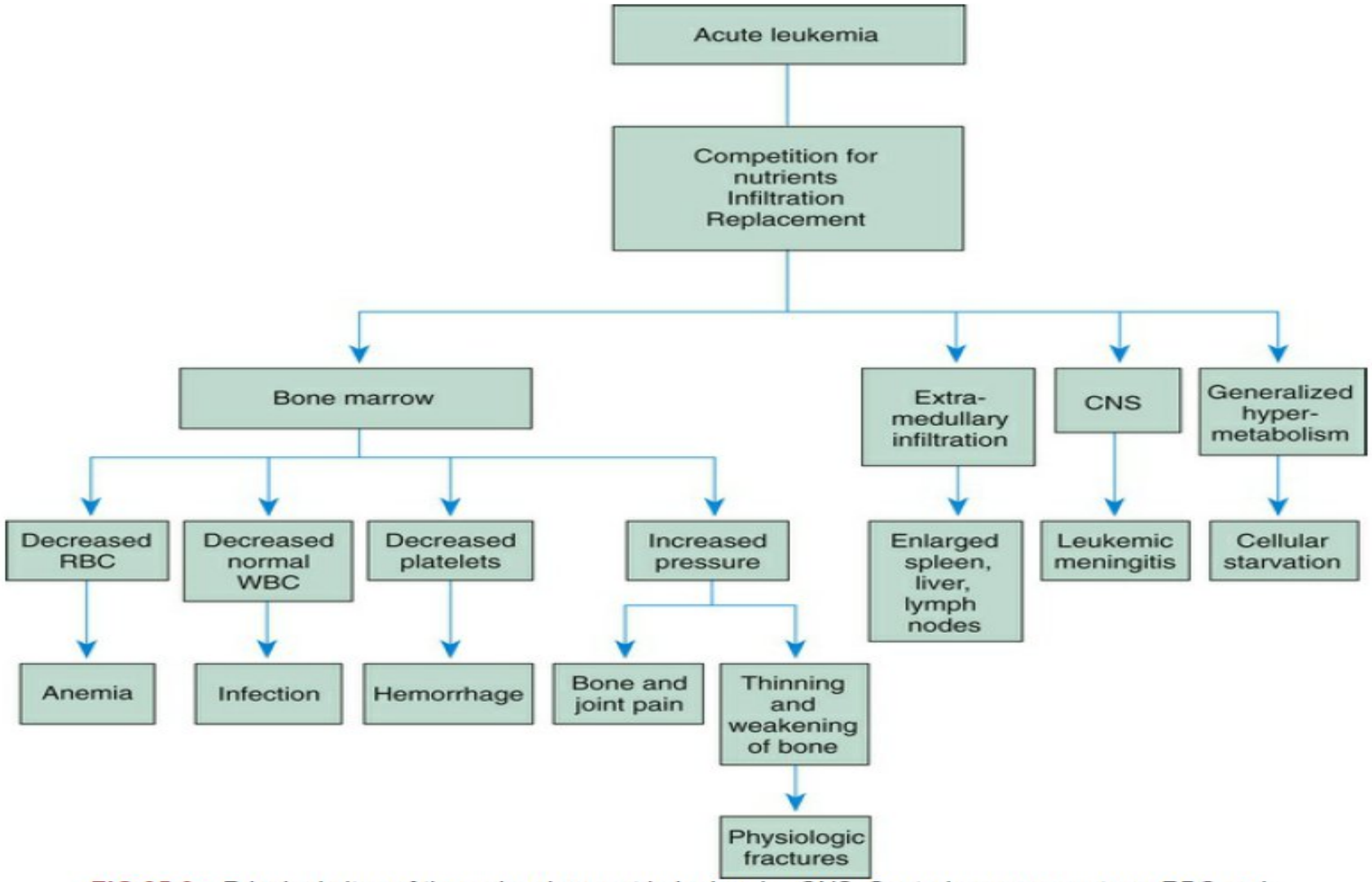


FIG 25-3 Principal sites of tissue involvement in leukemia. *CNS,* Central nervous system; *RBC,* red blood cell; *WBC,* white blood cell.

Acute lymphocyte leukemia

Primary disorder of bone marrow in which the normal marrow elements are replaced by immature or undifferentiated blast cell

It is more common among white and boys

Exact cause is unknown

Viral, genetic and chromosomal factors are suspected

All results from the growth of an abnormal type of non-granular, fragile leukocyte in the blood forming tissues, particularly in the bone marrow, spleen and lymph node

RBC precursors and megakaryocytes are decreased that lead to anemia and tendency to bleeding

Normal WBC are decreased predisposing to infection

Leukemic cells may infiltrate into lymph node, spleen and liver causing adenopathy and hepatosplenomegaly

Expansion of marrow cause bone and joint pain

Invasion of the CNS by leukemic cell causing headache, vomiting, cranial nerve palsies (smell, sight, hearing, equilibrium, and taste), convulsion, coma, papilledema and blurred and double vision

Weight loss, muscle wasting

Diagnosis

Blood tests (Low Hb, RBC, platelet), WBC (Decrease, normal, elevated)

Radiology exams (x-rays, CT, MRI)

Pathology (biopsy of mass)

Bone marrow examination (large number of lymphocyte)

Diagnostic procedures

- Bone marrow aspirate
- Bone marrow biopsy
- Lumbar puncture

Lumber puncture determines CNS involvment

Renal, liver function

Management

Treatment of leukemia involves the use of IV and intrathecal chemotherapeutic agents. Radiation is sometimes used for resistant CNS disease or testicular relapse

Supportive therapy to control such disease complications

Specific therapy is warranted to eradicate malignant cell

Chemotherapy is used to achieve complete remission

- Side effect : pain injection site, allergic reaction, nausea, vomiting, headache, hair loss, bone marrow suppression

Complications

Infection

Hemorrhage

CNS involvement

Bone involvement

Testicular involvement

- Lymphomas

  The lymphomas, a group of neoplastic diseases that arise from the lymphoid and hematopoietic systems, are divided into Hodgkin disease and NHL. These diseases are further subdivided according to tissue type and extent of disease (staging). In children NHL is more common than Hodgkin disease. Although Hodgkin disease is extremely rare before 5 years of age, there is a striking increase in children 15 to 19 years old, when it occurs with almost the same frequency as leukemia.

- Hodgkin Disease

  Hodgkin disease affects about 29 in 1 million children, mostly adolescents. The malignancy originates in the lymphoid system and primarily involves the lymph nodes.

- Clinical Manifestations

  Hodgkin disease is characterized by painless enlargement of lymph nodes. The most common finding is enlarged, firm, nontender, movable nodes in the supraclavicular or cervical area. In children, the sentinel node located near the left clavicle may be the first enlarged node. Enlargement of axillary and inguinal lymph nodes is less frequent.

- Therapeutic Management

  The primary modalities of therapy are chemotherapy and irradiation. Each may be used alone or in combination based on the clinical staging. The goal of treatment is obviously a cure; however, aggressive therapy increases the chances of complications in the disease-free state and can seriously compromise the quality of life

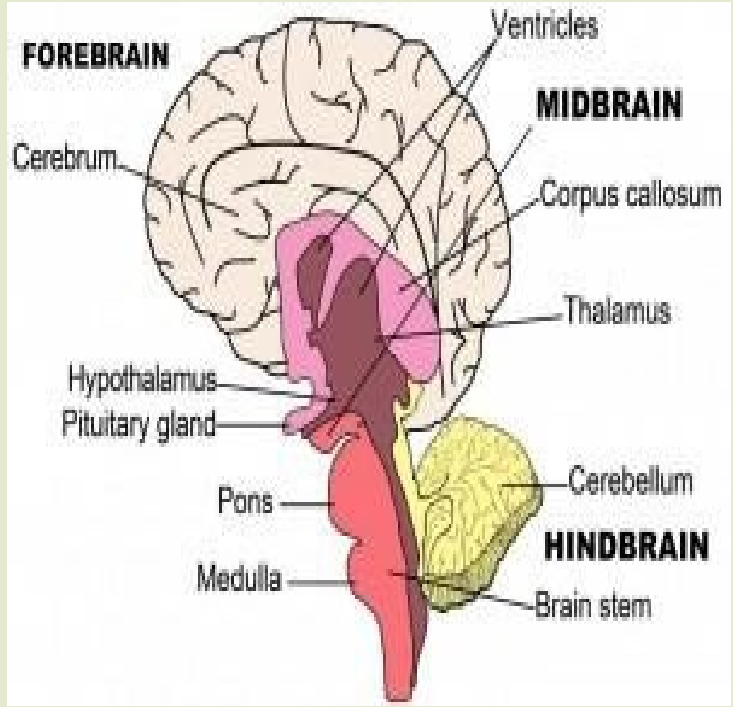
Nervous System Tumors
Brain Tumors

Tumors of the CNS are the most common solid tumor in children and account for about 25% of all childhood cancers, with an annual incidence of 5 per 100,000 children younger than 20 years old.
Expanding lesions within the skull
Cerebellar astrocytoma: cystic tumor of cerebellum

Medulloblastoma: is highly malignant

Brain stem glioma: tumor of brain stem(arises from the glial cells in the brain)

Ependymoma: arise from ependymal cells that line the ventricles of the brain and the center of the spinal cord.



Clinical manifestation cerebellar astrocytoma

The signs and symptoms of brain tumors are directly related to their anatomic location and size and to some extent the child's age

- Increased ICP
- Cerebellar signs: ataxia (gait abnormality), dysmetria (inability of control muscular movement), nystagmus (involuntary eye movement)
- Behavioral change
- Seizures
- Precocious puberty

Clinical Manifestations and Assessment of Brain Tumors

Signs and Symptoms	Assessment
Headache	
Recurrent and progressive In frontal or occipital areas Usually dull and throbbing Worse on arising, less during day Intensified by lowering head and straining, such as during bowel movement, coughing, sneezing	Record description of pain, location, severity, and duration. Use pain rating scale to assess severity of pain (see Chapter 5). Note changes in relation to time of day and activity. Observe changes in behavior in infants (e.g., persistent irritability, crying, head rolling).
Vomiting	
With or without nausea or feeding Progressively more projectile More severe in morning upon arising Relieved by moving about and changing position	Record time, amount, and relationship to feeding, nausea, and activity.
Neuromuscular Changes	
Incoordination or clumsiness Loss of balance (e.g., use of wide-based stance, falling, tripping, banging into objects) Poor fine motor control Weakness Hyporeflexia or hyperreflexia Positive Babinski sign Spasticity Paralysis	Test muscle strength, gait, coordination, and reflexes (see Chapter 4).
Behavioral Changes	
Irritability Decreased appetite Failure to thrive Fatigue (frequent naps) Lethargy Coma Bizarre behavior (e.g., staring, automatic movements)	Observe behavior regularly. Compare observations with parental reports of normal behavioral patterns. Monitor growth and food intake. Monitor activity and sleep.
Cranial Nerve Neuropathy	
Cranial nerve involvement varied according to tumor location Most common signs: • Head tilt • Visual defects (e.g., nystagmus, diplopia, strabismus, episodic "graying out" of vision, visual field defect)	Assess cranial nerves, especially VII (facial), IX (glossopharyngeal), X (vagus), V (trigeminal, sensory roots), and VI (abducens) (see Chapter 4). Assess visual acuity, binocularity, and peripheral vision (see Chapter 4).
Vital Sign Disturbances	
Decreased pulse and respiration Increased blood pressure Decreased pulse pressure Hypothermia or hyperthermia	Measure vital signs frequently. Monitor pulse and respirations for 1 full min. Record pulse pressure (difference between systolic and diastolic blood pressure).
Other Signs	
Seizures Cranial enlargement* Tense, bulging fontanel at rest* Nuchal rigidity Papilledema (edema of optic nerve)	Record seizure activity (see Chapter 27). Measure head circumference daily (infant and young child). Perform funduscopic examination if skilled in procedure.

Medulloblastoma

Similar to manifestation of cerebellar but more rapid

Anorexia, vomiting and headache

Papilledema, drowsiness, increased head circumstances

Cranial nerve palsies

Obstructive hydrocephalus

Diagnosis

Computed tomography (CT)

MRI

Myelogram (uses a contrast medium to detect pathology of the spinal cord)

Positron emission tomography (hree-dimensional images of tracer concentration within the body are then constructed by computer analysis)

Lumber Puncture (LP)

Angiography

Management

Surgery

Radiation therapy as soon as surgical wound healed

Chemotherapy for children less than 4 yr to avoid radiation and for ependymomas

A ventriculopertoneal shunt who develop hydrocephaly

Immunotherapy or gene transfer for future treatment

Complications: brain stem herniation and hydrocephalus

Neuroblastoma

cancer in certain types of nerve tissue.

It most frequently starts from one of the adrenal glands, but can also develop in the neck, chest, abdomen, or spine

Malignant tumor arises from sympathetic CNS

Tumor arises embryonic neural crest cells

Neuroblastoma is one of rare tumor that may demonstrate spontaneous remission

Surgery is for diagnostic and therapeutic

Metastasis to liver, soft tissues, lymph node, none marrow, skin

Rhabdomyosarcoma

Is highly malignant soft tissue tumor

Most common sites are head, neck, GU tract and extremities

Tumor spread by local extension, metastasis by venous or lymph

The lung is the most common site of metastasis

Surgery, radiation and chemotherapy are mixed as treatment

Wilms’ tumor

Malignant renal tumor

Most commonly a unilateral disease

wilms’ tumor has the capacity for rapid growth size before it is diagnosed

Children with wilms’ tumor may have associated with anomali

Surgery, radiation are treatment and chemotherapy

postoperative

Questions ???

Thank You

Pediatric Nursing & Newborn

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Pediatric Primary Care

Pediatric Primary Care

- Pediatric Primary Care includes:
  - Health promotion
  - Disease prevention: immunization, proper nutrition, counseling

- Goal of primary care
  - √ Child wellbeing
  - √ To achieve physical, emotional & developmental health

Health promotion

- Is the process of enabling people to increase control over their health and its determinants, and thereby improve their health

Immunizations

One of the most dramatic advances in pediatrics has been the decline of infectious diseases during the twentieth century because of the widespread use of immunization for preventable diseases.

Although many of the immunizations can be given to individuals of any age, the recommended primary schedule begins during infancy and, with the exception of boosters, is completed during early childhood.

Therefore, health promotion during infancy includes a discussion of childhood immunizations for diphtheria, tetanus, and acellular pertussis (DTaP); poliovirus; measles, mumps, and rubella (MMR); Haemophilus influenzae type b (Hib); hepatitis B virus (HBV); hepatitis A virus (HAV); meningococcal; pneumococcal conjugate vaccine (PCV); influenza (and H1N1); and varicella-zoster virus (VZV; chickenpox)

Reduce morbidity & mortality from infectious diseases

The patient, parent, or legal guardian should be informed about the benefit & risk of immunization

Routine immunization are started in infancy.

Immunization may be started at any age and slightly different schedule may be followed according to age

Interrupted immunizations doesn’t need to be restarted

Key Immunization Terms

Immunity: An inherited or acquired state in which an individual is resistant to the occurrence or the effects of a specific disease, particularly an infectious agent

Immunization: Inclusive term denoting the process of inducing or providing active or passive immunity artificially by administering an immunobiologic

Natural immunity: Innate immunity or resistance to infection or toxicity

Passive immunity: Temporary immunity obtained by transfusing immunoglobulins or antitoxins either artificially from another human or an animal that has been actively immunized against an antigen or naturally from the mother to the fetus via the placenta

Herd immunity: A condition in which the majority of the population community is vaccinated and the spread of certain diseases is stopped, because the population that has been vaccinated protects those in the same population who are unvaccinated

Acquired immunity: Immunity from exposure to the invading agent, either bacteria, virus, or toxin

Active immunity: A state where immune bodies are actively formed against specific antigens, either naturally by having had the disease clinically or sub clinically or artificially by introducing the antigen into the individual

Contraindications & precautions

Anaphylactic reaction

Moderate to severe illness with or without fever

All live vaccine (OPV, MMR, Varicella) are contraindicated in pregnancy, immunosuppression

IPV: anaphylactic to neomycin, streptomycin, polymyxcin B

Influenza: anaphylactic to egg protein

HBV: anaphylactic reaction to baker's yeast

Conditions can still be immunized

Mild acute illness

Reaction to previous DPT with only soreness, redness, swelling immediately with fever lower than 40C

Prematurity

Malnutrition

Allergy to penicillin

Person using steroid less than 14 days

Recent exposed to infectious disease

Vaccine administration consideration

Strict adherence to manufacturing storage

Health care provider should be immunized against measles, mumps, rubella, HBV, DPT.

Wear gloves

Sterile & disposable needle should be discarded promptly.

Don’t recap needle

Before administrating a subsequent dose, question parent about adverse reaction of previous dose

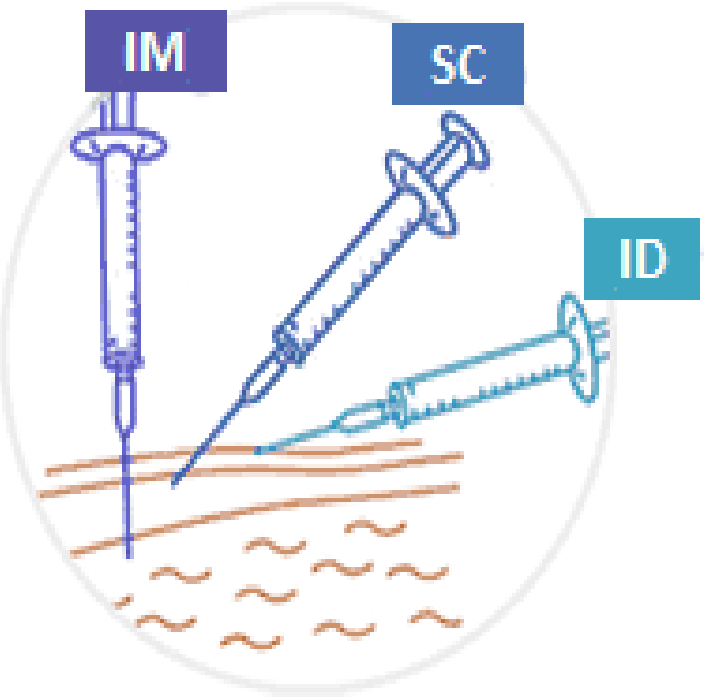
Vaccinations

- BCG (bacille Calmette-Guerin)
- DPT, DTaP
- Measles
- MMR
- Polio
- HBV
- Pneumococcal
- Meningococcal

- Tuberculin skin test ??

- **Oral**
  - OPV
  - Rotavirus
- **Intramuscular (IM)**
  - DTwP, DTaP, DT, Td, TT
  - Hepatitis B
  - IPV
  - Hib
  - PCV-7
- **Subcutaneous (SC)**
  - Measles
  - Yellow fever
- **Intradermal (ID)**
  - BCG





Hepatitis B Virus

HBV is a significant pediatric disease because HBV infections that occur during childhood and adolescence can lead to fatal consequences from cirrhosis or liver cancer during adulthood

It is recommended that newborns receive HepB before hospital discharge if the mother is hepatitis B surface antigen (HBsAg) negative

Both full-term and preterm infants born to mothers whose HBsAg status is positive or unknown should receive HepB and hepatitis B immune globulin (HBIG), 0.5 ml, within 12 hours of birth at two different injection sites.

The vaccine is given intramuscularly in the vastus lateralis in newborns or in the deltoid for older infants and children

Diphtheria

Respiratory manifestations include respiratory nasopharyngitis or obstructive laryngotracheitis with upper airway obstruction.

Diphtheria vaccine is commonly administered (1) in combination with tetanus and pertussis vaccines (DTaP) or DTaP and Hib vaccines for children younger than 7 years old, (2) in combination with a conjugate Hib vaccine, (3) in a combined vaccine with tetanus (DT) for children younger than 7 years old who have some contraindication to receiving pertussis vaccine, (4) in combination with tetanus and acellular pertussis (Tdap) for children 11 years old and older, or (5) as a single antigen when combined antigen preparations are not indicated.

Polio

An all-IPV (inactivated poliovirus vaccine) schedule for routine childhood polio vaccination is now recommended for children. All children should receive four doses of IPV at 2 months old, 4 months old, 6 to 18 months old, and 4 to 6 years old.

The change from the exclusive use of oral polio vaccine (OPV) to the exclusive use of IPV is related to the rare risk of vaccine-associated polio paralysis (VAPP) from OPV.

MMR, Measles, Mumps, Rubella

Given between age 12-15 months

Transient skin rash and fever up to wks

Women should avoid pregnancy within 3 months of vaccination

Haemophilus influenzae Type B

Hib conjugate vaccines protect against a number of serious infections caused by H. influenza type b, especially bacterial meningitis, epiglottitis, bacterial pneumonia, septic arthritis, and sepsis (Hib is not associated with the viruses that cause influenza, or “flu”)

Administered in a four dose series at 2, 4, 6, and 12 to 15 months old.

All Hib vaccines are administered by intramuscular injection using a separate syringe and at a site separate from any concurrent vaccinations.

BCG

Is a vaccine for tuberculosis (TB) disease.

Contraindications in immunosuppression, pregnancy.

Testing for TB in BCG-Vaccinated Persons

There are common side effect:

- Accumulation of pus
- peeling or scaling of the skin
- sores at place of injection

Tuberculin skin test (Mantoux tuberculin test)

- Is done to see if you have ever been exposed to tuberculosis (TB). The test is done by putting a small amount of TB protein (antigens) under the top layer of skin on your inner forearm. If you have ever been exposed to the TB bacteria (Mycobacterium tuberculosis), your skin will react to the antigens by developing a firm red bump at the site within 2 days.

Breast-feeding

- Natural and ideal nourishment
- Immunologic and anti-infection properties
- Prevent GI disturbance
- Reduce allergies development
- Provide psychological and emotional satisfaction (child-mother bond)
- Breast milk cab be frozen up to 6 months

Causes of infant and neonatal mortality

Infant Mortality

The infant mortality rate is the number of deaths during the first year of life per 1000 live births. (Number of deaths among infants under one year old per 1,000 live births in a given year)

It may be further divided into neonatal mortality (<28 days of life) and post neonatal mortality (28 days to 11 months).

Worldwide, approximately 10 million infants die each year, more than 90% of these infant deaths occur in the developing world

Worldwide, approximately 10 million infants die each year. In poorer countries, estimated 10-20% of all infants die before their first birthday

Only a small proportion occur in the U.S.

- 2007 - 6.4 per 1,000;
- 2004 - 6.8 per 1,000;
- 2002 - 7.0 per 1,000;
- 28,000 infant deaths and 4,059,000 live births in the U.S.

Infant Mortality Rate and Percentage of Total Deaths for 10 Leading Causes of Infant Death in 2013 (Rate per 1000 Live Births)

Rank	Cause of Death (Based on International Classification of Diseases, 10th Revision)	Percent	Rate
	All races, all causes	100.00%	596.1
1	Congenital anomalies	20.3	121
2	Disorders relating to short gestation and unspecified low birth weight	17.9	106.9
3	Newborn affected by maternal complication of pregnancy	6.7	39.7
4	Sudden infant death syndrome	6.1	40.6
5	Accidents (unintentional injuries)	4.9	29.4
6	Newborn affected by complications of placenta, cord, and membranes	4.1	24.2
7	Bacterial sepsis of newborn	2.5	14.7
8	Respiratory distress of newborn	2.2	13.3
9	Diseases of circulatory system	2.0	11.6
10	Neonatal hemorrhage	2.1	9.9

Activate Windows

Causes of Neonatal Mortality

- Developed
- Neonatal
    - LBW
    - Congenital anomalies
    - Maternal complications

- Developing
- Neonatal
    - LBW
    - Trauma/asphyxia
    - Tetanus
    - Pneumonia

Causes of Infant Mortality

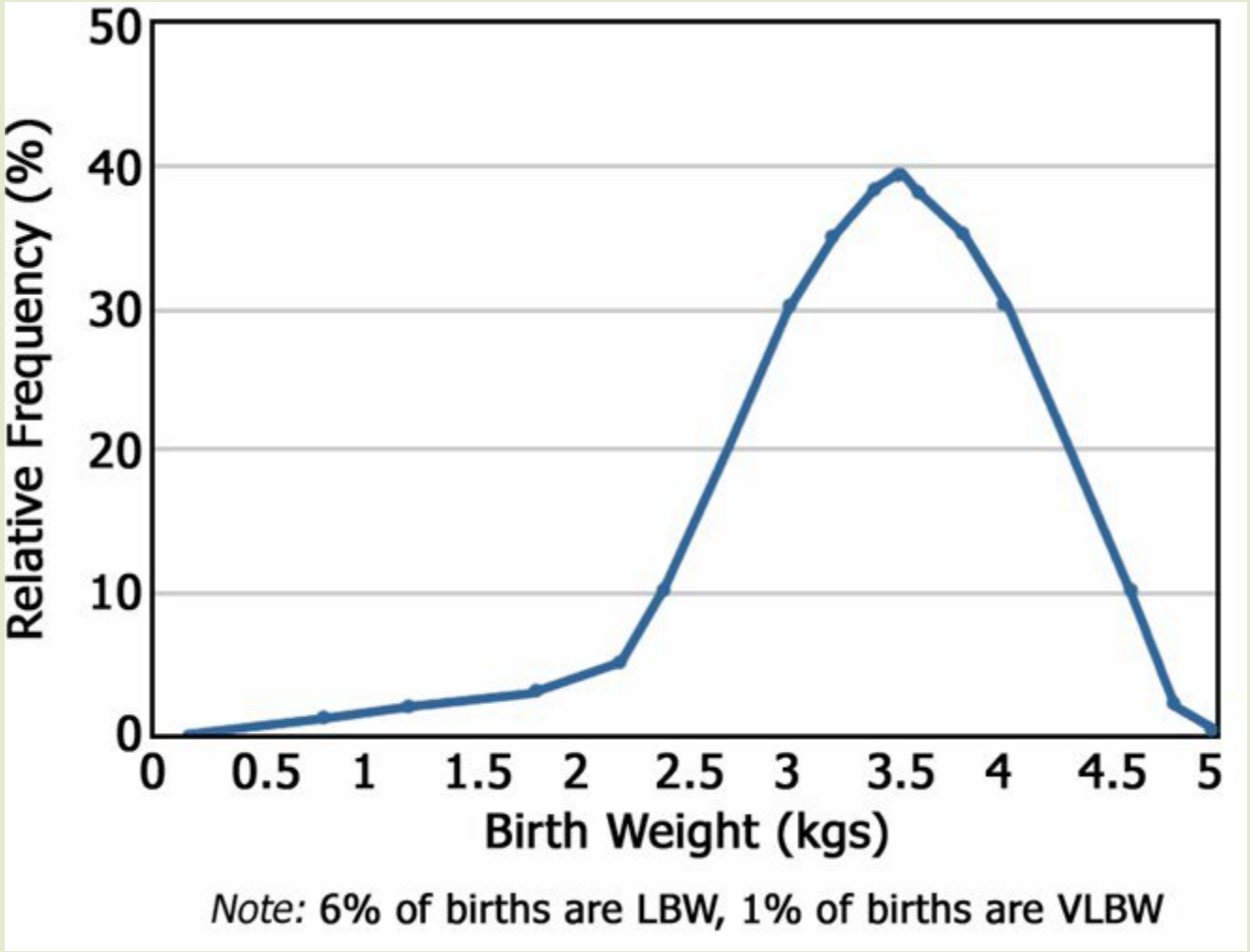
- Developed
- Post neonatal
  - SIDS
  - Congenital anomalies
  - Injuries
  - Infection

- Developing
- Post neonatal
  - Infection
  - Malnutrition
  - Injury

Five Leading Causes of Death in Children in the United States: Selected Age Intervals, 2013 (Rate per 100,000 Population)

	1 TO 4 YEARS OLD		5 TO 9 YEARS OLD		10 TO 14 YEARS OLD		15 TO 19 YEARS OLD	
Rank	Cause	Rate	Cause	Rate	Cause	Rate	Cause	Rate
	All causes	25.5	All causes	11.8	All causes	14.1	All causes	44.8
1	Accidents	8.3	Accidents	3.6	Accidents	3.8	Accidents	17.3
2	Congenital anomalies	3.0	Cancer	2.2	Cancer	2.2	Suicide	8.3
3	Homicide	2.1	Congenital anomalies	0.9	Suicide	1.9	Homicide	6.6
4	Cancer	2.1	Homicide	0.6	Congenital anomalies	0.8	Cancer	3.0
5	Heart disease	1.1	Respiratory Disease	0.4	Homicide	0.7	Heart disease	1.4

Distribution of Birth Weight in the U.S.



Characteristics of Prematurity

Weight < 2500 gram

Little body fat

Little scalp hair

Lots of lanugo

Weak cry and tone

Causes of Prematurity

- Maternal factors
  - Preeclampsia
  - Chronic medical illness
  - Infection (B streptococcus, UTI,--)
  - Drug use
  - Abnormal structure of uterus
  - Cervical
- Pregnancy factors
  - Placenta previa
  - Abruption placenta
  - Premature rupture of membrane
- Factors involving the fetus
  - Multiple gestation
  - Chromosomal abnormalities
  - Anatomic abnormalities
  - Feto-placetal dysfunction

Problems may be experience perterm

- Temperature instability
- Respiratory problem
- Respiratory distress syndrome due to lack of surfactant lead to air sac cant stay open
- Incomplete lung development
- Cardiovascular disorder
- Jaundice
- Difficulty feeding (unable to suck & swallow properly before 35 wk)

- Intra ventricual hemorrhage (brain bleeding)
- Poor muscle tone
- Reflexes are weak
- Susceptible to infection
- Less active
- Anemia

Preterm Management

Monitoring vital signs and oxygen level

Mechanical ventilation if needed

IV fluid when feeding cant be given

Special feeding

Medication for any complication, antibiotic

Categories of Gestational Age

- Pre-term

  Infants born through the 37th week

- Term infants

  Infants born between the first day of the 38th week and the last day of the 42nd week of gestation

- Post-term

  Infants born from the beginning of the first day of the 43rd week onward

Post Term Babies

Infants born after 42 weeks

Primigravida, history of previous Post trem or high parity are predisposing factors

Post-mature may suffer from intrauterine and hypoxia

Clinical manifestation of post-mature

Wasted physical appearance

Long thin appearance

Long fingernail and toenails

Minimum vernix caseosa

Abundant scalp hair

Cracked skin & desquamation

Absence of lanugo hair

Complications of post-mature

Meconium aspiration

Hypoglycemia

Hypocalcemia

Pneumonia

Polycythemia

loss of subcutaneous fat occur in some.

Congenital Neuromuscular or Muscular Disorders

Cerebral palsy

A “group of permanent disorders of the development of movement and posture, causing activity limitation, that are attributed to nonprogressive disturbances that occurred in the developing fetal or infant brain”.

In addition to motor disorders, the condition often involves disturbances of sensation, perception, communication, cognition, and behavior; secondary musculoskeletal problems; and epilepsy

CP is the most common permanent physical disability of childhood

The insult to the brain is believed to occur between the time of conception and age 2 years, at which time a significant amount of motor development has already occurred.

In general, infants exposed to maternal and perinatal infections are at increased risk for the development of CP as a result of the effects on the developing brain.

Etiology

- Prenatal

fetus (most commonly genetic disorders),

mother (seizure disorders, mental retardation, and previous pregnancy loss),

pregnancy itself (Rh incompatibility, polyhydramnios, placental rupture, and drug or alcohol exposure).

External factors, such as TORCH (toxoplasmosis, other agents, rubella, cytomegalovirus, herpes simplex).

- Perinatal

asphyxia or trauma that occurs during labor.

Oxytocin augmentation, umbilical cord

prolapse breech presentation.

Low-birth-weight infants (<1500 g) incidence of 60 per 1000 births compared with two per 1000 births in normal-weight infants.

Pregnancies involving multiple births

- Postnatal

Hypoxic-ischemic encephalopathy,

Infections such as encephalitis and meningitis

Traumatic brain injury

Box 30-1

Clinical Classification of Cerebral Palsy

Spastic (Pyramidal)

Characterized by persistent primitive reflexes, positive Babinski reflex, ankle clonus, exaggerated stretch reflexes, eventual development of contractures

- 70% to 80% of all cases of cerebral palsy (CP)
- Diplegia: All extremities affected; lower more than upper (30% to 40% of spastic CP)
- Tetraplegia: All four extremities involved—legs and trunk, mouth, pharynx, and tongue (10% to 15% of spastic CP)
- Triplegia: Three limbs involved
- Monoplegia: Only one limb involved
- Hemiplegia: Motor dysfunction on one side of the body; upper extremity more affected than lower (20% to 30% of spastic CP)

Other features:

- Hypertonicity with poor control of posture, balance, and coordinated motion
- Impairment of fine and gross motor skills

Dyskinetic (Nonspastic, Extrapyramidal)

Athetoid: Chorea (involuntary, irregular, jerking movements); characterized by slow, wormlike, writhing movements that usually involve the extremities, trunk, neck, facial muscles, and tongue

Dystonic: Slow, twisting movements of the trunk or extremities; abnormal posture

Involvement of the pharyngeal, laryngeal, and oral muscles causing drooling and dysarthria (imperfect speech articulation)

Ataxic (Nonspastic, Extrapyramidal)

Wide-based gait

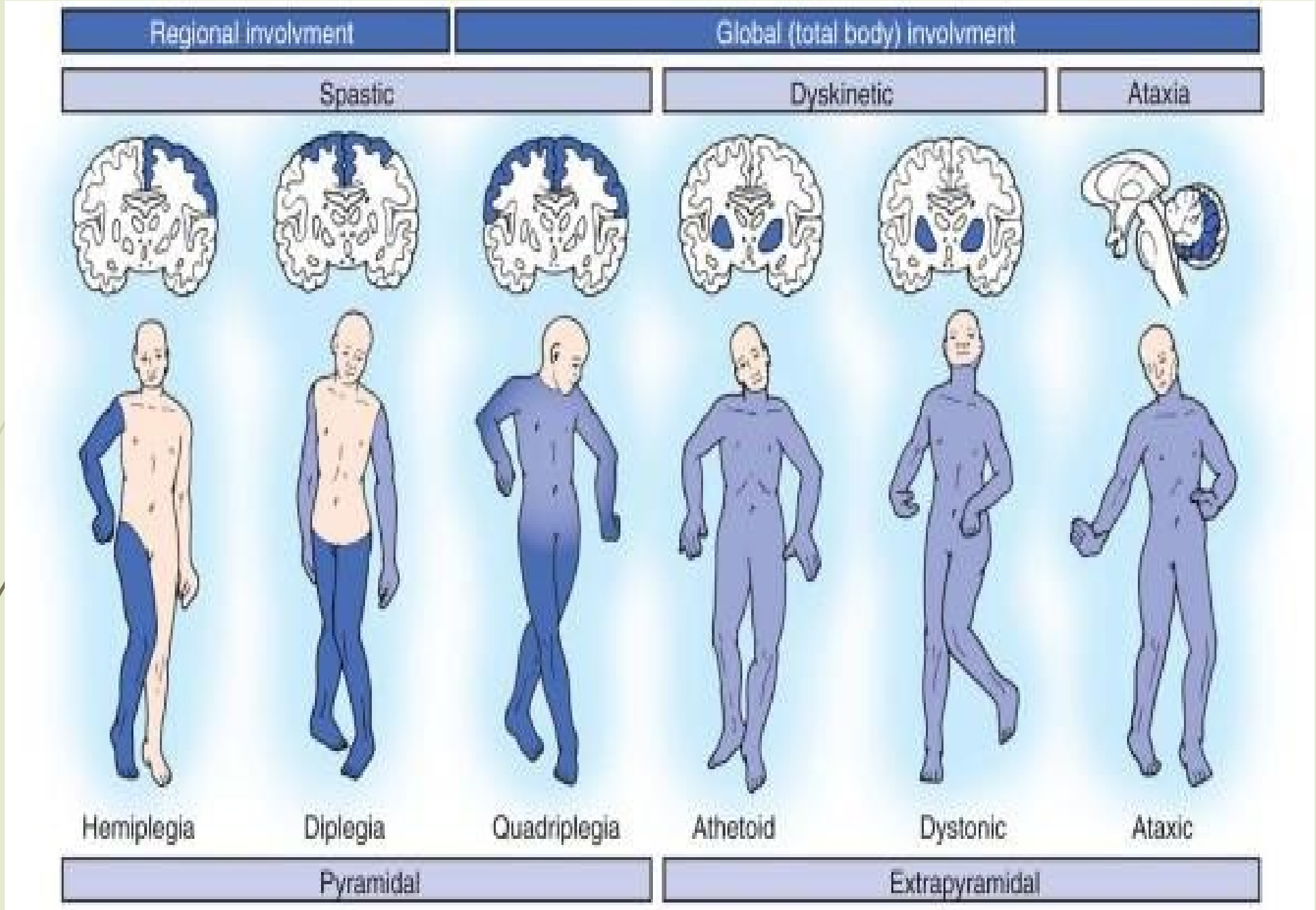
Rapid, repetitive movements performed poorly

Disintegration of movements of the upper extremities when the child reaches for objects

Mixed Type

Combination of spastic CP and dyskinetic CP

May be labeled *mixed* when no specific motor pattern is dominant; however, this term is losing favor to more precise descriptions of motor function and affected area of brain involved


Regional involvment
Global (total body) involvment
Spastic
Dyskinetic
Ataxia
Hemiplegia
Diplegia
Quadriplegia
Athetoid
Dystonic
Ataxic
Pyramidal
Extrapyramidal

Box 30-3

Early Signs of Cerebral Palsy

- Failure to meet any developmental milestones, such as rolling over, raising head, sitting up, crawling
- Persistent primitive reflexes, such as Moro, atonic neck
- Poor head control (head lag) and clenched fists after 3 months old
- Stiff or rigid arms or legs; scissoring legs
- Pushing away or arching back; stiff posture
- Floppy or limp body posture, especially while sleeping
- Inability to sit up without support by 8 months old
- Using only one side of the body or only the arms to crawl
- Feeding difficulties
- Persistent gagging or choking when fed
- After 6 months old, tongue pushing soft food out of the mouth
- Extreme irritability or crying
- Failure to smile by 3 months old
- Lack of interest in surroundings

ASSOCIATED DISABILITIES

Seizures- generalized tonic-clonic;more in postnatally acquired hemiplegia

Drooling- may occur and lead to wet clothing/skin irritation

Feeding- alterations in muscle tone lead to difficulties chewing, swallowing, talking, etc.

Address nutritional concerns.

Coughing, choking may lead to aspiration.

Altered respiratory patterns may lead to inadequate gas exchange.

Diagnostic Studies

Physical Assessment

Observe LBW, preterm, and those with low Apgar scores at 5 minutes.

Observe infants who have seizures, intracranial hemorrhage, metabolic disturbances

Therapeutic Management

The goals of therapy for children with CP are early recognition and promotion of optimal development to enable affected children to attain normalization and realize their potential within the limits of the existing health problems. The disorder is permanent, and therapy is primarily preventive and symptomatic.

Each child is evaluated and managed on an individual basis. The plan of therapy may involve a variety of settings, facilities, and specially trained persons. The scope of the child's needs requires multidisciplinary planning and care coordination among professionals and the child's family.

The outcome for the child and family with CP is normalization and promotion of self-care activities that empower the child and family to achieve maximum potential.

Surgical Intervention

Correct contractures or spastic deformities
Provide stability for uncontrolled joint
Tendon-lengthening procedures

Hydrocephalus



Hydrocephalus is a condition caused by an imbalance in the production and absorption of CSF in the ventricular system.

An abnormal increase in the amount of cerebrospinal fluid within the cranial cavity that is accompanied by expansion of the cerebral ventricles, enlargement of the skull and especially the forehead, and atrophy of the brain

Normal CSF production: 20 ml/h.

The CSF volume of an average adult ranges from 80 to 160 ml

The causes of hydrocephalus are varied, include (1) impaired absorption of CSF fluid within the subarachnoid space, (2) obstruction to the flow of CSF through the ventricular system

Any imbalance of secretion and absorption causes an increased accumulation of CSF in the ventricles, which become dilated

Etiology

Obstruction in the system

Congenital

Acquired: infection, trauma, hemorrhage, tumor

Failure in absorption system

Excessive CFS production

Clinical manifestation

In infancy, before closure of the cranial sutures, head enlargement (increasing occipitofrontal circumference) is the predominant sign, but in older infants and children, the lesions responsible for hydrocephalus produce other neurologic signs through pressure on adjacent structures.
Excessive head growth

Delayed closure of anterior fontanel

Sign of increased ICP

Vomiting

Restlessness

Increased systolic pressure, decreased pulse

Seizure

Therapeutic Management

The treatment of hydrocephalus is directed toward relief of ventricular pressure, treatment of the cause of the ventriculomegaly, treatment of associated complications, and management of problems related to the effect of the disorder on psychomotor development.

Most children require placement of a shunt that provides primary drainage of the CSF from the ventricles to an extracranial compartment, usually the peritoneum (ventriculoperitoneal [VP] shunt)

The two most commonly used shunt systems are:

- Ventriculoatrial (VA)
- Ventriculoperitoneal (VP) shunts.

  The VP shunt is most commonly used as it is simpler to place, extra tubing may be placed in the peritoneum and the consequences of infection are less.

Medication Therapy

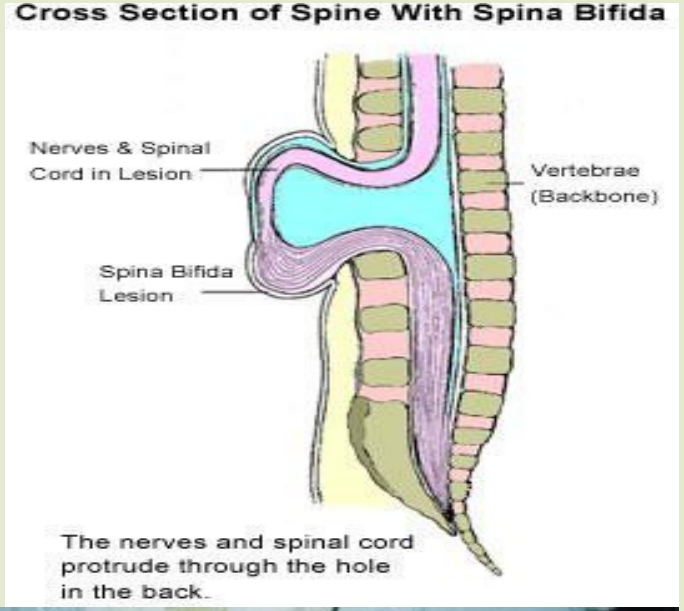
Little usefulness

Anti-anxiety agents may relieve excessive motion and tension (child with athetosis)

- Skeletal muscle relaxants (methocarbamol (Tobaxin), dantrolene (Dantrium), Baclofen, may be used short-term for older children and adolescents.
- Diazepam (Valium) for older children and adolescents, may relieve stiffness and ease motion

Spina bifida

- Malformation of the spine in which posterior portion of lamina fail to close





Types of Spina Bifida

1. Spina Bifida Occulta
   - An abnormality is confined to the vertebrae only and is due to an unclosed posterior vertebral arch.
   - Refers to a defect that is not visible externally. It occurs most frequently in the lumbosacral area (L5 and S1)

2. Spina Bifida Cystica
   - Refers to a visible defect with an external saclike protrusion. The two major forms of SB cystica are meningocele, which encases meninges and spinal fluid but no neural elements and myelomeningocele (or meningomyelocele), which contains meninges, spinal fluid, and nerves.
   - A more severe type of spina bifida
   - Meningocele – Where the meninges protrude through the defect. (4%)
   - Myelomeningocele – Elements of the cord also protrude through the defect, resulting in severe neural deficits. (96%)

.

A, Myelomeningocele with an intact sac. B, Myelomeningocele with a ruptured sac.

Causes of Spina Bifida

Unknown cause

Genetics and environmental factors.

Valporic acid during pregnancy

Low folic acid consumption by the mother during pregnancy.

Diagnostic Evaluation

The diagnosis of SB is made on the basis of clinical manifestations examination of the meningeal sac. Diagnostic measures used to evaluate the brain and spinal cord include MRI, ultrasonography, and CT.

A neurologic evaluation will determine the extent of involvement of bowel and bladder function as well as lower extremity neuromuscular involvement.

Flaccid paralysis of the lower extremities is a common finding with absent deep tendon reflexes.

Signs and Symptoms

Swelling
Dimple in skin
Truft of hair
Muscle weakness
Paralysis
Loss of a sensation
Fluid build up (hydrocephalus)
Brain damage
Seizures
Blindness
Possibly none at all

Therapeutic Management



Management of the child who has a myelomeningocele requires a multidisciplinary team approach involving the specialties of neurology, neurosurgery, pediatrics, urology, orthopedics, rehabilitation, physical therapy, occupational therapy, and social services, as well as intensive nursing care in a variety of specialty areas.

A variety of neurosurgical and plastic surgical procedures are used for skin closure without disturbing the neural elements or removing any portion of the sac.

No cure

Regular check ups with physician

Surgery (24 hours after birth)

Physiotherapy

Prevention

The Centers for Disease Control and Prevention (2009) continues to affirm that 50% to 70% of NTDs can be prevented by daily consumption of 0.4 mg of folic acid among women of childbearing age.

Meningitis

- Bacterial Meningitis

Bacterial meningitis is an acute inflammation of the meninges and CSF. Suspected bacterial meningitis is a medical emergency, and immediate action must be taken to identify the causative organism and to initiate prompt treatment.

The advent of antimicrobial therapy has had a significant effect on the course and prognosis of children with bacterial meningitis

The most common types of bacteria are:

- Streptococcus pneumoniae,

  found in the resp. tract, sinuses, and nasal cavity “pneumococcal meningitis”

- Neisseria meningititis

  spread through saliva and other respiratory fluids and causes what’s called “meningococcal meningitis”

  Severe meningococcal and pneumococcal infections may result in hemorrhaging of the adrenal glands, leading to Waterhouse-Friderichsen syndrome, which is often fatal

- Haemophilus influenza

  cause not only meningitis but infection of the blood, inflammation of the windpipe, cellulitis, and infectious arthritis

- Listeria monocytogenes

  which is a foodborne bacteria

Clinical Manifestations of Bacterial Meningitis

Children and Adolescents

Usually abrupt onset

Fever

Chills

Headache

Vomiting

Alterations in sensorium

Seizures (often the initial sign)

Irritability

Agitation

May develop:

- Photophobia
- Delirium
- Hallucinations
- Aggressive behavior
- Drowsiness

Infants and Young Children

Classic picture (above) rarely seen in children between 3 months and 2 years old

Fever

Poor feeding

Vomiting

Marked irritability

Frequent seizures (often accompanied by a high-pitched cry)

Bulging fontanel

Nuchal rigidity possible

Brudzinski and Kernig signs not helpful in diagnosis

Difficult to elicit and evaluate in this age group

Subdural empyema (*H. influenzae* infection)

Neonates

Specific Signs

Child well at birth but within a few days begins to look and behave poorly

Refuses feedings

Poor sucking ability

Vomiting or diarrhea

Poor tone

Lack of movement

Weak cry

Full, tense, and bulging fontanel may appear late in course of illness

Neck usually supple

Nonspecific Signs That May Be Present

Hypothermia or fever (depending on the infant's maturity)

Jaundice

Irritability

Drowsiness

Seizures

Respiratory irregularities or apnea

Cyanosis

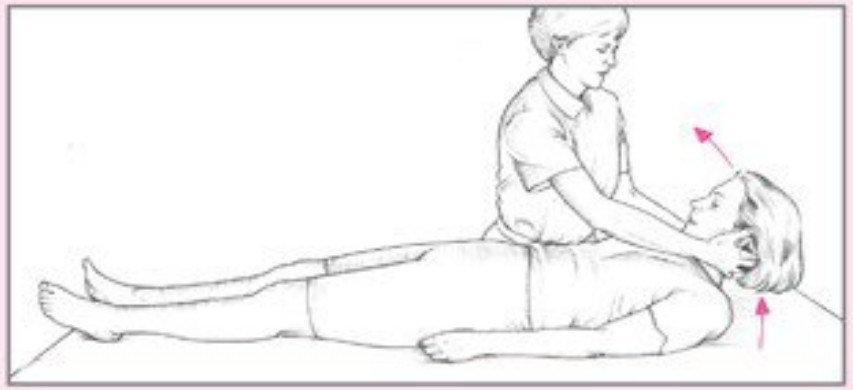
Weight loss

ASSESSMENT TIP

Testing for Brudzinski's sign

Here's how to test for Brudzinski's sign when you suspect meningeal irritation:

With the patient in a supine position, place your hands behind her neck and lift her head toward her chest (as shown at right).



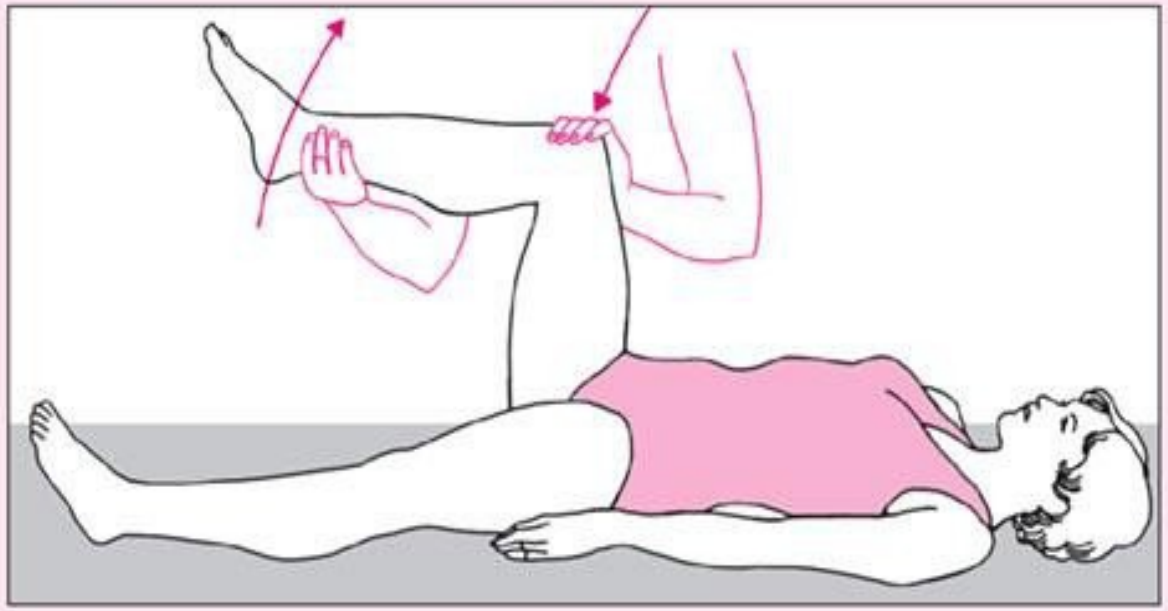
If your patient has meningeal irritation, she'll flex her hips and knees in response to the passive neck flexion (as shown at right).



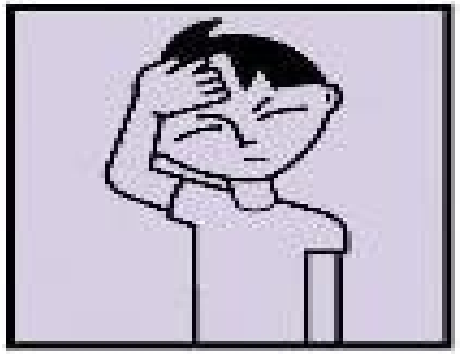
EXAMINATION TIP

Eliciting Kernig's sign

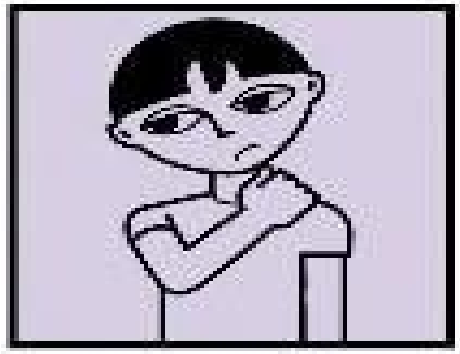
To elicit Kernig's sign, place the patient in a supine position. Flex her leg at the hip and knee, as shown here. Then try to extend the leg while you keep the hip flexed. If the patient experiences pain and possibly spasm in the hamstring muscle and resists further extension, you can assume that meningeal irritation has occurred.



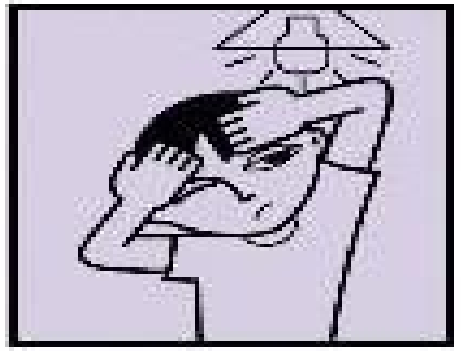
Meningitis: older children



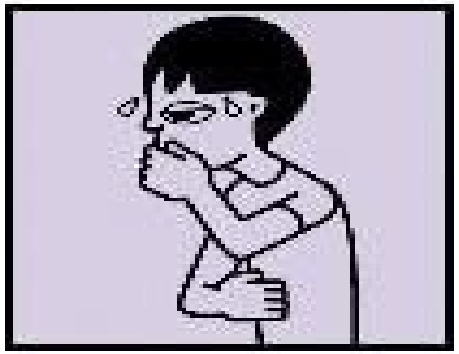
Severe headache



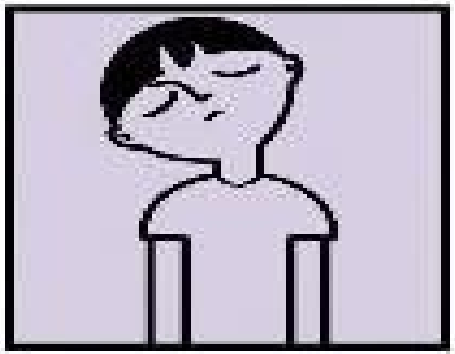
Stiff neck



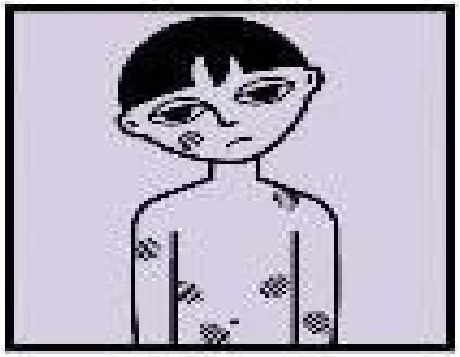
Dislike of bright lights



Fever/vomiting

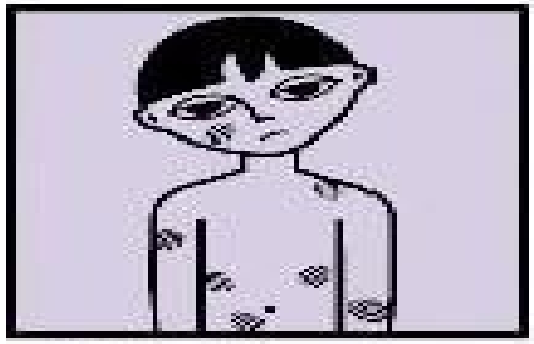


Drowsy and less responsive/ vacant

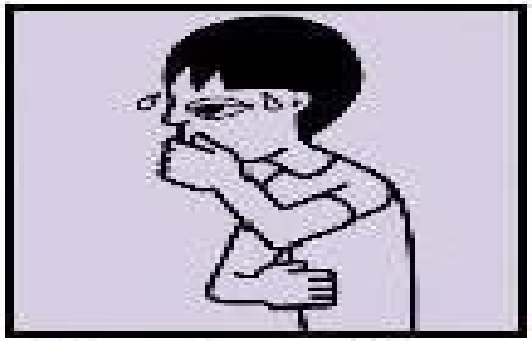


Rash (develops anywhere on body)

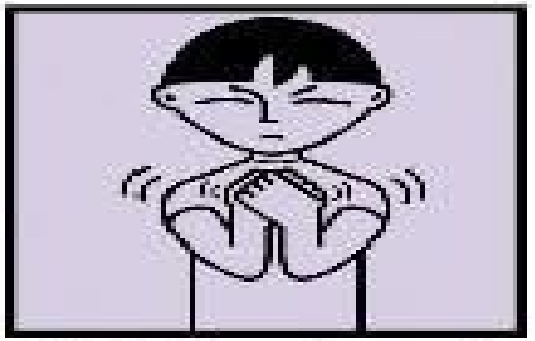
Septicaemia



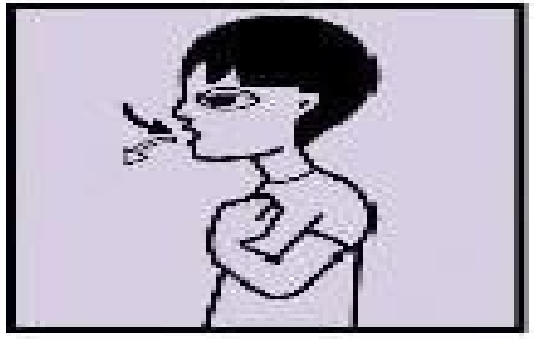
Rash (develops anywhere on body)



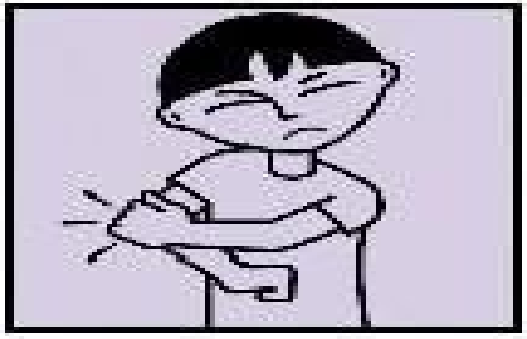
Fever/vomiting



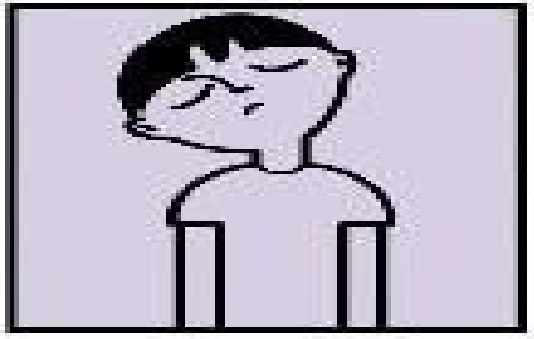
Cold hands and feet/shivering



Rapid breathing (or unusual breathing)



Stomach/joint/ muscle pain



Drowsy and less responsive/ vacant

Diagnostic Evaluation



A lumbar puncture is the definitive diagnostic test for meningitis. The fluid pressure is measured, and samples are obtained for culture, Gram stain, blood cell count, and determination of glucose and protein content. These findings are usually diagnostic. Culture and sensitivity testing are needed to identify the causative organism. Spinal fluid pressure is usually elevated, but interpretation is often difficult when the child is crying

Lumber puncture (CSF)

- Bacterial: Turbid, increased CSF pressure, high cell acount,low glucose level, culture positive
- Viral: lymphcyte

CBC

- increased WBC
- Platelet count
- Hyponatremia, hypochloraemia
- Serum glucose reduced
- T.B skin test

A blood culture is advisable for all children suspected of having meningitis and occasionally will be positive when CSF culture is negative. Nose and throat cultures may provide helpful information in some cases.

Therapeutic Management

- Acute bacterial meningitis is a medical emergency that requires early recognition and immediate therapy to prevent death and avoid residual disabilities. The initial therapeutic management includes:

  Isolation precautions

  Initiation of antimicrobial therapy

  Maintenance of hydration

  Maintenance of ventilation

  Reduction of increased ICP

  Management of systemic shock

  Control of seizures

  Control of temperature

  Treatment of complications

Bacterial Meningitis Management

Medical emergency

Early diagnosis essential

Immediate optimum treatment

Intensive supportive therapy

Rehabilitation

Prophylaxis to family

Antibiotics

- <u>Less than 2 months of age:</u>
  Ampicillin + Cefotaxime+/-
  Gentamicin
  Treat for 3 weeks (neonate)

- <u>Over 2 months:</u>
  Cefotaxime
  Treat for 7-10 days
  Dexamethasone to avoid hearing loss

Prophylaxis

- Rifampicin:

  Children 5mg/kg bd x 2/7

  Adults: 600 mg bd x 2/7

- Pregnant contact:

  Cefuroxime IM x 1 dose

Meningitis - Complications

Septic shock - DIC

Cerebral oedema

Seizures

Arteritis/venous thrombosis

Subdural effusions

Hydrocephalus .

Abscess

Brain damage

Deafness

Nonbacterial (Aseptic) Meningitis (Viral meningitis)

The term aseptic meningitis refers to the onset of meningeal symptoms, fever, and pleocytosis without bacterial growth from CSF cultures. Aseptic meningitis is caused by many different viruses, including arbovirus, herpes simplex virus (HSV), cytomegalovirus, adenovirus, and human immunodeficiency virus (HIV). Enteroviruses are the most common cause of aseptic meningitis.

The onset may be abrupt or gradual, and many of the presenting signs and symptoms are the same as bacterial meningitis, including headache, fever, photophobia, and nuchal rigidity.

Most common infection of CNS especially in <1yr

Mononuclear lymphocytes in CSF

Symptomatic treatment. Complications associated with encephalitis and ICP

Variation of Cerebrospinal Fluid Analysis in Bacterial and Viral Meningitis

Manifestations	Bacterial	Viral
White blood cell count	Elevated; increased neutrophils	Slightly elevated; increased lymphs
Protein content	Elevated	Normal or slightly increased
Glucose content	Decreased	Normal
Gram stain; bacteria culture	Positive	Negative
Color	Turbid or cloudy	Clear or slightly cloudy
Opening pressure	Elevated	Normal

Management

Treatment is primarily symptomatic, such as acetaminophen for headache and muscle pain, maintenance of hydration, and positioning for comfort.

Until a definitive diagnosis is made, antimicrobial agents may be administered and isolation enforced as a precaution against the possibility that the disease might be of bacterial origin.

Nursing care is similar to the care of the child with bacterial meningitis. The clinical course of viral meningitis is much shorter and typically without any significant complications.

Hydrocele

A hydrocele is the presence of peritoneal fluid in the scrotum between the parietal and visceral layers of the tunica vaginalis and is the most common cause of painless scrotal swelling in children and adolescents, along with nonincarcerated inguinal hernia.

Hydrocele is a collection of peritoneal fluid that accumulates in the scrotum through a small passage called the processus vaginalis.

This processus is a finger-like projection in the inguinal canal through which the testes descend.

Usually, the processus closes soon after birth; if the processus does not close, fluid from the peritoneal cavity passes through, causing hydrocele.

This is the same passage through which intestines may slip, causing inguinal hernia.

If the hydrocele remains by the end of the first year, corrective surgery is performed.

Nursing Care Management

Surgical correction is an outpatient procedure. Advise parents that there may be temporary swelling and discoloration of the scrotum that resolves spontaneously. Straddle toys are avoided for 2 to 4 weeks and strenuous activities in older boys may be avoided for 1 month. If a dressing is used, it is removed in 2 to 3 days and typically the child can bathe in 3 days.

Kawasaki Disease

Kawasaki disease is an acute systemic vasculitis of unknown cause

Kawasaki disease (mucocutaneous lymph node syndrome) is an acute, febrile disease that is most often seen in boys younger than 5 years.

The etiology is unknown, but the disease appears to be caused by an infectious agent.

After an infection, an alteration in the immune system occurs.

Most cases occur in the late winter or early spring.

The major concern for the child is development of cardiac involvement.

The acute disease is self-limited; however, without treatment, approximately 20% to 25% of children develop coronary artery dilation or aneurysm formation

The illness is not spread by person-to-person contact; however, several factors support an infectious etiologic trigger, possible in a genetically susceptible host.

Clinical Manifestations

An elevated temperature from first day of the illness and may continue 1 to 3 weeks.

Irritability; lethargy

inflammation in the conjunctiva in both eyes

strawberry colored tongue; dry, red, cracked lips

edema in the hands and feet; and red, swollen joints are seen. The skin on the fingers and toes peels in layers, and a rash covers the trunk and extremities.

Cervical lymph nodes may be enlarged.

Inflammation of the arteries, veins, and capillaries occurs, and this inflammation can lead to serious cardiac concerns, including aneurysms and thrombus.

The child may report abdominal pain.

- Kawasaki disease manifests in three phases: acute, subacute, and convalescent

1. Acute—high fever that does not respond to antibiotics or antipyretics; child is irritable.
2. Subacute—fever resolves, irritability continues; greatest risk for aneurysms.
3. Convalescent—symptoms are gone; phase continues until lab values are normal and child's energy, appetite, and temperament have returned.

Diagnostic Criteria for Kawasaki Disease

Child must have fever for more than 5 days along with four of five clinical criteria* (diagnosis may be made on day 4 by an experienced clinician if child has all the clinical criteria):

1. Changes in the extremities: In the acute phase edema, erythema of the palms and soles; in the subacute phase, periungual desquamation (peeling) of the hands and feet

2. Bilateral conjunctival injection (inflammation) without exudation

3. Changes in the oral mucous membranes, such as erythema of the lips, oropharyngeal reddening; or "strawberry tongue" (large papillae are exposed)

4. Polymorphous rash

5. Cervical lymphadenopathy (one lymph node >1.5 cm)

Diagnosis

For Kawasaki disease to be diagnosed, the child must have an elevated temperature and four of the following symptoms:

- cervical lymphadenopathy; conjunctivitis; dry, swollen, cracked lips;
- strawberry tongue; aneurysm; abdominal pain; peeling of hands and feet; trunk rash; or red, swollen joints.

The white blood cell count (WBC) and erythrocyte sedimentation rate (ESR) are elevated.

During the subacute stage, the platelet count increases, which may lead to blood clotting and cardiac problems.

Echocardiograms may show cardiac involvement.

Therapeutic Management

The current treatment of children with Kawasaki disease includes high-dose intravenous immunoglobulin (IVIG) along with salicylate therapy.

IVIG has been demonstrated to be effective at reducing the incidence of coronary artery abnormalities when given within the first 10 days of the illness and ideally in the first 7 days of illness.

Aspirin is used in an anti-inflammatory dose (80 to 100 mg/kg/day in divided doses every 6 hours) to control fever and symptoms of inflammation. However, after the fever has subsided, aspirin can be reduced to an antiplatelet dose (3 to 5 mg/kg/day).

Additional anticoagulation (e.g., clopidogrel [Plavix], enoxaparin [Lovenox], or warfarin) may be indicated in children who have medium-sized or giant coronary artery aneurysms.

Burns

Burn injuries are usually attributed to extreme heat sources but may also result from exposure to cold, chemicals, electricity, or radiation.

Most burns are relatively minor and can be treated in an outpatient setting. However, burns involving a large body surface area, critical body parts, or the geriatric or pediatric population often benefit from treatment in specialized burn centers.

More than 70% of burn accidents happen to children younger than age 5 years.

Nearly all childhood burns are preventable, and this causes considerable guilt for families and the child. Adult carelessness, the child's exploring and curious nature, and failure to supervise the child adequately all contribute to the high incidence of burns in children. In addition, burns are a common form of child abuse.

When burns are categorized according to the patient's age and type of injury, the following patterns become apparent

- Burns may result from various causes including:

1. Scalds from hot liquids, which are common in small children and result from a dangling electric coffee-maker cord, pans of hot liquid on the stove with handles turned out, cups of hot tea or coffee, bowls of soup or other hot liquids, or small children left alone in bathtubs. Dangerous, sometimes fatal, burns can occur from these conditions.
2. Burns from fire are the second most common kind of burn, resulting from children playing with matches or being left alone in buildings that catch fire.
3. Careless use of smoking materials is a common cause of house fires. Although cigarette lighters are currently being produced with a “child-safe” lighting mechanism, they should still be kept away from children.
4. Electricity can cause severe facial or mouth burns to infant and toddlers who bite on electrical cords plugged into a socket; such burns may require extensive plastic surgery.

Characteristics of Burn Injury

The physiologic responses, treatment modalities, prognosis, and disposition of the injured child are all directly related to the amount of tissue destroyed. Therefore the severity of the burn injury is assessed on the basis of the percentage of total body surface area (TBSA) burned and depth of the burn.

Among children in the school-age group or younger age groups, a burn that is 10% TBSA can be life threatening if not treated correctly.

Other important factors in determining the seriousness of the injury are the child's age and general health, the causative agent, the location of the wounds, the presence of respiratory involvement, and any associated injury or condition.

Type of Injury

The majority of burns result from contact with thermal agents, such as a flame, hot surfaces, or hot liquids. Of those children who die from fire or burns, 44% were ages 4 and under; and of all children's deaths due to fire and burns, 87% were involved in a residential home fire.

Electrical injuries caused by household current have the greatest incidence in young children, who insert conductive objects into electrical outlets and bite or suck on connected electrical cords.

Chemical burns are seen in the pediatric population and can cause extensive injury because noxious agents exist in many cleaning products commonly found in the home.

Extent of Injury

The extent of a burn is expressed as a percentage of the TBSA. This is most accurately estimated by using specially designed age-related charts. It is more efficient to use a chart designed to assign body proportions to children of different ages.

Depth of Injury

A burn is a three-dimensional wound that is also assessed in relation to depth of injury. Traditionally, the terms first, second, and third degree have been used to describe the depth of tissue injury. However, with the current emphasis on wound healing, these have been replaced by more descriptive terms based on the extent of destruction to the epithelializing elements of the skin.

- Superficial (first-degree) burns are usually of minor significance. This type of injury involves the epidermal layer only. There is often a latent period followed by erythema. Tissue damage is minimal, and there is no blistering. The protective functions of the skin (such as bacterial and vapor barrier) remain intact, and systemic effects are rare. Pain is the predominant symptom, and the burn heals in 5 to 10 days without scarring. A mild sunburn is an example of a superficial burn.

- Partial-thickness (second-degree) burns involve the epidermis and varying degrees of the dermal layer. These wounds are painful, moist, red, and blistered. With superficial partial-thickness burns, dermal elements are intact, and the wound should heal in approximately 14 to 21 days with variable amounts of scarring. The wound is extremely sensitive to temperature changes, exposure to air, and light touch. Although classified as second-degree or partial-thickness burn, deep dermal burns resemble full-thickness injuries in many respects except that sweat glands and hair follicles remain intact.

- Full-thickness (third-degree) burns are serious injuries that involve the entire epidermis and dermis and extend into subcutaneous tissue. Nerve endings, sweat glands, and hair follicles are destroyed. The burn varies in color from red to tan, waxy white, brown, or black. It is distinguished by a dry, leathery appearance and texture since the elasticity of the dermis is compromised.

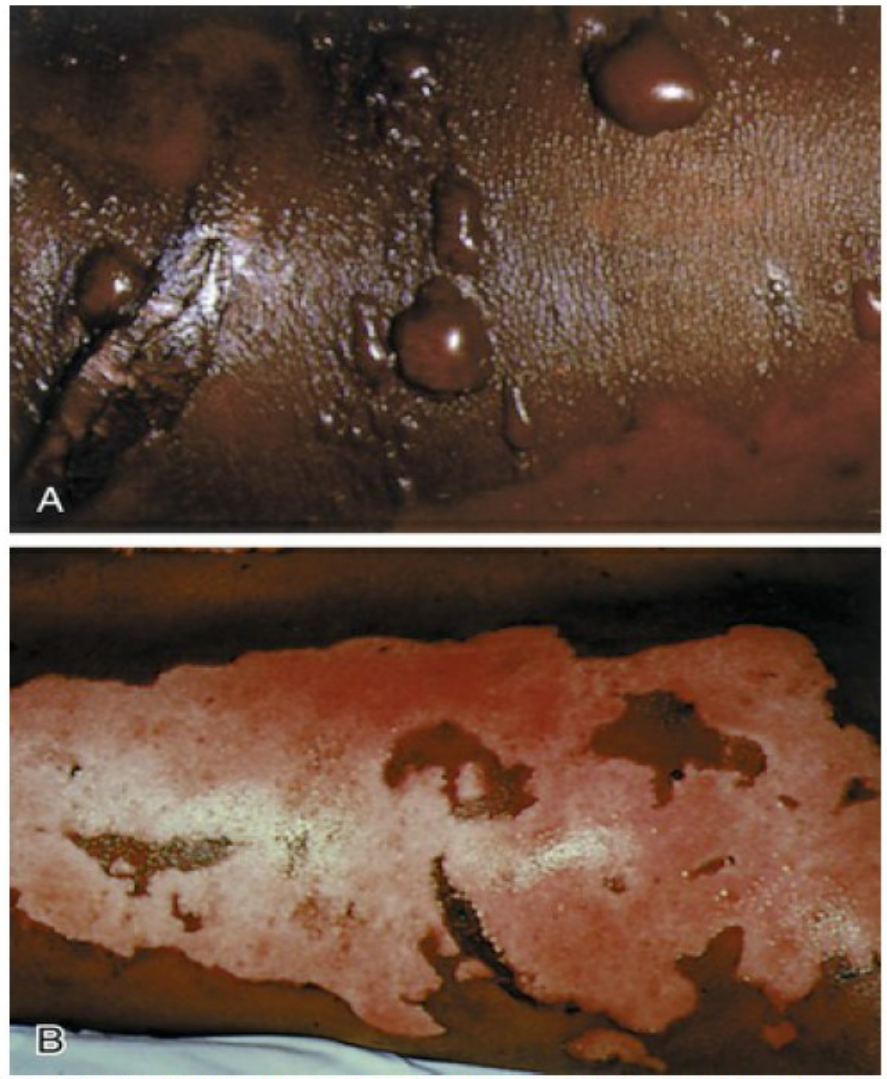


FIG 13-5 Superficial partial-thickness burns on an African-American child. **A,** Blisters intact. **B,** Blisters removed. (Courtesy of Hillcrest Medical Center, Tulsa, OK.)



FIG 13-6 *Bottom to top:* Deep partial-thickness burn *(red area)*, full-thickness burn *(white area)*, and full-thickness burn with eschar *(brown area)*. (Courtesy of Hillcrest Medical Center, Tulsa, OK.)

- Fourth-degree burns are full-thickness burns that involve underlying structures, such as muscle, fascia, and bone. The wound appears dull and dry, and ligaments, tendons, and bone may be exposed.



FIG 13-7 Full-thickness burn with muscle and fascia involved. (Courtesy of Hillcrest Medical Center, Tulsa, OK.)

Therapeutic Management
Emergency Care

The initial management of the burn patient begins at the scene of injury. The first priority is to stop the burning process. The child should then be transported immediately to the nearest medical facility for treatment and evaluation. The child and the family are usually extremely frightened and anxious; sensitivity to their emotional state and reassurance should be provided during the transport process.

Emergency Treatment

Burns

Minor Burns

Stop the burning process:

- Remove burned clothing and jewelry.
- Apply cool water to the burn or hold the burned area under cool running water.
- Do not use ice.

Do not disturb any blisters that form unless the injury is from a chemical substance.

Do not apply anything to the burn.

Cover with a clean cloth if risk of damage or contamination.

Major Burns

Stop the burning process:

- Flame burns—smother the fire.
- Place victim in the horizontal position.
- Roll victim in a blanket or similar object; avoid covering the head.
- Remove burned clothing and jewelry.

Assess for an adequate airway and breathing.

If a child is not breathing, begin mouth-to-mouth resuscitation.

Cover burn with a clean cloth.

Keep victim warm.

Begin intravenous (IV) and oxygen therapy as prescribed.

Transport to medical aid.

Treatment of Moderate to Severe

Burns: First Phase—48 to 72 Hours

- Hypovolemic shock is the major manifestation in the first 48 hours in massive burns. As extracellular fluid pours into the burned area, it collects in enormous quantities, which dehydrates the body. Edema becomes noticeable, and symptoms of severe shock appear. Intense pain is seldom a major factor. Symptoms of shock are low blood pressure, rapid pulse, pallor

Airway. Inhalation injury is a leading cause of complications in burns. If there are burns around the face and neck or if the burns occurred in a small enclosed space, inhalation injury should be suspected. In fires, toxic substances and the heat produced cause damage to respiratory tract.

Intravenous Fluids, The primary concern is to replace body fluids that have been lost or immobilized at the burn area, most often lactated Ringer ’s should relieve the child’s thirst, which is usually severe, and sips of water may be allowed.

- Oral feedings can be started when bowel sounds are heard. Nasogastric feedings may be needed to supplement intake.

Urine Output and Diuresis. Urinary output, which may be decreased because of the decrease in blood volume

Infection control

Management of Burns

- Minor Burns

Minor burns can usually be managed adequately on an outpatient basis when it is determined that the parent can be relied on to carry out instructions for care and observation.

Close follow-up to ensure adherence with treatment.

The burn is cleansed with a mild soap and tepid water. Debridement of the burn includes removal of any embedded debris, chemicals, and devitalized tissue

The dressing consists of non-adherent fine-mesh gauze placed over the ointment and a light wrap of gauze dressing that avoids interference with movement, this helps keep the burn clean and protect it from trauma.

- Major Burns

The first priority is airway maintenance. The inhalation of noxious agents or respiratory burns is suggested when there is a history of injury in an enclosed space; edema of the oral and nasal membranes; burn injury to the face, nares, and upper torso; hyperemia; and blisters or evidence of trauma to the upper respiratory passages.

If the child exhibits air hunger, or other signs of respiratory distress, an endotracheal tube is inserted to maintain the airway.

Controlled intubation is preferred to an emergency intubation. Intubation allows for the delivery of humidified oxygen, the removal of secretions from respiratory passages, and the provision of ventilatory support.

Fluid replacement therapy

The objectives of fluid therapy are to (1) compensate for water and sodium lost to traumatized areas and interstitial spaces, (2) reestablish sodium balance, (3) restore circulating volume, (4) provide adequate perfusion, (5) correct acidosis, and (6) improve renal function.

Crystalloid solutions are used during this initial phase of therapy.

However, children with burns usually require fluids in excess of their calculated maintenance and replacement volume.

Nutrition

The enhanced metabolic requirements and catabolism in severe burns make nutritional needs of paramount importance and often difficult to satisfy. To avoid protein breakdown, the diet must provide sufficient calories to meet the increased metabolic needs and enough protein.

Hypoglycemia can result from the stress of the burn because the liver glycogen stores are rapidly depleted.

A high-protein, high-calorie diet is encouraged. Many children have poor appetites and are unable to meet energy requirements solely by oral feeding.

Medication

Antibiotics are usually not administered prophylactically. The administration of systemic antibiotics to control wound colonization is not indicated because decreased circulation to the burned area prevents delivery of the medication to areas of deepest burn injury.

Surveillance cultures and monitoring of the clinical course provide the most reliable indicators of developing infection. Appropriate antibiotics are instituted to treat the specific identified organism population

Some form of sedation and analgesia is required in the care of burned children. Morphine sulfate is the drug of choice for severe burn injuries

Biologic skin coverings “Grafting”

Permanent coverage of extensive burns is a prolonged process that requires repeated operative procedures using general anesthesia for atraumatic care in debridement and grafting. Early closure shortens the period of metabolic stress and decreases the likelihood of burn wound sepsis.

In the acute phase, biologic dressings cover and protect the burn from contamination, reduce fluid and protein loss, increase the rate of epithelialization, reduce pain, and facilitate movement of joints to retain range of motion.

Allograft (homograft) skin is obtained from human cadavers that are screened for communicable diseases. Allograft is particularly useful as a temporary skin covering of surgically excised deep partial- and full-thickness burns and extensive burns when available donor sites are limited.

Xenograft from a variety of species, most notably pigs, is commercially available. In large burns, the porcine xenograft is commonly applied when extensive early debridement is indicated to cover a partial-thickness burn; this provides a temporary covering for the burn until an available autograft can be applied to the full-thickness areas.

A homograft consists of skin taken from another person, which is eventually rejected by the recipient tissue and sloughed off after 3 to 6 weeks. Skin from cadavers is often used in a procedure called an allograft; this skin can be stored and used up to several weeks, and permission for this use is seldom refused.

A heterograft is skin obtained from animals, usually pigs (porcine). Both homografts and hetero grafts provide a temporary dressing after débridement and have been proven to be lifesaving measures for children with extensive burns.

An autograft, consisting of skin taken from the child's own body, is the only kind of skin accepted permanently by recipient tissues, in addition to the skin from an identical twin. Obtaining enough healthy skin to cover a large area is usually impossible; there- fore, homografts are of great value for immediate covering. If the donor site is kept free from infection and grafts of sufficient thinness are taken, the site should be ready for use again in 10 to 12 days. After grafting, the donor and the graft sites are kept covered with sterile dressings.

Complications

Curling’s ulcer (also called a stress ulcer) is a gastric or duodenal ulcer that often occurs after serious skin burns. Ulcers appear during the first 6 weeks.

The health care team must guard carefully against the complication of contractures. If the burn extends over a movable body part, fibrous scarring that forms in the healing process can cause serious deformities and limit movement.

Appendicitis

Appendicitis refers to an inflammation of the appendix.
Inflammation of the vermiform appendix (blind sac at the end of the cecum), is the most common cause of emergency abdominal surgery in childhood.

Obstruction of the lumen of the appendix is the primary cause of appendicitis.
The obstruction usually is caused by hardened fecal matter or a foreign body. Swollen lymphoid tissue, frequently occurring after a viral infection, can also obstruct the appendix.

This obstruction causes circulation to be slowed or interrupted, resulting in pain and necrosis of the appendix.

The necrotic area can rupture, causing escape of fecal matter and bacteria into the peritoneal cavity and resulting in the complication of peritonitis.

Most cases of appendicitis in childhood occur in the school-age child.

Clinical Manifestations

Right lower quadrant abdominal pain
Fever
Rigid abdomen
Decreased or absent bowel sounds
Vomiting (typically follows onset of pain)
Constipation or diarrhea
Anorexia
Tachycardia
Rapid, shallow breathing
Pallor
Lethargy
Irritability
Stooped posture

Treatment

Surgical removal of the appendix is necessary and should be performed as soon as possible after diagnosis.

If the appendix has not ruptured before surgery, the operative risk is nearly negligible. Even after perforation has occurred, the mortality rate is less than 1%.

Food and fluids by mouth are withheld before surgery. If the child is dehydrated, IV fluids are ordered. If fever is present, the temperature should be reduced to below 102 F (38.9 C).

Recovery is rapid. The child is ambulated early and can leave the hospital a few days after surgery. If peritonitis or a localized abscess occurs, gastric suction, parenteral fluids, and antibiotics may be ordered.

Diabetes Mellitus

DM is a chronic disorder of metabolism characterized by hyperglycemia and insulin resistance. It is the most common metabolic disease, resulting in metabolic adjustment or physiologic change in almost all areas of the body.

Traditionally, DM had been classified according to the type of treatment needed. The old categories were insulin-dependent diabetes mellitus (IDDM), or type I, and non–insulin-dependent diabetes mellitus (NIDDM), or type II. In 1997, these terms were eliminated because treatment can vary (some people with NIDDM require insulin) and because the terms do not indicate the underlying problem. The new terms are type 1 and type 2, using Arabic symbols to avoid confusion

Characteristics of Type 1 and Type 2 Diabetes Mellitus

Characteristic	Type 1	Type 2
Age at onset	<20 years	Increasingly occurring in younger children
Type of onset	Abrupt	Gradual
Sex ratio	Affects males slightly more than females	Females outnumber males
Percentage of diabetic population	5% to 8%	85% to 90%
Heredity:		
Family history	Sometimes	Frequently
Human leukocyte antigen	Associations	No association
Twin concordance	25% to 50%	90% to 100%
Ethnic distribution	Primarily whites	Increased incidence in American Indians, Hispanics, African Americans
Presenting symptoms	Three Ps common—polyuria, polydipsia, polyphagia	May be related to long-term complications
Nutritional status	Underweight	Overweight
Insulin (natural):		
Pancreatic content	Usually none	>50% normal
Serum insulin	Low to absent	High or low
Primary resistance	Minimum	Marked
Islet cell antibodies	80% to 85%	<5%
Therapy:		
Insulin	Always	20% to 30% of patients
Oral agents	Ineffective	Often effective
Diet only	Ineffective	Often effective
Chronic complications	>80%	Variable
Ketoacidosis	Common	Infrequent

Type 1 Diabetes Mellitus

Type 1 diabetes mellitus is estimated to affect about 1 in 600 children between the ages of 5 and 15 years.

Management of diabetes in children is different from that in adults and requires conscientious care geared to the child's developmental stage.

Pathogenesis

It is known to result from dysfunction of the beta (insulin-secreting) cells of the islets of Langerhans in the pancreas which produce insulin

Some researchers believe that the presence of an acute infection during childhood may trigger a mechanism in genetically susceptible children, activating beta-cell dysfunction and disrupting insulin secretion.

Pancreatic tumors, pancreatitis, and long-term corticosteroid use are contributing factors.

In children, type 1 diabetes causes an abrupt pronounced decrease in insulin production, resulting in decreased ability to derive energy from the food eaten. Large amounts of protein and fat are used to supply the child's energy needs, causing loss of weight and slowed growth.

This combination of failure to gain weight and lack of energy may be the initial reason the child is brought to the health care provider 's attention.

However, a health care provider may not see the child until symptoms of ketoacidosis are evident.

It typically starts in children or young adults who are slim, but it can arise in adults of any age.

Clinical Manifestations of Type 1 Diabetes Mellitus

Polyphagia

Polyuria

Polydipsia

Weight loss

Enuresis or nocturia

Irritability; "not himself" or "not herself"

Shortened attention span

Lowered frustration tolerance

Dry skin

Blurred vision

Poor wound healing

Fatigue

Flushed skin

Headache

Frequent infections

Hyperglycemia

- Elevated blood glucose levels
- Glucosuria

Diabetic ketosis

- Ketones and glucose in urine
- Dehydration in some cases

Diabetic ketoacidosis (DKA)

- Dehydration
- Electrolyte imbalance
- Acidosis
- Deep, rapid breathing (Kussmaul respirations)

Clinical Manifestations

Classic symptoms of type 1 diabetes mellitus are

- polyuria (dramatic increase in urinary output, probably with enuresis)
- polydipsia (increased thirst)
- polyphagia (increased hunger and food consumption).

These symptoms are usually accompanied by weight loss or failure to gain weight and lack of energy, even though the child has increased food consumption. is likely to progress to diabetic ketoacidosis, because of inadequate insulin production, carbohydrates are not converted into fuel for energy production. Fats are then mobilized for energy but are incompletely oxidized in the absence of glucose. Ketone bodies (acetone, diacetic acid, and oxybutyric acid) accumulate.

Diabetic ketoacidosis is characterized by drowsiness, dry skin, flushed cheeks and cherry-red lips, acetone breath with a fruity smell, and Kussmaul breathing (abnormal increase in the depth and rate of the respiratory movements).

Nausea and vomiting may occur.

If untreated, the child lapses into coma and exhibits dehydration, electrolyte imbalance, rapid pulse, and subnormal temperature and blood pressure.

Diagnosis

Early detection and control are critical in postponing or minimizing later complications of diabetes.

The nurse should observe carefully for any signs or symptoms in all members of a family with a history of diabetes.
The family also should be taught to observe the children for frequent thirst, urination, and weight loss. All relatives of diabetics are considered a high risk group and should have periodic testing.

a fingerstick glucose test and for ketones in the urine using a urine dipstick test.
a fasting blood sugar (FBS) is performed. An FBS result of 200 mg/dL or higher almost certainly is diagnostic for diabetes when other signs such as polyuria and weight loss, despite polyphagia, are present.
Although glucose tolerance tests are performed in adults to confirm diabetes, they are not commonly used in children. The traditional oral glucose tolerance test is often unsuccessful in children because they may vomit the concentrated glucose that must be swallowed.

Treatment

Management of type 1 diabetes in children includes insulin therapy and a meal and exercise plan.

Treatment of the diabetic child involves the family and child and a number of health team members, such as the nurse, the pediatrician, the nutritionist, and the diabetic nurse educator. After diabetes is diagnosed, the child may be hospitalized for a period of time. This allows the condition to be stabilized under supervision.

The child's teacher, the school nurse, and others who supervise the child during daily activities must be informed of the diagnosis.

Insulin Therapy. Insulin therapy is an essential part of the treatment of diabetes in children. The dosage of insulin is adjusted according to blood glucose levels so that the levels are maintained near normal.

- Insulin reaction (insulin shock, hypoglycemia) is caused by insulin overload, resulting in too-rapid metabolism of the body's glucose. This may be attributable to a change in the body's requirement, carelessness in diet (such as failure to eat proper amounts of food), an error in insulin measurement, or excessive exercise. Because diabetes in children is very labile (unstable, fluctuating), the child is subject to insulin reactions.

- Insulin Regimen. Most children with newly diag- nosed diabetes show a decreased need for insulin during the first weeks or months after control is estab- lished. This is often referred to as the "honeymoon period," and it should be explained to the family in advance to avoid false hope. As the child grows, the need for insulin increases and continues to do so until the child reaches full growth. Again, family caregivers need to know that this is normal and that the child's condition is not getting worse.

Enuresis

Enuresis, or bed-wetting, is involuntary urination beyond the age when control of urination commonly is acquired.

Many children do not acquire complete nighttime control before 5 to 7 years of age, and occasional bed-wetting may be seen in children as late as 9 or 10 years of age.

Boys have more difficulty than do girls, and in some instances enuresis may persist into the adult years.

Enuresis may have a physiologic or psychological cause

Physiologic causes may include a small bladder capacity, urinary tract infection, and lack of awareness of the signal to empty the bladder.

Persistent bed- wetting in a 5- or 6-year-old child may be a result of rigorous toilet training before the child was physically or psychologically ready. Emotional stress can be a precipitating factor. The health care team also needs to consider the possibility that enuresis can be a symptom of sexual abuse.

If a physiologic cause has been ruled out, efforts should be made to discover possible psychological causes, including emotional stress. If the child is interested in achieving control, waking the child during the night to go to the toilet or limiting fluids before retiring may be helpful.

However, these measures should not be used as a replacement for searching for the cause. Help from a pediatric mental health professional may be needed.

The family caregiver may become extremely frustrated about having to deal with smelly wet bedding every morning.

The child may go to great efforts to hide the fact that the bed is wet.

Health care personnel must take a supportive understanding attitude toward the problems of the caregiver and the child, allowing each of them to ventilate feelings and providing a place where emotions can be freely expressed.

Encopresis

Encopresis is chronic involuntary fecal soiling beyond the age when control is expected (about 3 years of age).

Speech and learning disabilities may accompany this problem.

If no organic causes (e.g., worms, megacolon) exist, encopresis indicates a serious emotional problem and a need for counseling for the child and the family caregivers.

Some experts believe that overcontrol or undercontrol by a caregiver can cause encopresis.

Recommendations for treatment differ; however, the most important goal is recognition of the problem and referral for treatment and counseling.

Treatment includes:

Cleaning the hard stool out of the lower colon, Keeping bowel movements soft so the stool will pass easily, diet & exercise is very important.

Muscular Dystrophies

Muscular dystrophies (MDs) constitute the largest and most important single group of muscle diseases of childhood.

The MDs have a genetic origin in which there is gradual degeneration of muscle fibers, and they are characterized by progressive weakness and wasting of symmetric groups of skeletal muscles, with increasing disability and deformity. In all forms of MD, there is an insidious loss of strength, but each type differs in regard to the muscle groups affected, age of onset, rate of progression, and inheritance pattern.

1860's Guillaume Duchenne described boys who grew progressively weaker and died in their teens, Named after him.

The most common form, Duchenne muscular dystrophy (DMD),

Duchenne (Pseudohypertrophic) Muscular Dystrophy

DMD is the most severe and the most common MD of childhood.

It is inherited as an X-linked recessive trait, and the single-gene defect is located on the short arm of the X chromosome.

DMD has a high mutation rate, with a positive family history in about 65% of cases.

The incidence is approximately 1 in 3600 male births for the Duchenne form

The diagnosis of DMD is primarily established by blood polymerase chain reaction (PCR) for the dystrophin gene mutation

DMD is characterized by the wasting away of muscles.

Due to deficiency of the muscle protein Dystrophin

Diagnosis in boys usually occurs between 16 months and 8 years.

Parents are usually the first to notice problem.

Without intervention, average age survival 19yrs

Improved use of respiratory support can see survival into 40’s

Characteristics of Duchenne Muscular Dystrophy

i. Early onset, usually between 3 and 7 years old
ii. Progressive muscular weakness, wasting, and contractures
iii. Calf muscle pseudohypertrophy in most patients
iv. Loss of independent ambulation by 9 to 12 years old
v. Slowly progressive, generalized weakness during teenage years

Therapeutic Management

No curative treatment exists for childhood MD. The use of the corticosteroids prednisone has been evaluated as a treatment for DMD

Several clinical trials demonstrated increased muscle strength and improved performance and pulmonary function, with significant decrease in the progression of weakness, when prednisone was administered for 6 months to 2 years

Maintaining optimal function in all muscles for as long as possible is the primary goal; secondary is the prevention of contractures

Maintenance of function often involves stretching exercises, strength and muscle training, breathing exercises to increase and maintain vital lung capacity, range-of-motion exercises, surgery to release contracture deformities, bracing, and performance of ADLs.

Parents should always be involved in making decisions about the child's care, and teaching regarding home safety and prevention of falls is important as well. Parents should also be encouraged to have the child keep follow-up appointments for medical care and physical and occupational therapy

Respiratory and cardiac problems become the central focus of the debilitating illness. Children with neuromuscular disease develop abnormal breathing patterns, and hypoxia may occur as a result of inadequate oxygenation

Several devices are available for children with neuromuscular disease to assist in clearing the airway when the cough reflex is ineffective or diminished.

The use of routine chest physiotherapy (postural drainage) for DMD has not been adequately evaluated for its effectiveness in clearing the airway of mucus except when there is focal atelectasis and mucus plugging the airways

Questions ???

Thank You